The Macroeconomic Consequences of Political Instability

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Abstract

Do coups d'état lower productivity, and does the regime a coup produces shape the damage? Using the Bjørnskov–Rode panel of the world's regimes and Penn World Table productivity, we follow coups two ways: clean recurrent coup episodes with a stacked, entropy-balanced event study, and countries' first coups with local projections. After reweighting a clean risk set to match coup and comparison countries on predetermined characteristics, coups are followed by a productivity contraction of about three percent on impact, deepening to four percent a year later. This immediate drop is the most robust result: it barely moves across comparison designs, and it is similar whether the coup ends a democracy, replaces one autocrat with another, or leaves democracy standing. The medium-run path is more design-sensitive, and longer horizons provide no robust evidence of permanent scarring. Among first coups the damage looks larger where a coup ends a democracy, though imprecisely. The clearest structural channel runs through investment.

Keywords: Coups d'état, Institutional change, Local projections, Total factor productivity, Option-value framework, Regime transitions

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1 Introduction

A coup d'état is, for the people who live through it, a sudden contraction of the future. For the economy it is harder to price, and this paper asks two questions. First, what happens to productivity after a coup, taken across the broad run of coup events, not a selected few? Second, among the initial institutional shocks that coups deliver, does the regime trajectory a coup produces help organise the size and composition of the response, and what does an option-value model imply about the channels involved?

On the first question we use a stacked event study that draws on all clean coup episodes in the Bjørnskov–Rode regime panel. To build a credible comparison we take a clean risk set of countries with no coup in the event window and reweight it by entropy balancing to match the coup countries on region, decade, pre-coup regime, income, and pre-event productivity. On this balanced design, productivity contracts about three percent in the year of a coup and about four percent a year later. This immediate drop is the most robust result in the paper. It barely moves across control constructions, from never-treated comparisons to the balanced risk set, and it is much the same whether the coup ends a democracy, replaces one autocrat with another, or leaves a democracy standing.

Balancing removes most of the pre-event differences. Leads not used to construct the weights do not jointly reject in the primary window, though a deeper eight-year pre-period reveals some residual sensitivity, so we read the pre-event path as attenuated, not as validated parallel trends. The medium-run path is more design-sensitive than the impact: in the balanced design the year-two-through-five coefficients are jointly detectable, but that persistence weakens once we require a longer pre-event history, and longer horizons show no robust evidence of permanent scarring.

The regime trajectory still matters for the *initial* shock, and a first-event decomposition isolates it. There the democracy-destroying class carries a larger impact-year contraction, about four percent. This first-event estimate rests on few countries and a pre-event trend that the recurrent-event design substantially attenuates, so we do not lean on it.

We read the pattern through the option-value adoption framework of Dixit and Pindyck (1994), in which irreversible productivity-enhancing investment responds to the dispersion of the institutional environment a firm expects to face. The framework says things about the response to the coup *hazard* that this design cannot test, because a falling pre-coup path is also what reverse causality produces, and we set those aside instead of claiming them. What it says about the response to the *realised* regime change is testable: it yields two restrictions and permits one directional investment pattern that a generic account of worse coups does not. Applied to the initial institutional shock, the framework predicts that the first-event productivity contraction is larger where the dispersion of post-coup institutional outcomes widens and smaller where institutions survive. The model permits investment to *rise* in the democracy-to-autocracy class and fall where the post-coup regime remains democratic, the opposite ordering from the one a worse-coups-are-worse story implies. The observed reversal is a directional point-estimate pattern, not a sign the model derives. And productivity and growth should load on the same two channels up to a scale factor, a cross-outcome restriction a reduced-form description does not impose.

In the first-event decomposition these patterns line up, and the productivity contrast between democracy-destroying and democracy-preserving coups holds. That contrast is about the composition of the initial shock, and it does not carry over to the recurrent-event design, where the impact does not differ across transition classes, so the two designs are consistent once read as answers to different questions. We report the investment reversal as directional, since its contrast is imprecise, while the cross-outcome proportionality restriction is formally tested and not rejected, though non-rejection is not evidence for it.

The paper makes three contributions. The first is an empirical fact that survives hard scrutiny: coups are followed by an immediate productivity contraction of about three percent that barely moves across control constructions and coup types. The second is on identification: balancing a clean risk set on predetermined characteristics attenuates the pre-event differences without establishing parallel trends, and the medium-run path, though detectable in the preferred design, is sensitive to the comparison window. The third is a structural interpretation: an option-value model organises the initial institutional shock into coalition and uncertainty channels, with the most robust empirical support on the investment coalition margin. Two objects should be kept distinct: the common productivity disruption that coups bring, and the institutional composition of the initial shock. The paper's two designs speak to each in turn. Where the first generation of this literature priced political instability as a single class of shock and the second established that coups are endogenous to the economies they strike, the contribution here is to the third phase, which asks how the institutional content of a coup organises a productivity cost that is, in the aggregate, general.

The remainder of the paper proceeds as follows. Section 2 situates the paper in the literature on the economic consequences of coups d'état. Section 3 develops the theoretical framework that motivates the empirical classification: a regime-uncertainty mechanism that generates distinct predictions for productivity and growth as functions of the institutional distance the coup traverses. Section 4 describes the data, the Bjørnskov–Rode coup typology, and the implications of the first-event-only restriction for sample composition. Section 5 specifies the two dynamic designs and the transition decomposition of Bennett et al. (2025), which we refer to throughout as BBG after its authors. The first-event analysis uses horizon-specific Jordà local projections, one regression per horizon, while the recurrent-event analysis uses a stacked difference-in-differences event-study design that better handles the overlapping event windows and prior treatment that recurrent coups create. Section 6 reports the reduced-form results in two designs: a recurrent-event design that establishes the broad productivity cost and its dynamics, and a first-institutional-transition design that decomposes the initial shock by regime trajectory, sets the democracy-destroying class against the democracy-preserving counterfactual, and submits the focal cell to its robustness architecture. Section 7 gives the structural interpretation: the two-block joint estimates, the investment reversal, and the cross-outcome restriction, with the nested single-outcome step and the cumulative test in OA Appendix B. Section 8 discusses the implications.

2 Related Literature

The literature on the economic consequences of coups d'état has evolved through three identifiable phases, and the contribution of this paper is located in the third.

The first phase of the literature treated coups and the broader category of *political instability* as a single class of shock and estimated its effects on aggregate growth. Alesina et al. (1996) established the canonical finding that countries with higher political instability grow more slowly, using cross-country panels and government-turnover-based measures of instability. Aisen and Veiga (2013) extended the analysis to identify mechanisms through productivity and capital accumulation. Jong-A-Pin (2009) contributed an important methodological observation: "political instability" as measured in this literature is multidimensional, and different operationalizations capture distinct phenomena (violence-driven, mass-protest-driven, within-regime-competition-driven, and regime-change-driven instability), with potentially different economic consequences. The first-phase literature, however, typically pooled across these dimensions, treating a coup, a mass protest, a government turnover, and a regime-change event as conceptually equivalent shocks.

The second phase, beginning in the 2010s, brought identification concerns to the foreground. Bjørnskov (2022) provided the central result: economic conditions themselves predict coup occurrence, and the cross-country correlation between past growth and current coup probability is strong enough to confound naive estimates of the reverse effect. Bjørnskov and Rode (2020) simultaneously assembled the Regime Dataset that this paper uses, providing carefully coded coup events and pre- and post-coup regime classifications for 192 sovereign countries and 16 self-governing territories from 1950 to 2018, extended through 2025 in the version we use, with the 204-country merged sample of Section 4 the active estimation window. The combination (a curated event dataset and an explicit acknowledgment that the events are endogenous to the macro environment) shifted the literature toward identification strategies that account for the reverse channel. Local-projection and event-study designs with pre-event falsification tests (Jordà, 2005; Callaway and Sant'Anna,

2021) became the natural tools for this work.

The third phase, of which the present paper is part, focuses on the *institutional channels* through which coups affect the economy and on the heterogeneity of coup events themselves. The motivating result is the working paper of Bennett et al. (2025), who use labor-productivity data from the Groningen Growth and Development Centre 10-sector database to estimate the effects of coups on productivity in a panel of 39 countries between 1950 and 2012. They find that the productivity effects of coups concentrate in coups that produce a transition from democracy to civilian autocracy, and they articulate the theoretical mechanism in terms of institutional restructuring: changes in property-rights regimes, judicial constraints, and the formal post-coup environment alter the conditions under which firms invest in productivity-enhancing innovation. The BBG framework is distinguished from earlier work in two respects. First, it isolates a specific class of coup events (democracy-destroying transitions) instead of pooling across heterogeneous classes. Second, it identifies a specific channel, formal institutional change, where earlier work invoked generic political uncertainty.

Related strands in the broader political-economy literature support the institutional-channel framing. Acemoglu and Robinson (2008) model the persistence of elite political power across regime changes, showing that investment in de facto power can offset a loss of de jure power so that economic institutions persist, with implications for the institutional environment a regime change leaves behind. Meyersson (2014) uses a regression-discontinuity design on close municipal elections in Turkey to estimate the consequences of a change in political control for human capital, and is cited here as an example of credible identification of the effects of who holds political power, not for its treatment or its outcome.

The earlier literature on the macroeconomic effects of political instability through inflation (Aisen and Veiga, 2007; Cukierman et al., 1992), fiscal policy (Darby et al., 2004), and foreign investment (Chen and Feng, 1996; Sabir et al., 2019) remains relevant for the broader set of outcomes we examine. In the local-projection results of Section 6.2, foreign direct investment (FDI), inflation and unemployment show no detectable impact-year response in the first-event-only sample, and government spending contracts at the impact year without being treated as a focal result. These nulls do not contradict the earlier literature, which used different identification strategies and different operationalizations of instability. They indicate that under the specification we adopt (first-event local projections on coup events with country and year fixed effects) the empirical signal for these outcomes does not survive. The paper is therefore most directly positioned as a productivity-channel and institutional-channel contribution, with the broader macro-outcome nulls reported for completeness, not as the focal claims.

3 Theoretical Framework

3.1 Overview

A single mechanism runs through this section. Political instability is a stochastic hazard, the instantaneous arrival rate, or intensity, of a coup, and every forward-looking agent in the model must act on a future that a coup may cut short. Firms choose when to commit irreversible capital, and incumbents choose how much to inflate. Each optimises against the same hazard: for the firm it inflates the option value of waiting, for the incumbent it shortens the horizon over which the cost of inflation is borne. The framework is closest to Bennett et al. (2025), who treat regime uncertainty as a distortion of entrepreneurial judgement. The option-value apparatus is that of Dixit and Pindyck (1994).

The model decomposes a coup's policy consequences into two moments. The *mean* of the post-coup policy distribution captures the coalition channel: which group writes the rules after the event determines the level of expropriation, the seigniorage policy, the public-goods bundle. The *variance* captures the uncertainty channel: how dispersed the possible post-coup outcomes are determines the option value of waiting on investment, technology adoption, and hiring. The two channels load differently on different outcomes, and it is that differential loading, not the size of any single response, which the empirical sections confront. The full formal derivation is given in OA Appendix A.

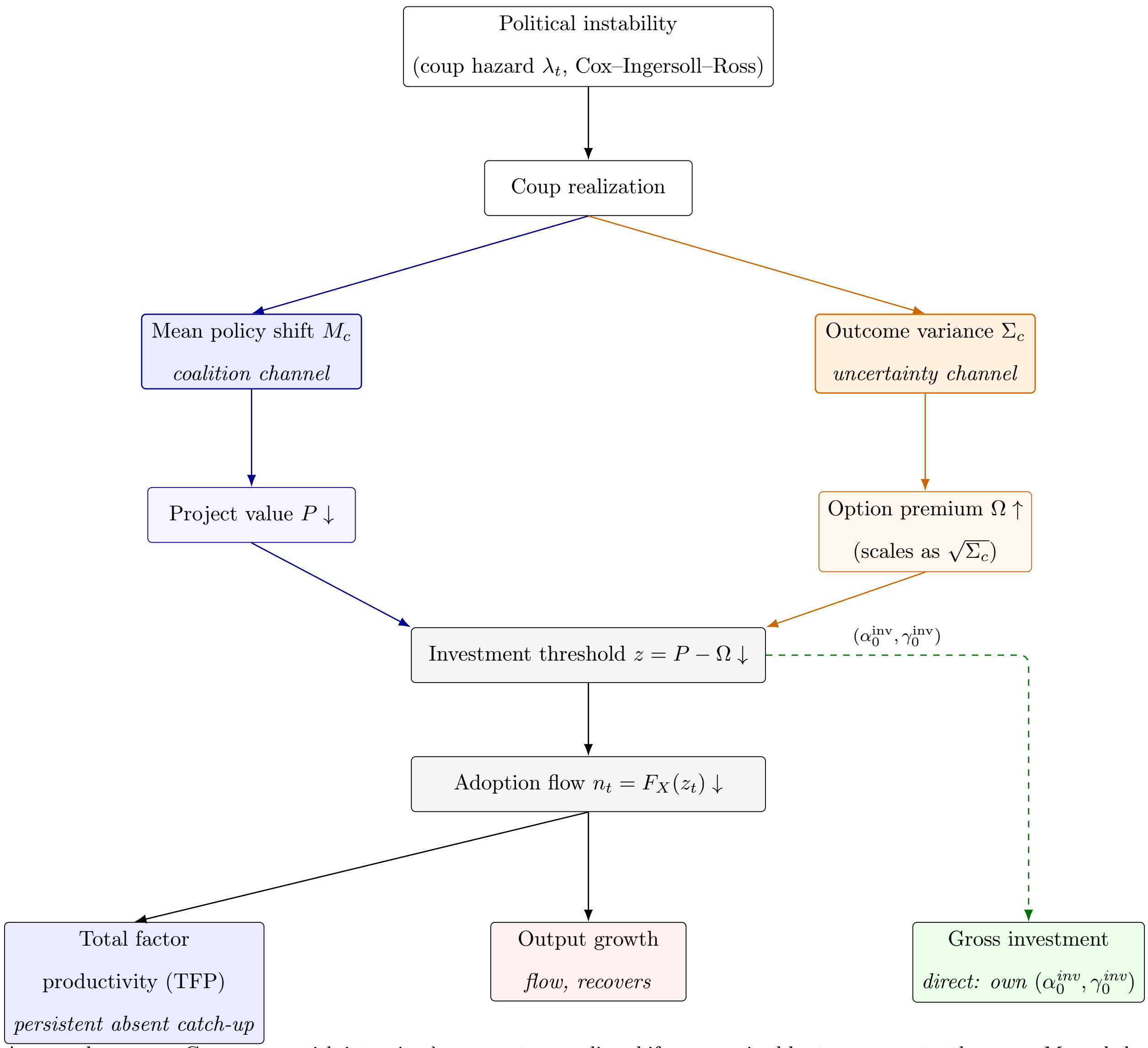
Index a coup by the class c of institutional transition it produces, and write M_c for the mean policy shift a coup of class c delivers and Σ_c for the variance of the post-coup outcomes it can bring. The impact-year productivity response of class c loads on the two channels as

$$\beta^A_{c,0} = \alpha_0\, M_c + \gamma_0\, \sqrt{\Sigma_c}, \tag{1}$$

where the superscript A denotes total factor productivity, α_0 is the coalition loading and γ_0 the uncertainty loading. The square root is the standard option-value scaling: the premium on waiting rises with the standard deviation of the outcomes, not with their variance. Output growth loads on the same pair up to a scale factor κ_{growth}, the fifth parameter estimated in Section 7, and gross investment carries its own pair $(\alpha_0^{\text{inv}}, \gamma_0^{\text{inv}})$, because investment is the firm's direct decision while productivity and growth are downstream consequences of the adoption flow that decision generates. Section 7 estimates these five parameters. OA Appendix A derives the shared productivity block. The separate gross-investment loadings are introduced as a reduced-form extension for measured aggregate investment.

Figure 1 sketches the transmission the rest of this section formalizes: political instability enters as a stochastic coup hazard, a coup realization is summarized by two policy moments, each moment loads on a separate channel of the firm's option-value decision, the firm's adoption-cum-investment decision responds, and it maps to the three macroeconomic outcomes that the empirical section adjudicates.

Figure 1: Transmission from political instability to macroeconomic outcomes. F_X denotes the adoption-cost distribution. The threshold path $z = P - \Omega$ governs the firm's *private* adoption decision. The gross-investment block is a separate reduced-form aggregate proxy that carries its own loadings $(\alpha_0^{\text{inv}}, \gamma_0^{\text{inv}})$ and does not inherit the sign of the displayed z path, since the measured series also contains government and residential investment.



A coup, drawn as a Cox process with intensity λ_t, generates a policy shift summarized by two moments: the mean M_c and the dispersion Σ_c of the class c of transition it produces. The mean loads on the project's expected value through the net-present-value (NPV) channel, the dispersion loads on the option premium of waiting through the standard Black–Scholes $\sqrt{\Sigma_c}$ scaling. Their combination determines the threshold $z = P - \Omega$, which drives the adoption flow $n_t = F_X(z_t)$. The flow maps to two of the outcomes, TFP as the integral of the adoption flow (stock-like persistence) and output growth as the adoption flow itself (flow-like recovery). In the model, investment is instead the firm's direct decision variable, loading on both channels through its own pair of parameters $(\alpha_0^{\text{inv}}, \gamma_0^{\text{inv}})$, which capture the direct effect of post-coup policy shifts on the project value and on the option premium. The full derivation is in OA Appendix A.

The framework delivers more implications than this design can adjudicate, and we separate them here instead of letting the reader discover the difference later. It implies an *ordering* across outcome families, a *persistence* contrast between stock and flow variables, a *crossing* in which a self-coup behaves like a normal successful coup on variance-driven outcomes and like a failed coup on coalition-driven ones, an asymmetry by *coup type*, and an *institutional* scaling in which the per-coup effect rises with ex-ante dispersion. Of these, the crossing and the coup-type asymmetry turn on cells that the first-event-only restriction leaves too thin to distinguish, and the institutional scaling is proxied but not estimated. What the design does confront is set out in Section 3.6, and it is a narrower set.

3.2 The entrepreneur and the option-value channel

A coup is, for the people who build factories, a risk to be priced. Forward-looking firms observe the coup hazard λ_t in continuous time and decide *when* to commit irreversible capital. A higher hazard does two things at once. It depresses the net present value of an operating project, because the expected post-coup change in the capital tax is unfavourable. And it raises the option value of waiting, because the dispersion of the possible post-coup tax outcomes makes the bad realisations costly enough that the option of avoiding them gains value.

The relative weight of these two effects determines which outcomes already respond in the years before a coup occurs, while the hazard is merely elevated, and which respond only once the coup arrives. For outcomes whose value depends on the *variance* of post-coup policy (foreign direct investment, total factor productivity, output growth), the option-value channel dominates: a higher hazard reduces investment even before any coup occurs. For outcomes whose value depends on the *level* of post-coup policy (inflation, government spending), the response works through the incumbent-horizon channel instead of being concentrated at the event: optimal inflation rises with the coup hazard ($\partial m^*/\partial \lambda > 0$, Section 3.4), and a normal successful coup may also produce an unsigned coalition-preference response at the event.

Proposition 1 (Investment and the coup hazard)**.** *For an interior investment threshold and an expected institutionally degrading policy jump $M > 0$, aggregate investment is strictly decreasing in the coup hazard λ_t through both the NPV (coalition) and the option-value (uncertainty) channels. We impose, and do not derive, the ordering that foreign direct investment loads more heavily than domestic investment on both channels, taking foreign capital to be disproportionately exposed to expropriation. The model carries no separate foreign-investment decision rule, so that ordering is a qualitative extension of the framework, not a result of it (OA Appendix A.2).*

Proposition 1 is a property of the model, not a prediction this design tests. Its empirical signature is a decline in investment and productivity *before* the coup, as the hazard rises. That signature is observationally equivalent to the reverse-causality channel of Bjørnskov (2022), in which deteriorating economic conditions raise the probability of a coup: both predict a falling pre-event path, with the same sign. A pre-event coefficient therefore cannot separate the anticipatory channel from the confound that most threatens the design, and we do not use it as evidence for the model. We retain the hazard response because it is what generates the disrupted adoption flow the realised-event predictions rest on, and we treat the pre-event coefficients purely as a pre-trend diagnostic (Section 6.6). The predictions the design does adjudicate are all statements about the response to the *realised* regime change, not to the hazard.

The full derivation, the Cox–Ingersoll–Ross (CIR) specification of the hazard process, the regime-specific calibration to the Bjørnskov–Rode panel, and the formal statement of Proposition 1 are given in OA Appendix A.

Two consequences follow, and they concern different objects. The hazard response of Proposition 1 operates *before* the coup, and for the reason just given we do not test it. The response to the *realised* regime change is a separate matter, and it is where the framework commits itself: within the democracy-destroying class, coalition redistribution toward favoured sectors can *raise* measured investment, while the intra-democracy class preserves the formal regime type, though its two successful coups mean it is not strictly a no-coalition-change class. The separate investment loading permits this cross-class reversal instead of deriving it: the observed ordering is a directional point-estimate pattern, not a sign the model pins down. It remains discriminating, since a generic account in which worse coups are worse predicts the opposite ordering. We hold the model to that pattern, and not to the sign of any single cell.

The two signs are not in conflict, though they look as though they are. Proposition 1 signs the derivative of investment with respect to the *hazard*, $\partial I/\partial\lambda < 0$: a country in which a coup becomes more likely invests less. The realised-response prediction signs a different derivative, the response to the *mean policy shift* a coup delivers, $\partial I/\partial M$, which the coalition loading α_0^{inv} carries. A positive investment coefficient in the democracy-destroying class is therefore not evidence against Proposition 1. It is evidence about a derivative Proposition 1 does not sign. Reading the realised coefficient as a test of the anticipatory proposition would confuse the two, and would let the model claim support from either sign, which is precisely what we are refusing to do.

3.3 Total factor productivity and persistence

Productivity rises when firms adopt better technology. Adoption, like investment, is irreversible and carries a sunk cost, so it is governed by the same option logic: a firm adopts when the value of the new technology, net of the option of waiting, clears the sunk cost. The adoption flow n_t of Figure 1 is therefore decreasing in the coup hazard, $n'(\lambda) < 0$, by Proposition 1 applied to the technology-choice margin.

The difference between productivity and investment is not in the decision but in the *accumulation*. Investment and FDI are flows: when λ reverts after a coup, the flow returns to its normal level. Total factor productivity is a stock, the accumulated body of adopted technology,

$$\ln A_t = \ln A_0 + \int_0^t g\, n_s\, ds + \mu_0 t. \tag{2}$$

Here g is the log-productivity increment per adopted upgrade, n_s the adoption rate, which responds to the tax state and the hazard through the threshold of OA Appendix A.2, and μ_0 a baseline drift unrelated to the coup, all as derived in OA Appendix A.2.3. A coup episode depresses the adoption flow for the duration of the high-λ window. When λ reverts, n returns toward its baseline. Absent a subsequent period of above-baseline adoption that makes up the missed upgrades, $\ln A_t$ inherits a persistent level gap equal to the cumulative shortfall $\int g\,(\bar{n} - n_s)\,ds$ accrued during the episode, where $\bar{n}$ is the flow that would have prevailed absent the coup. If adoption later runs above baseline, part of that gap is recovered. The drift μ_0 is common to both paths and cancels in the difference, which is why it appears in the level but in none of the responses reported below.

This is the model's persistence prediction. The performance flows (investment, FDI, output growth) trace V-shaped responses that return toward trend once the hazard subsides, while total factor productivity, the integral of a disrupted flow, traces a downward step that persists absent subsequent catch-up in adoption. This is consistent with the medium-run persistence we estimate, while the attenuation of the longer-horizon point estimates is consistent with partial catch-up. The contrast can be read off the ratio of the cumulative response over the five years after a coup to the response at the coup itself. A perfectly persistent stock accumulates the same shortfall in every one of the six years, so its ratio approaches six, while a flow that recovers immediately contributes only in the first year and returns a ratio near one. We use that ratio as a description and not as a test. A level that persists more than its own growth rate is close to definitional, and any model with a persistent productivity shock delivers it, so the ratio discriminates weakly between this framework and the alternatives.

3.4 The incumbent and the level channel

One further mechanism earns its place in the main text because it motivates the level-versus-dispersion branch distinction the empirical work rests on. It does not motivate the investment-versus-productivity block split, which turns on the direct-decision-versus-downstream distinction of OA Appendix A.2.3. The incumbent's seigniorage problem generates a *level-channel* response: the coup hazard enters the incumbent's effective discount rate (Cukierman et al., 1992), so inflation responds through the incumbent-horizon channel, not through the dispersion of outcomes, the uncertainty channel. A normal successful coup may also produce an unsigned coalition-preference shift, which the model does not sign. That level-versus-dispersion split does not separate the investment block from the productivity block in the structural specification. Both blocks load on the mean and on the dispersion. What separates them is that investment is the firm's direct decision while productivity and growth are its downstream consequences, which is why investment carries its own pair of loadings.

The model has more branch structure than this. The three coup branches, failed, self-coup and normal successful, can generate a crossing in which the self-coup behaves like a normal coup on variance-driven outcomes and like a failed coup on level-driven ones, and the military or civilian character of the installed coalition refines the successful branch further (Bennett et al., 2021). We do not develop either here, because we cannot test them: the self-coup cell that identifies the crossing holds four events in the first-event-only sample, and the military cell is thin for a structural reason, since military dictatorships typically emerge *from* coups instead of preceding them (Section 4.2). The derivations, the branch prediction table and the crossing diagram are in OA Appendix A, and the reader should treat them as the model's machinery, not as claims this paper advances.

3.5 Institutions as a shock absorber

The cost of a coup is not the same everywhere, and the model says why. A coup's damage is the option-value response of Section 3.2, and two regime features scale it. The first is surprise. A coup is measured against its pre-event baseline. Where the long-run hazard is low, that baseline carries little priced-in coup risk, so a coup is a large and unanticipated shock. Where the long-run hazard is high, coup risk is chronically priced in, the baseline is already depressed, and the same coup moves the economy less. The second feature is the variance of post-coup policy: the more dispersed the outcomes a coup can bring, the larger the option-value response.

The framework also predicts how institutions absorb the shock. Separate the macroeconomic burden of political instability into a per-coup effect and a frequency. Holding post-coup dispersion fixed, or when the surprise margin dominates, strong institutions deliver a *larger* per-coup effect, because a coup is a rarer surprise, but a *low* frequency of coups. The aggregate burden is the product of the two, and it is smaller than under weak institutions whenever the frequency gap dominates the per-coup gap. Weak institutions deliver a muted per-coup effect but a high frequency, and under that same condition a larger aggregate burden. The two factors move in opposite directions, so the model orders the products only under that restriction. Where it holds, the aggregate cost of political instability is lower where institutions are stronger, not because each coup is milder there, but because coups are rare. This resolves an apparent paradox: a parliamentary democracy is politically stable, yet a coup in one is unusually damaging.

Empirical proxy for the variance margin. The variance margin is not directly observable, but the transition decomposition we estimate below provides an empirical proxy:[1] the post-coup regime trajectory captures the institutional distance a coup traverses, with democracy-destroying transitions on one end and intra-democracy outcomes (failed coups and no-change successful coups) on the other. The decomposition is the empirical counterpart of the institutions prediction employed in this paper.[2]

[1] The classification uses the Bjørnskov and Rode (2020) regime codes, and the decomposition design follows Bennett et al. (2025). It is set out in Section 6.3.

[2] A full structural estimation of the variance margin from the Bjørnskov–Rode transition matrix is not used as the primary calibration. OA Appendix B.1 reports it as a secondary one.

3.6 Empirical scope

The design does not test everything the model implies, and we separate the two. The anticipatory investment decline of Proposition 1 is a prediction about the response to the coup *hazard*, and it is not adjudicated here, being observationally equivalent to reverse causality as set out above. The prediction that foreign direct investment loads more heavily on both channels than domestic investment cannot be tested at the hazard either, but a realised regime change moves the same two channels, so it admits an indirect check. We run that check in OA Appendix G and report what it shows instead of reinterpreting it.

Three predictions concern the response to the *realised* regime change, and these the design does confront. First, the impact-year productivity contraction should concentrate in coups that widen the dispersion of post-coup institutional outcomes, and be absent where institutions survive. Second, the coalition channel is permitted to carry opposite signs across the productivity and investment blocks, so that investment can *rise* in the democracy-to-autocracy class and fall where the post-coup regime remains democratic. This is a directional point-estimate pattern the model allows, not a sign it derives, and a generic account in which worse coups are worse predicts investment more negative in the democracy-destroying class, not less. Third, productivity and growth should load on the same two channels up to a scale factor, a cross-outcome restriction the model imposes and a reduced-form description does not.

The three fare differently under the inference standard we adopt, which is the 90% level. The procedure differs by layer: reduced-form contrasts use the wild cluster bootstrap, while the structural estimates use the country-cluster two-step bootstrap of Section 7, which resamples the impact-year coefficients and the calibrated moments jointly. Where each is adjudicated, and what the verdict is, we state at the point where it is reported: the concentration contrast in Section 6.4, the investment contrast and the cross-outcome proportionality restriction in Section 7. The persistence contrast is close to definitional, and we flag it here as illustration, not as a test, so that it is not read as one. The remaining implications of the framework, the institutional-variance ordering, the self-coup crossing and the military–civilian asymmetry, are underpowered in a first-event-only sample and we do not claim them as evidence.

4 Data

We construct a merged country-year panel covering 204 countries from 1960 to 2022, or 12,852 country-years. Of these countries, 119 carry Penn World Table (PWT) version 11.0 productivity data (Feenstra et al., 2015) and 117 enter the total-factor-productivity local projections (LP), so PWT coverage, not the merge, is what binds the estimation sample. The window closes in 2022 because that is the last year with coverage in the main outcomes: PWT 11.0 reports productivity through 2023, but GDP per capita, inflation and foreign direct investment all end in 2022, and we do not want the productivity projections estimated over a year the output-growth projections cannot reach. The underlying Bjørnskov–Rode (BR) dataset covers 208 entities (192 sovereign countries and 16 self-governing territories) through 2025 and is summarized over its full range in Table 3. The merged 204-country panel is the active sample for outcome-based estimation. We draw on the World Bank's World Development Indicators, PWT 11.0, and the Bjørnskov–Rode Regime Dataset. The outcome variables measure macroeconomic and welfare performance. The treatment variable is the incidence of coups d'état.

4.1 Dependent Variables

The analysis focuses on total factor productivity as the primary outcome of theoretical interest, with output growth as a related secondary outcome. Seven additional macroeconomic outcomes are examined for completeness and reported in tables and the appendix: foreign direct investment, gross investment, total trade, inflation, government spending, unemployment, and net migration. The empirical results in Section 6 concentrate on the first two outcomes. Log GDP per capita, used elsewhere as a control, also appears as a supplemental level outcome in OA Figure F3 (OA Appendix F), estimated under the same transition-decomposed local-projection specification. It is not part of the principal nine-outcome table set. Six of the seven are reported in Table 4. Gross investment is not, and it enters the structural analysis of Section 7 as the outcome the investment block is estimated on. OA Appendix G revisits five of them. None is a focal claim.

Total factor productivity is the `rtfpna` index from the PWT 11.0 release, the most recent revision of the dataset incorporating updated price benchmarks and capital stock corrections. The series enters in natural logarithms, so that estimated effects read as log changes and, for effects of the size encountered here, approximately as proportional changes. The choice of release matters for coverage, not for measurement: pre-1972 productivity data for several Latin American countries exist only under PWT 11.0, which widens the set of countries available to the multi-period estimators reported in OA Appendix K. Gross investment is the share of output devoted to investment, and it enters as $\ln(1 + x/100) \times 100$ on the same criterion as the other ratio variables. It is the outcome the structural investment block is estimated on, and it is a different series from the World Bank gross fixed capital formation measure listed among the controls in Table 2, which is not used as an outcome anywhere. Output growth is the annual rate of real GDP growth. FDI and total trade measure external linkages, both expressed in nominal terms. Government spending is expressed as a share of GDP. Inflation is the first difference of the log consumer price index, $\Delta \ln(\text{CPI}_{it}) \times 100$. Unemployment is the World Bank's unemployment rate. Net migration captures the difference between immigration and emigration (Gebremedhin and Mavisakalyan, 2013).[3]

To make effects comparable across outcomes, we place all dependent variables on a logarithmic scale using a consistent criterion. Rate and ratio variables (growth, unemployment, government spending) enter as $\ln(1 + x/100) \times 100$, a transformation that is well defined at

[3]All variables except TFP and gross investment are from the World Bank's World Development Indicators.

zero. Coefficients on this scale are changes in $100\ln(1+x/100)$ and are only locally approximate percentage-point changes: the exact original-unit change of a coefficient b at baseline x_0 is $\Delta x = (100+x_0)(e^{b/100}-1)$. Strictly positive variables (total trade, TFP) enter in natural logarithms. In the recurrent multi-outcome table (OA Table E1) total trade is reported on a 100× log scale, so its coefficients are in the same units as the rate outcomes, while TFP is kept in log points. Variables that can take non-positive values (FDI and net migration) are shifted by the absolute value of the sample minimum plus 0.001 before taking logs. This is an author-chosen transformation. Chen and Roth (2024) is cited for the interpretive limitations of log-like transformations in the presence of zeros, not as the source of this shift. Inflation is measured as the first difference of the log consumer price index. All dependent variables are winsorized at the 1st and 99th percentiles.

4.2 Coups d'État as Events of Political Instability

We measure political instability with coups d'état, drawing on the Bjørnskov–Rode Regime Dataset, version 6.2 (Bjørnskov and Rode, 2020). A coup is a discrete, dated, and externally observable attempt by political or military insiders either to remove the sitting executive or, in a self-coup, for the sitting executive to seize unconstitutional powers. This is a deliberate departure from perception-based indices of instability, whose content is difficult to separate from general institutional quality (Langbein and Knack, 2010): a coup either occurs in a given country-year or it does not, and its timing is recorded, not inferred. The dataset records 569 coup attempts in 114 countries between 1950 and 2025. After the Powell–Thyne re-timing reconciliation (footnote 6 below shifts two coups originally dated 1950 to 1949, before the start of the BR panel proper), the constructed coup panel contains 567 events (Table 3), of which 310 fail and 257 succeed. The mode breakdown on the 569 raw count is 414 military, 143 civilian, and 12 royal. The two re-timed coups are excluded from the constructed panel and therefore from the 257 / 310 success-failure totals in Table 3. Merged into the country-year panel by ISO country code, coups become the treatment whose macroeconomic consequences the remainder of the paper traces.

Instead of treating every coup alike, we separate coups into three arms entered as parallel indicators with non-coup country-years as the omitted category. *Failed coups* (298 failed-coup cell assignments, distinct from the 310 failed individual attempts counted in Table 3) are attempts that are defeated, leaving the incumbent regime in place. A failed coup realizes the turmoil of an openly contested executive without producing any change in leadership or formal institutions, and so supplies political disruption without an institutional shift. *Self-coups* (17) are successful coups in which the sitting head of state remains in office and consolidates power, the classic *autogolpe*. Uncertainty is resolved through entrenchment, not replacement. *Normal successful coups* (230) are successful coups that install a new leader or ruling coalition, where the uncertainty shock is compounded by an actual transfer of power. The three arms are defined on coup-year cells, not on individual events. Twenty cells carry more than one arm in the full record, and seventeen do in the merged 1960–2022 panel, so the arm counts above are assignments and not a partition of the 567 events. The arm counts above are assignments on the full 1950–2025 coup record of Table 3, not on the estimation panel. The classification is descriptive background on the data. Table 2 instead reports its incidence over the merged 1960–2022 panel that the estimation uses. The main empirical specification (Section 5) uses a four-way BBG-style transition decomposition by the pair (pre-coup regime, post-coup regime), which is the empirical operationalization of the institutional-replacement channel central to the paper. Twenty country-years contain coups of more than one arm. Table 1 reports the arms in both samples.

Table 1: Coup-arm assignments by sample

Coup arm	Full record 1950–2025	Merged panel 1960–2022
Any coup (cells)	525	460
Failed	298	260
Normal successful	230	200
Self-coup	17	17

Notes. Coup-year cell assignments in the Bjørnskov–Rode record. Twenty cells carry more than one arm in the full record (column 1) and seventeen in the merged panel (column 2), so the arms do not partition the any-coup total. Column 1 is the full record and is the source of the counts in Section 4. Column 2 is the merged 1960–2022 estimation panel of 12,852 observations and reproduces the arm incidence in Table 2, Panel C (failed 260/12,852 = 0.0202).

A self-coup is identified mechanically. A successful coup is coded as a self-coup when the Bjørnskov–Rode head-of-state record names the same individual as the executive in the year before, the year of, and the year after the event. Because the record is annual, the three-year window distinguishes an *autogolpe* only from restorations that span a calendar year. A deposition and restoration inside the same calendar year leaves the annual head-of-state entries unchanged and is coded as a self-coup.[4]

Because the local-projection design in Section 5 dates every response relative to the event year, the year assigned to each coup is consequential. We therefore cross-check the Bjørnskov–Rode datings against an independent catalogue of coups, that of Powell and Thyne (2011), in its updated release covering events through 2023. The reconciliation is run on the

[4] The seventeen self-coups are: the Democratic Republic of the Congo (1961), Nigeria (1967), Syria (1969), Turkey (1971), Burundi (1972), Uruguay (1973), Cyprus (1974), Maldives (1975), Ghana (1976), Madagascar (1976), Bangladesh (1977), Fiji (1988), Peru (1992), São Tomé and Príncipe (2003), Fiji (2006), Tunisia (2021), and Sudan (2022). Peru's 1992 *autogolpe* under Alberto Fujimori is the motivating case. The strict three-year rule excludes coup-plus-restoration episodes such as Sierra Leone (1997).

Bjørnskov–Rode source file, which records 526 coup country-years.[5] Of these, 439 match a Powell–Thyne coup within one year: 290 are dated to the same year, 131 are dated one year later by Bjørnskov–Rode, and 18 one year earlier. The remaining 87 have no Powell–Thyne counterpart within one year, two of them because they postdate the catalogue's coverage. Of these, 43 differ by two years or more. The asymmetry of the one-year cases (131 late against 18 early) indicates a systematic tendency in the integrated file to assign certain coups to a year adjacent to the one in the source catalogue. Those 131 late country-years contain 137 individual coup events, and for each we adopt the Powell–Thyne year. The country-years that Bjørnskov–Rode dates one year earlier are left unchanged, a conservative choice that avoids manufacturing agreement. All results use this reconciled timeline.[6]

An important descriptive implication of the first-event-only restriction adopted in Section 5 is that military dictatorships become comparatively rare pre-coup states. In the full Bjørnskov–Rode coup panel, the regime the year before the coup is a military dictatorship in 38.7% of coup-year cells (203 of 525), in the first-event-only sample, only 9 of the 110 first-coup cells occur in a country whose regime is classified as military the year before. This compositional shift suggests that many military autocracies in repeated-coup samples are themselves endogenous products of prior coups, with the repeated-coup cycles in countries such as Bolivia, Chad, Sudan, and Burkina Faso emerging from a first coup against a democracy or civilian dictatorship that installs a military regime which then experiences subsequent military-on-military coups.

Coups are the measure of political instability throughout the main analysis.

[5]The constructed country-year panel contains 525 coup-year cells, one fewer than the source file, the difference arising in the ISO merge. The reconciliation is reported on the 526 source cells, and the counts below sum to that total.

[6]The two coups shifted from 1950 to 1949 predate the 1960–2022 estimation window and affect no estimate. The 567 / 310 failed / 257 successful totals in Table 3 reflect the post-re-timing panel, while the raw Bjørnskov–Rode record contains 569 coups (310 failed, 259 successful), the basis for the mode breakdown quoted above.

4.3 Control Variables

The local projections control for the lagged outcome and lagged log GDP per capita, following Acemoglu et al. (2008) and Aisen and Veiga (2013), together with country and year fixed effects. The remaining covariates in Table 2, years of schooling (Barro and Lee, 2010), population, urbanization, natural resource rents, gross fixed capital formation and R&D expenditure, are reported descriptively and do not enter the estimating equations. R&D expenditure in particular is available for far fewer country-years than any outcome.

4.4 Descriptive Statistics

Table 2 reports summary statistics for the outcome and control variables, grouped by family. Economic growth averages 3.6 percent per year. Coverage varies across the panel: R&D expenditure is the thinnest series (2,237 country-years) and unemployment the next thinnest (6,168). All dependent variables are winsorized at the 1st and 99th percentiles. Table 3 summarizes the coup treatment: its distribution across decades, and the split between failed and successful events. Coups are concentrated in Sub-Saharan Africa and Latin America but reach every region except North America, and they are spread across the sample period, with a peak in the 1960s and 1970s and a gradual decline thereafter.[7]

[7] These three figures count three different things and do not form a single decomposition. The panel records 567 coup *events*, which fall into 525 coup-year *cells*, since some country-years contain more than one coup. The arm indicators are defined on cells, and a cell may carry more than one arm: 505 cells carry exactly one and 20 carry exactly two, so the arm *assignments* number $505 + 2 \times 20 = 545$. No coup cell is left without an arm. The first-event-only restriction adopted in Section 5 further reduces the sample to 110 first-coup country-year cells (one treated year per country). All form the eligible first-event sample, and outcome- and horizon-specific regressions use the subset with complete required cells.

Table 2: Descriptive Statistics

Variable	Obs.	Mean	Std. Dev.	Min	Max
Panel A: Macroeconomic Performance					
Economic Growth (%)	10,096	3.61	5.19	−14.14	20.80
Total Factor Productivity	6,697	1.05	0.42	0.35	3.45
Foreign Direct Investment (mn USD)	9,136	4,298	14,697	−2,138	105,293
Total Trade (mn USD)	7,317	115,578	299,353	57	1,844,136
Net Migration (millions)	13,888	−0.00	0.08	−0.37	0.44
Panel B: Macroeconomic Stability					
Inflation Rate (%)	8,506	9.25	22.98	−19.41	477.49
Unemployment Rate (%)	6,168	8.17	5.92	0.60	27.70
Government Spending (% GDP)	8,295	16.15	6.94	4.84	45.78
Panel C: Coup Treatment Indicators					
Any coup	12,852	0.0358	0.186	0	1
Failed coup	12,852	0.0202	0.141	0	1
Self-coup (*autogolpe*)	12,852	0.0013	0.036	0	1
Normal successful coup	12,852	0.0156	0.124	0	1
Military coup	12,852	0.0257	0.158	0	1
Civilian coup	12,852	0.0106	0.102	0	1
Panel D: Control Variables					
GDP Per Capita (USD)	10,337	11,029	12,337	713	59,120
Years of Schooling	7,042	6.48	3.37	0.04	13.74
Population (millions)	13,641	24.89	104.30	0.003	1,420
Urban Population (%)	13,545	51.63	25.71	2.08	100
Gross Fixed Capital Formation (% GDP)	7,720	22.20	8.11	−2.42	93.55
R&D Expenditure (% GDP)	2,237	0.95	0.96	0.01	5.71
Natural Resource Rents (% GDP)	9,235	6.83	11.05	0	88.59

Notes: This table reports summary statistics for selected outcome, treatment, and control variables available in the source panel, organized by outcome family. All dependent variables are winsorized at the 1st and 99th percentiles. Panel C and the outcome/control panels are summarised over different samples, and their N are not directly comparable. Coup treatment indicators (Panel C) are observed for the merged country-year panel, 12,852 observations over 1960–2022. The outcome and control variables are summarised over the full source panel of 14,259 country-years before the merge with the coup record, so their N reflect both missing data and country-years that carry no coup observation. Coverage is uneven: R&D expenditure is the thinnest series at 2,237 country-years. The local-projection estimation samples in Section 6 apply additional restrictions from the first-event identification design and are smaller than the descriptive coverage reported here. Negative FDI values reflect net disinvestment. For variables that can take non-positive values (FDI and net migration), the series is shifted by the absolute value of the minimum plus 0.001 before taking logs. This is an author-chosen shift, with Chen and Roth (2024) cited for the limitations of log-like transforms, not as its source.

The full coup dataset spans 1950 to 2025 and contains 567 events. The decade distribution is reported in Table 3. Coups peak in the 1960s and 1970s and decline thereafter, with the regional distribution weighted toward Sub-Saharan Africa and Latin America. The estimation sample in the local-projection analysis below covers 1960 to 2022, the 1950s row is reported for completeness only.

Table 3: Coups d'État by Decade, 1950–2025

Decade	Failed	Successful	Total
1950s	30	28	58
1960s	50	61	111
1970s	55	62	117
1980s	65	40	105
1990s	54	28	82
2000s	27	18	45
2010s	16	10	26
2020s	13	10	23
Total	310	257	567

Coup events from the Bjørnskov–Rode Regime Dataset v6.2, dated on the timeline reconciled with Powell and Thyne (2011). The 567 events are the 569 recorded coups less two re-timed to 1949, before the start of the panel. Of the 569 recorded coups, Sub-Saharan Africa accounts for 243, Latin America and the Caribbean for 127, the Middle East and North Africa for 67, East Asia and the Pacific for 51, South Asia for 27, and Europe and Central Asia for 23, the remaining 31 are in small states and territories not classified to a region.

5 Empirical Strategy

The empirical strategy uses two designs for two questions. Design A, the recurrent-event design of Section 6.1, estimates the average productivity response to coups on all clean episodes with a covariate-balanced weighted stacked difference-in-differences event study, and is the source of the paper's first result. Design B, developed in the following subsections, restricts attention to the first coup observed in each country, estimates it with horizon-specific local projections, and decomposes the response by the regime trajectory the coup produces (Bennett et al., 2025), with a robustness architecture concentrated on the focal cell the decomposition identifies. The strategy is associational throughout. We do not impose strict parallel-trends assumptions as identification restrictions, and we state separately for each design what the diagnostics do and do not establish in Section 5.6.

5.1 Local-projection specification

For each outcome Y_{it} and horizon $h \in \{-3, -2, 0, +1, \ldots, +8\}$ we estimate the regression

$$Y_{i,t+h} - Y_{i,t-1} = \mu_i^h + \delta_t^h + \beta_h \cdot \mathrm{shock}_{it} + \phi^h \, Y_{i,t-1} + \psi^h \, \ln \mathrm{GDP}_{i,t-1} + \varepsilon_{i,t+h}^h, \tag{3}$$

where shock_{it} is an indicator equal to one in the year of the first coup observed in country i and zero elsewhere. The coefficient β_h estimates the average conditional dynamic response to a first coup, evaluated at horizon h, with the year preceding the event ($h = -1$) implicitly serving as the reference. Country fixed effects μ_i^h absorb time-invariant level differences across countries, year fixed effects δ_t^h absorb common shocks. The lagged outcome and lagged log GDP per capita are included as controls. Standard errors are clustered at the country level. We estimate (3) separately at each horizon, following Jordà (2005) and the more recent applied discussion in Plagborg-Møller and Wolf (2021).

The pre-event horizons $h = -3$ and $h = -2$ are included as falsification. Under the conditional-associations framing we adopt, the pre-event coefficients are part of the empirical pattern, not an identifying restriction, we report them, test them for nullity, and acknowledge the implication for interpretation. The horizon $h = +5$ in the post-event window is reported as a summary of medium-run persistence, horizons $h = +6$ through $h = +8$ are estimated but the reported focal-cell persistence discussion ends at $h = +5$.

5.2 First-event restriction

The estimation sample is restricted to country-years prior to the second coup in country i. The restriction has two parts. First, the shock_{it} indicator equals one only in the year of the first coup observed for country i, subsequent coups in the same country do not enter the regression as additional shocks. Second, the country's panel is truncated at the year of its second coup, so that no country-year observation contributes to the regression after the second coup has occurred. The reason for both restrictions is the same: in repeated-coup countries, the post-event horizons of the first coup overlap with the pre-event horizons of subsequent coups, contaminating the estimated β_h. The indexing by first-treatment cohort is that of the staggered difference-in-differences literature (Callaway and Sant'Anna, 2021). The truncation at the second coup is specific to this recurrent-event setting: that literature assumes treatment is absorbing, an assumption the cycling documented in OA Appendix K.1 violates.

Beyond the econometric motivation, the first-event design isolates the initial observed institutional shock, which is the cleanest counterpart to the option-value experiment of Section 3: a firm facing a hazard that has not yet realised. The formal model itself permits recurring coups, since they arrive as a Cox process, so the restriction is an empirical design choice to avoid overlapping event windows, not something the structure imposes. A multi-period staggered comparison reported in OA Appendix K compares the magnitude under both absorbing (Callaway–Sant'Anna) and non-absorbing (de Chaisemartin–D'Haultfœuille) designs.

The restriction has consequences for the sample. The Bjørnskov–Rode panel contains 525 coup-year cells across 114 countries, the first-event-only sample contains 110 first-coup country-year cells, one treated year per country. Four of the 114 coup countries leave no first-coup cell in the estimation panel. Cuba and Jordan have no coup inside the 1960–2022 window, and Anguilla and the Cook Islands carry no matched outcome data in the merge. All form the eligible first-event sample, and outcome- and horizon-specific regressions use the subset with complete required cells. The truncation at the second coup reduces the country-year sample from 12,852 to 9,019 observations. The compositional shift in pre-coup regime types implied by the restriction is discussed in Section 4.2, in particular, military dictatorships, which account for 38.7% of coup-year cells in the full panel, account for only 9 first-coup cells in the first-event-only sample, reflecting the historical regularity that military regimes are typically the consequence of an earlier coup, not a stable starting state. By Bjørnskov–Rode coup type, the 110 first-coup cells carry 116 coup-arm assignments: 53 normal-successful, 59 failed attempts, and 4 self-coups by the incumbent executive, six cells carrying more than one arm. These assignment counts sum to 116, not 110, because six first-coup years contain more than one classified coup. The outcome variables are total factor productivity, real output growth, gross investment, foreign direct investment, total trade, inflation, government spending, unemployment, and net migration, all in logs (inflation in log first differences) and winsorized at the 1% level.

5.3 Recurrent-event design

The first-event restriction buys clean event windows at the cost of treated support and, as Section 6.5 shows, a pre-event trend in the democracy-destroying cell. Design A relaxes it. It uses all clean coup episodes, where a clean episode is a coup with no other coup in the window $[t-5, t+8]$. Around each episode we stack the treated country together with a clean risk set of controls, that is, any country with no coup inside the focal window, whether or not it has coups elsewhere in its history, that has complete outcome data over the window. We reweight the controls by pooled entropy balancing (Hainmueller, 2012) on predetermined characteristics, region, decade, pre-coup regime, lagged GDP per capita, and pre-event TFP at $h = -5, -3, -1$, and apply the corrective stack weights of Wing, Freedman, and Hollingsworth (Wing et al., 2024), which make each event's controls contribute in proportion to its treated mass. The specification carries episode-by-unit and episode-by-year fixed effects, clusters on country, and normalizes the year before the coup to zero. The coefficients on the event-time indicators $\{g_{-5}, \ldots, g_{-2}, g_0, g_1, \ldots, g_8\}$ trace the response. Because the balancing weights use pre-event TFP at $h = -5, -3, -1$, those leads are not independent pre-trend tests, and the remaining leads serve as holdout diagnostics. Inference is the restricted wild cluster bootstrap with Webb weights. Because treatment is not absorbing here, a country re-enters the risk set once its event window closes, which distinguishes this design from staggered estimators that assume an absorbing state. The full ladder of control constructions, the weight diagnostics, and the holdout tests are in OA Appendix D.

5.4 Transition decomposition

For the decomposition in Sections 6.3 and 6.5, we classify each first coup by the pair (pre-coup regime, post-coup regime), using the Bjørnskov–Rode regime classification. The pre-coup regime is measured at $t-1$, the year preceding the coup. The post-coup regime is measured at $t+1$ with a fallback to $t+2$ when the $t+1$ value is missing or transitional. We group the pre-coup and post-coup classifications into democratic (parliamentary, mixed, and presidential democracies, codes $\{0, 1, 2\}$) and autocratic (civilian, military, and royal dictatorships, codes $\{3, 4, 5\}$) and define four transition categories. DEM $\rightarrow$ AUT comprises events with pre-coup democracy and post-coup autocracy ($n = 14$ in the first-event-only sample), AUT $\rightarrow$ DEM comprises pre-coup autocracy and post-coup democracy ($n = 2$), INTRA_AUT comprises events that are autocratic both before and after ($n = 65$), and INTRA_DEM comprises events that remain democratic both before and after ($n = 18$), containing 16 failed coups and 2 successful coups that did not produce a regime change and functioning as the empirical counterfactual to the DEM $\rightarrow$ AUT class in Section 6.4. Throughout we abbreviate these transitions DA $\equiv$ DEM $\rightarrow$ AUT, AD $\equiv$ AUT $\rightarrow$ DEM, AA $\equiv$ INTRA_AUT, and DD $\equiv$ INTRA_DEM. In OA Table B2, AGG denotes the pooled first-coup coefficient. Pre-coup regime codes outside the democracy–autocracy range (colonies and trusteeships), and observations with a missing pre-coup regime at $t-1$, are assigned to an OTHER category ($n = 11$: 2 and 9 respectively). None of the eleven has the complete outcome and control cells required for the impact-year TFP or growth projections. Missing Penn World Table coverage specifically explains its exclusion from the TFP projection, and the OTHER indicator is collinear and dropped in those specifications. Nine of the eleven are first coups dated 1960, the opening year of the panel, so their lagged regime and lagged outcome both fall outside it. The four-class breakdown therefore covers $14+2+65+18 = 99$ events, and the OTHER category accounts for the remaining 11 first-coup events, summing to the 110 first-coup events in the panel.

The decomposed local projection replaces the single shock_{it} indicator in (3) with four focal transition-class indicators and a residual OTHER indicator, entered jointly:

$$
\begin{aligned}
Y_{i,t+h} - Y_{i,t-1} = \mu_i^h + \delta_t^h \\
&+ \beta_h^{DA}\,\mathbb{1}[\text{DEM}\rightarrow\text{AUT}]_{it} + \beta_h^{AD}\,\mathbb{1}[\text{AUT}\rightarrow\text{DEM}]_{it} \\
&+ \beta_h^{AA}\,\mathbb{1}[\text{INTRA\_AUT}]_{it} + \beta_h^{DD}\,\mathbb{1}[\text{INTRA\_DEM}]_{it} \\
&+ \beta_h^{O}\,\mathbb{1}[\text{OTHER}]_{it} \\
&+ \phi^h\, Y_{i,t-1} + \psi^h\, \ln \text{GDP}_{i,t-1} + \varepsilon_{i,t+h}^h.
\end{aligned}
\tag{4}
$$

The OTHER indicator is included in the estimating command, but none of the eleven events it flags has a complete impact-year cell in the productivity or growth samples, so the indicator is identically zero there and β_h^O is dropped, not estimated. It is estimated only for net migration, where one such event has the cells the specification requires. The four focal coefficients are estimated jointly within the same regression, which preserves the comparison of magnitudes across classes and permits Wald tests of pairwise equality. The empirical counterfactual underlying the identification logic of Section 6.4 is the pairwise contrast $\beta_h^{DA} - \beta_h^{DD}$: the difference between democracy-destroying coups and democracy-preserving coup attempts in the same outcome at the same horizon.

Using democracy-preserving coup attempts as the counterfactual for regime-destroying ones is in the spirit of Jones and Olken (2009), who contrast successful and failed assassination attempts on national leaders. We read $\beta_h^{DA} - \beta_h^{DD}$ as an associational contrast conditional on the controls in (4), not as the effect of an as-good-as-random regime change.

5.5 Robustness architecture

The robustness exercises in Section 6.6 are concentrated on the DEM → AUT TFP coefficient at $h = 0$, the focal cell that the decomposition identifies. Four checks are reported. The first is a *country jackknife*: the decomposition (4) is re-estimated 14 times, dropping each of the 14 DEM → AUT countries one at a time, to test whether any single country drives the focal estimate. The second is *leave-one-region-out*: the decomposition is re-estimated seven times, dropping each World Bank region in turn, with particular attention to Latin America and the Caribbean, which accounts for eight of the fourteen DEM → AUT events. The third is the *alternative regime-classification timing*: the decomposition is re-estimated with the post-coup regime measured at $t + 2$ instead of $t + 1$, testing whether the focal estimate depends on the particular year used to classify the post-coup regime and distinguishing transient classification artifacts from a durable institutional effect. The fourth is the *joint pre-event nullity test*: a stacked specification with horizon-interacted fixed effects estimates β_{-3}^{DA} and β_{-2}^{DA} jointly and tests their joint nullity, assessing whether the focal estimate reflects a response to the event or the continuation of a pre-existing trajectory.

The robustness battery is targeted at the focal cell, not distributed across all decomposition cells. This choice reflects the tradeoff between thoroughness and multiple-comparisons inflation in a sample with $n = 110$ first-coup events and four decomposition classes, of which the focal cell has 14 events.

5.6 Identification framing

Each design carries its own identification content. For Design A, the recurrent-event estimator, a causal reading would require conditional parallel trends between coup countries and the reweighted risk set after balancing. We do not assert that condition. The pre-event coefficients not used to build the weights, and the deeper holdout leads, are diagnostics consistent with that condition without establishing it, and the balancing attenuates but does not eliminate the pre-event differences. For Design B, the first-event local projections, we adopt a conditional-associations framing and have no quasi-experimental source of exogenous variation in coup occurrence. Neither design claims a structural causal parameter free of selection.

The local projections (3) and (4) estimate the response of each outcome to a first coup, conditional on country and year fixed effects, lagged outcome, and lagged log GDP per capita. We do not impose strict parallel-trends assumptions as identification restrictions. Where pre-event coefficients are non-zero we report them, and where the joint pre-event nullity test rejects we describe the implication for the interpretation of the focal estimate (Section 6.6).

For Design B, the empirical content of the strategy is the four robustness exercises in Section 5.5, taken together. We do not claim that the pattern identified by the decomposition is a structural causal parameter free of selection. What the strategy can establish is whether that pattern is stable under those exercises, and whether it is more consistent with institutional change than with the coup event itself or generic political uncertainty alone. It does not separately distinguish the dispersion channel of the option-value framework of Dixit and Pindyck (1994) from the class mean shift. Section 6.6 reports that assessment, including the exercises under which the pattern is not informative. The identification logic parallels that of Bennett et al. (2025) applied to the Groningen 10-sector labor-productivity panel.

6 Results

The empirical analysis uses two complementary designs. The first is a *recurrent-event* design that treats every clean coup episode as a separate event, with horizon-consistent risk sets and controls not contaminated by other coups within the event window. It asks how far productivity falls after a coup on the broadest available support, and whether that fall depends on the regime transition the coup produces. The second is a *first-institutional-transition* design that isolates the initial institutional shock a country's first coup delivers, decomposes it by the regime trajectory it produces following Bennett et al. (2025), and sets democracy-destroying transitions against coups that leave democratic institutions standing. The two designs answer different questions. The first is the average productivity cost of a coup, and the second is the institutional composition of the initial shock. We present them in that order. The recurrent-event design lets a country re-enter the risk set once its event window closes. The first-event design instead isolates the initial coup and truncates the panel at the second, avoiding overlapping treatment histories.

6.1 Recurrent-event responses

We first estimate the productivity response to coups on all clean episodes. The two designs in this section use two estimators for two questions. The first-event analysis of the following subsections uses horizon-specific Jordà local projections, one regression per horizon. For recurrent coups, where overlapping event windows and prior treatment complicate the panel comparison, we use a stacked difference-in-differences event-study design with episode-specific unit and calendar-time fixed effects. A clean episode is a coup with no other coup in the window $[t-5, t+8]$, and the year before the coup is normalized to zero. We do not compare coups to never-treated countries alone. Instead we build a clean risk set: any country with no coup inside the focal window may serve as a control, whether or not it experiences coups elsewhere in its history. We then reweight that risk set by pooled entropy balancing

(Hainmueller, 2012), over all country-episode observations, so that the controls match the treated on predetermined characteristics: region, decade, pre-coup regime, lagged GDP per capita, and pre-event TFP at $h = -5, -3, -1$. To aggregate the event-specific comparisons correctly we apply the corrective stack weights of Wing, Freedman, and Hollingsworth (Wing et al., 2024), which make each event's controls contribute in proportion to its treated mass. As a diagnostic we also attempted stack-specific covariate balancing within each event, following the same logic, but local support is limited in this application: entropy balancing succeeds in 32 of 44 stacks under the broad donor pool and in none under region-by-regime exact stratification, so the preferred specification uses the pooled balancing with corrective aggregation. Inference is the restricted wild cluster bootstrap with Webb weights, clustered by country. The full ladder of control constructions, the weight diagnostics, and the holdout pre-period tests are in OA Appendix D.

On this balanced design, coups are followed by an immediate total factor productivity contraction of about three percent ($\hat{\beta}_0 = -0.030$), deepening to about four percent one year later ($\hat{\beta}_1 = -0.044$). This short-run response is the most robust result in the paper. The impact-year coefficient stays near -0.03 across every control construction we try, from never-treated controls to the risk set to pooled entropy balancing, with and without the corrective stack weights, and it survives a holdout design estimated on a longer pre-event history (OA Appendix D).

Entropy balancing substantially attenuates the pre-event differences. Because the weights balance TFP at $h = -5, -3, -1$, those horizons are not independent pre-trend tests. The remaining holdout pre-event coefficients are small and are not jointly rejected in the primary window, and the deeper extrapolation leads in an eight-year design are not jointly rejected either, though that longer window shows some residual sensitivity. We therefore describe the pre-event path as substantially attenuated, not as validated parallel trends. The weights are well behaved, with no single control dominating, and the full ladder of designs, weight diagnostics, and holdout tests is in OA Appendix D.

The medium-run path is more design-sensitive than the impact. In the preferred balanced design the year-two-through-five coefficients are jointly detectable (wild $p = 0.011$), but this medium-run persistence is sensitive to requiring a substantially longer pre-event history, where it is no longer detectable ($p = 0.25$ on the eight-year holdout sample). Longer-horizon estimates attenuate toward zero and provide no robust evidence of permanent scarring.

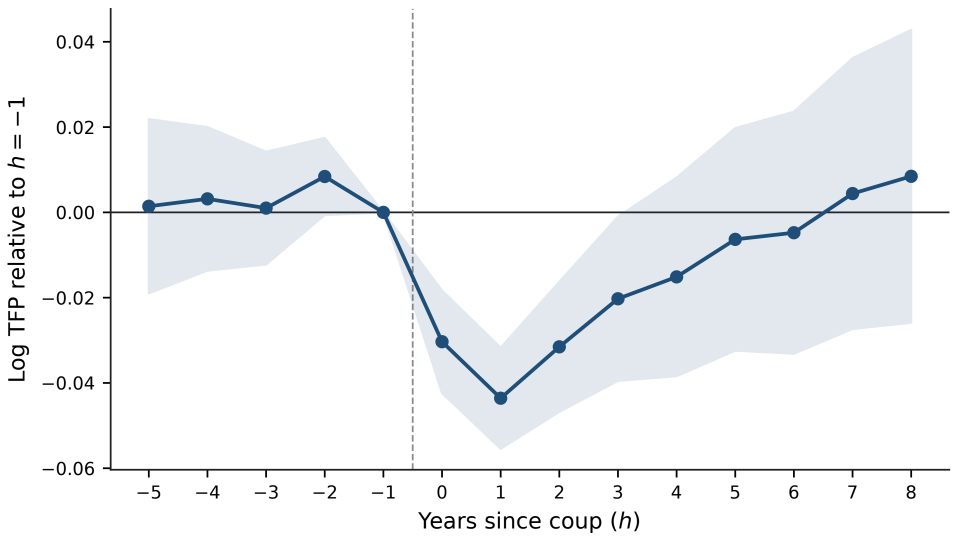


Figure 2: Recurrent-event TFP response, entropy-balanced risk-set stacked event study (window $[t-5, t+8]$, $h = -1$ normalized to zero). Controls are a clean risk set reweighted by global entropy balancing to match the treated on predetermined covariates: region, decade, pre-coup regime, lagged GDP per capita, and TFP at $h = -5, -3, -1$. Entropy weights balance predetermined covariates including TFP at $h = -5, -3, -1$; those horizons are therefore not independent pre-trend tests, and the remaining pre-event coefficients provide holdout diagnostics. Shaded bands are 90% intervals from country-clustered standard errors. Coups are followed by a contraction of about three percent on impact, deepening to about four percent at $h = +1$, with medium-run effects that attenuate toward zero. The full ladder of control constructions and the deep holdout are in OA Appendix D.

Is this contraction specific to coups that destroy democracy? A simpler never-treated version of the recurrent design, which gives a slightly smaller pooled impact of about two percent, lets us split the episodes by the regime transition they produce. The impact response is close to constant across the three classes: -0.019 for democracy-to-autocracy coups (wild $p = 0.023$), -0.022 for intra-autocratic coups ($p = 0.008$), and -0.021 for intra-democratic coups ($p = 0.003$). A joint test does not reject equality of the three impact responses (wild

$p = 0.68$), as Figure 3 shows. The immediate productivity cost of a coup is general across coup types, not specific to the democracy-destroying class, and it rests on far more treated support than the first democracy-destroying events alone. This revises the earlier reading that only democracy-destroying coups carry a measurable productivity cost.

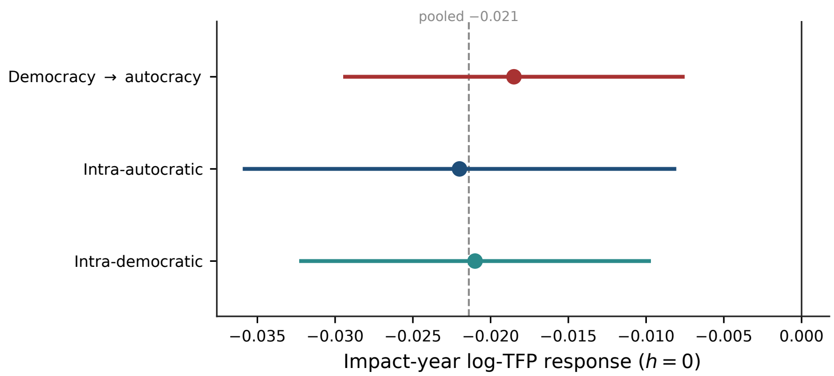
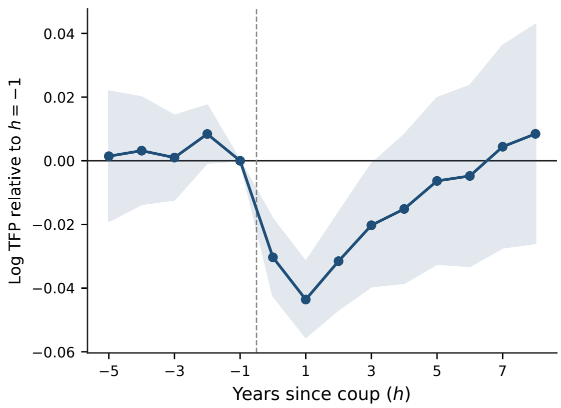
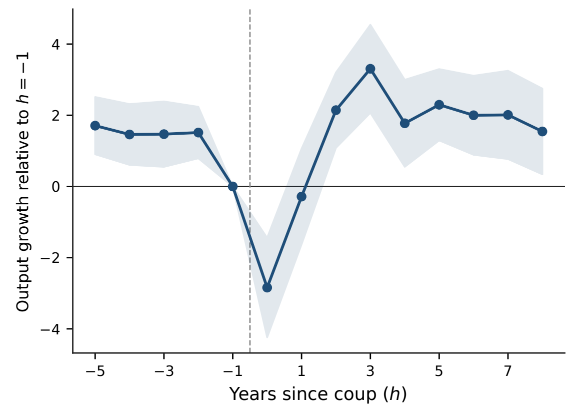


Figure 3: Impact-year TFP response by regime-transition class, recurrent-event design (strict per-class control states). Points are the class-specific impact coefficients. The dashed vertical line is the pooled impact (-0.021). Bars are conventional 90% intervals. The governing per-class inference is the restricted Webb wild cluster bootstrap (individual wild $p = 0.023$, 0.008, 0.003 for the three classes). A joint test does not reject equality across the three classes: $H_0 : \beta_{\mathrm{DA}} = \beta_{\mathrm{AA}} = \beta_{\mathrm{DD}}$, wild $p = 0.68$. The three responses cluster near -2% with no detectable institutional gradient. None of the three classes rejects a joint pre-event nullity test (wild $p = 0.58$, 0.16, and 0.84 for the democracy-to-autocracy, intra-autocratic, and intra-democratic classes, the intra-autocratic class carrying a marginally significant individual lead but no joint rejection).

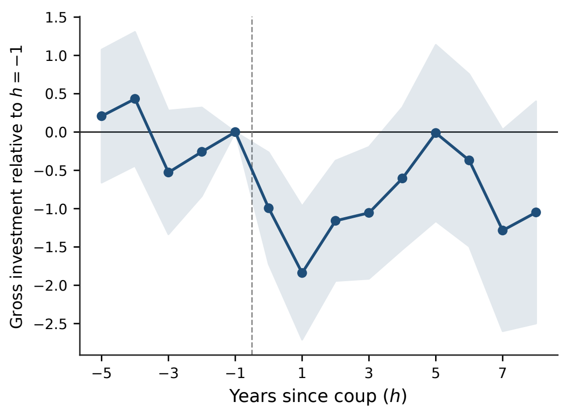
The same balanced design speaks to the margins the option-value model links to productivity. Figure 4 traces three of them on the identical estimator. Total factor productivity, a stock, steps down and recovers slowly (panel a). Output growth, the flow that feeds the stock, drops on impact, rebounds almost completely within a year, and subsequently overshoots its pre-coup path. For growth the pre-event leads held out of the balancing jointly reject (wild $p = 0.04$), a narrower test than the four-lead pre-trend statistic tabulated in OA Table E1, so we read it as a descriptive dynamic contrast, not a clean causal estimate (panel b). Gross investment, the firm's direct decision in the model, contracts on impact and through the first year and recovers gradually, but it is imprecisely estimated (panel c). Read together the panels trace a stock that falls and recovers, a flow that dips and rebounds, and an investment margin that contracts and recovers, a pattern consistent with the option-value account without requiring every margin to be equally precisely identified.



(a) Total factor productivity (stock)



(b) Output growth (flow)



(c) Gross investment

Figure 4: Dynamic responses across margins, recurrent-event design. All panels use the same entropy-balanced recurrent-event design and predetermined balancing variables, with weights re-estimated within each outcome-specific estimation sample (hence 44, 40, and 27 treated episodes for TFP, growth, and investment). Bands are 90% intervals from country-clustered standard errors. TFP exhibits a short-run contraction followed by gradual recovery. Output growth displays a larger transitory decline and subsequent rebound, although the pre-event leads held out of the balancing jointly reject for growth (wild $p = 0.04$). Gross investment contracts initially but is imprecisely estimated. The panels are therefore read jointly as dynamic patterns and not as equally strong causal estimates.

The remainder of the section turns to the first-institutional-transition design, which isolates the initial institutional shock and asks how its regime trajectory organizes the composition, not the mere occurrence, of the productivity response.

6.2 First-event aggregate responses

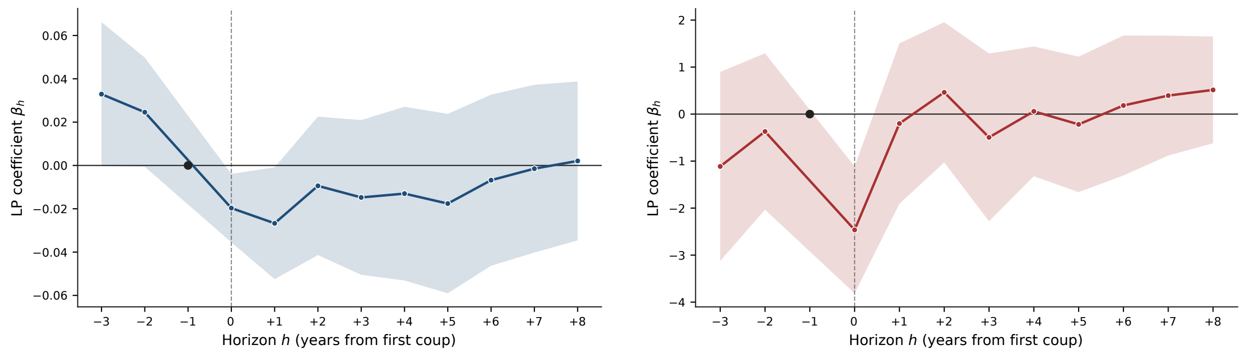
We now turn to the first-institutional-transition design, which answers a narrower and more heavily qualified question than the recurrent-event analysis. It restricts attention to the first coup observed in each country, truncated at the second coup, so that the estimated response is that of a country's initial institutional shock, separate from an average over recurring events. Table 4 reports $\hat{\beta}_h$ at the seven focal horizons. The two focal outcomes show impact-year contractions: total factor productivity ($\hat{\beta}_0 = -0.020$, $p < 0.05$) and output growth ($\hat{\beta}_0 = -2.463$, asymptotic $p < 0.01$, on the transformed growth scale). The TFP contraction extends to $h = +1$ ($\hat{\beta}_1 = -0.027$, $p < 0.05$) and fades by $h = +2$. The growth response is concentrated at the impact year and does not persist materially: $\hat{\beta}_h$ is statistically indistinguishable from zero at every post-impact horizon in the aggregate specification. Net migration is positive at $h = +5$ ($\hat{\beta}_5 = +0.20$, $p < 0.05$). Foreign direct investment, total trade, inflation and unemployment show no detectable impact at any reported horizon, and government spending is detectable only at the 10% level at the impact year ($\hat{\beta}_0 = -0.454$), which we do not treat as a result.

Table 4: Aggregate local projections, first-event-only sample

Outcome	$h = -3$	$h = -2$	$h = +0$	$h = +1$	$h = +2$	$h = +3$	$h = +5$
TFP	$+0.033^{*}$	$+0.025^{*}$	-0.020^{**}	-0.027^{**}	-0.009	-0.015	-0.018
	(0.017)	(0.013)	(0.008)	(0.013)	(0.016)	(0.018)	(0.021)
Growth	-1.113	-0.370	-2.463^{***}	-0.201	$+0.462$	-0.495	-0.219
	(1.026)	(0.849)	(0.689)	(0.870)	(0.761)	(0.910)	(0.735)
FDI	$+0.076$	-0.046	-0.020	$+0.028$	$+0.053$	$+0.101$	$+0.105$
	(0.064)	(0.069)	(0.041)	(0.056)	(0.064)	(0.071)	(0.090)
Total trade	$+0.038$	$+0.030$	-0.041	$+0.045$	$+0.098$	$+0.046$	-0.060
	(0.032)	(0.025)	(0.027)	(0.059)	(0.067)	(0.064)	(0.071)
Inflation	-0.273	-1.341	$+3.866$	-0.662	-1.164	-0.910	-1.284
	(1.387)	(1.414)	(2.636)	(2.328)	(1.808)	(3.383)	(1.795)
Gov. spending	-0.020	$+0.164$	-0.454^{*}	-0.192	$+0.132$	$+0.695$	-0.074
	(0.274)	(0.221)	(0.253)	(0.457)	(0.405)	(0.505)	(0.440)
Unemployment	-0.484	-0.045	-0.124	-0.264	$+0.000$	-0.212	$+0.172$
	(0.641)	(0.277)	(0.243)	(0.369)	(0.381)	(0.569)	(0.634)
Net migration	-0.021	-0.014	-0.190	-0.171	-0.100	-0.049	$+0.196^{**}$
	(0.028)	(0.020)	(0.202)	(0.182)	(0.169)	(0.156)	(0.097)

Notes. Coefficients on the first-coup indicator from horizon-h local projections of the form $Y_{i,t+h} - Y_{i,t-1} = \mu_i^h + \delta_t^h + \beta_h \cdot \text{coup}_{it}^{\text{first}} + \phi^h Y_{i,t-1} + \psi^h \ln \text{GDP}_{i,t-1} + \varepsilon_{i,t+h}^h$. Sample restricted to country-years prior to the second coup. $h{=}-3, -2$ are pre-event falsification horizons under the conventional causal reading, or part of the descriptive trajectory under the associations framing adopted here. Standard errors clustered by country. Significance stars in this table follow the *asymptotic* cluster-robust p-value: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 5 plots the TFP and growth responses with 95 percent confidence bands. The two outcomes that move at the impact year do so by economically sizable magnitudes (approximately a 2 percent TFP contraction and a 2.5-point output-growth contraction on the transformed scale) but the aggregate estimate averages across heterogeneous coup events of distinct institutional character. The decomposition in the next subsection unpacks that heterogeneity.



(a) Total factor productivity

(b) Output growth

Figure 5: Aggregate local-projection responses to the first coup. First-event-only sample, truncated at the second coup. Shaded bands are 95% confidence intervals from standard errors clustered at the country level. The reference point at $h = -1$ is normalized to zero by the LP specification.

6.3 Transition decomposition

Bennett et al. (2025) argue that the productivity-relevant consequences of coups operate primarily through institutional change rather than through the coup event itself. To assess this, we classify each first coup by the pair (pre-coup regime, post-coup regime), measured one year before and one year after the coup, with a $t + 2$ fallback where $t + 1$ is missing or transitional, using the Bjørnskov–Rode regime classification. Three of the four resulting transition categories are informative, the fourth (autocracy-to-democracy) has only two events and is reported for completeness only.

Regime at $t+1$

Regime at $t-1$	**Democracy**	**Autocracy**
Democracy	INTRA_DEM $n = 18$	DEM→AUT $\mathbf{n = 14}$
Autocracy	AUT→DEM $n = 2$	INTRA_AUT $n = 65$

Unclassified: $n = 11$

2 colony or trusteeship at $t-1$, 9 missing regime at $t-1$

classified 99 + unclassified 11 = 110 first-coup events

Figure 6: First-coup transition matrix. Each first coup is classified by the Bjørnskov–Rode regime one year before and one year after the event (with a $t+2$ fallback where $t+1$ is missing or transitional). Democracy pools the parliamentary, mixed and presidential categories, autocracy pools the civilian, military and royal categories. The shaded cell is the democracy-destroying class, highlighted not because it is the only transition that lowers productivity (the recurrent-event design of Section 6.1 shows the impact-year contraction is common across classes), but because it is the central institutional transition of the first-transition structural decomposition that follows. The eleven unclassified events carry a pre-coup regime outside the democracy–autocracy range (colonies and trusteeships) or no recorded pre-coup regime. None has the complete outcome and control cells required for the impact-year TFP or growth projections. Missing Penn World Table coverage specifically excludes them from the TFP projection. One contributes to the migration specification through the residual indicator.

Figure 6 shows how the 110 first coups distribute across the classification. The two intra-regime cells are the largest, and the focal democracy-destroying cell contains 14 events. Eleven coups cannot be placed in the matrix, because the country has no recorded regime in the year before the coup or is a colony or trusteeship in that year. None of the eleven has the complete outcome and control cells required for the impact-year TFP or growth projections. Missing Penn World Table coverage specifically excludes them from the TFP projection. One enters the migration specification through the residual indicator, which is estimated but not reported. The four focal classes account for all 99 classified first-coup events. Outcome- and horizon-specific subsets of them contribute to the TFP and growth projections.

Table 5: BBG-style transition decomposition: local projections by post-coup regime trajectory

	$h = -3$	$h = -2$	$h = +0$	$h = +1$	$h = +5$
TFP					
DEM→AUT (n=14)	+0.013	+0.025*	−0.042***	−0.030**	−0.031***
	(0.023)	(0.015)	(0.011)	(0.013)	(0.012)
INTRA_AUT (n=65)	+0.051**	+0.037*	−0.016	−0.037*	−0.007
	(0.026)	(0.021)	(0.012)	(0.020)	(0.035)
INTRA_DEM (n=18)	+0.004	+0.002	−0.011	+0.010	−0.016
	(0.010)	(0.015)	(0.010)	(0.015)	(0.017)
AUT→DEM (n=2)	−0.016	−0.024***	−0.042	−0.083	−0.108**
	(0.015)	(0.004)	(0.032)	(0.058)	(0.045)
Growth					
DEM→AUT (n=14)	+0.266	+1.905	−3.689**	+2.086	−0.674
	(1.349)	(1.174)	(1.485)	(1.516)	(1.316)
INTRA_AUT (n=65)	−1.375	−0.399	−2.522***	−0.998	+0.369
	(1.433)	(1.052)	(0.906)	(1.162)	(0.901)
INTRA_DEM (n=18)	−1.757	−3.879	−0.579	+1.758	−3.156
	(2.205)	(3.264)	(1.136)	(1.626)	(2.606)
AUT→DEM (n=2)	−0.550	+1.096	−2.022	−4.197	+1.910***
	(0.521)	(1.331)	(2.985)	(3.083)	(0.542)
Migration					
DEM→AUT (n=14)	−0.012	−0.002	−0.049	−0.047	+0.044
	(0.025)	(0.028)	(0.043)	(0.045)	(0.056)
INTRA_AUT (n=65)	−0.039	−0.026	−0.276	−0.248	+0.240*
	(0.039)	(0.027)	(0.293)	(0.266)	(0.132)
INTRA_DEM (n=18)	+0.029	−0.005	+0.001	−0.006	+0.086
	(0.061)	(0.054)	(0.031)	(0.031)	(0.099)
AUT→DEM (n=2)	+0.195**	+0.223*	+0.175*	+0.145	+0.512**
	(0.098)	(0.117)	(0.097)	(0.184)	(0.239)

Notes. Local-projection estimates from the BBG-style decomposition, first-event-only sample, truncated at the second coup year. Coups classified by pre-coup regime (parliamentary/mixed/presidential democracy = DEM, civilian/military/royal dictatorship = AUT) and post-coup regime (t+1, t+2 fallback). *DEM→AUT* = democracy-destroying coups (BBG focal class). *AUT→DEM* = transitions to democracy (n=2, for completeness). *INTRA_AUT* = autocracy-to-autocracy. *INTRA_DEM* = pre-coup democracy staying democratic (failed and no-change coups, the DEM→AUT counterfactual). The four classes cover 99 of the 110 first-coup events. The remaining 11 fall outside the democracy–autocracy range (colonies and trusteeships, 2 events) or lack a pre-coup regime at $t-1$ (9 events), and none of the 11 has the complete cells for the impact-year TFP or growth columns, with missing Penn World Table coverage explaining the TFP exclusion. One enters the migration column, estimated and not reported. Each regression includes lagged outcome, lagged log GDP per capita, country and year fixed effects, with country-clustered errors in parentheses. Stars follow the *asymptotic* cluster-robust p-value: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$, not the adopted standard given five countries carry the focal coefficient. OA Table I1 reports bootstrap p-values for TFP, growth and investment in the DA, AA and DD classes at $h = 0, \dots, 5$, not every cell here, so the coefficient carries more stars here and the bootstrap is the one to read.

The decomposition is reported in Table 5. The TFP contraction at the impact year is concentrated in coups that move a country from democracy to autocracy: $\hat{\beta}_0 = -0.042$ (bootstrap $p = 0.020$) in the DEM → AUT class versus -0.016 in the intra-autocracy class and -0.011 in the intra-democracy class. The output growth contraction is larger in magnitude but less concentrated: the -3.69 at DEM → AUT (bootstrap $p = 0.017$) exceeds the -2.52 at INTRA_AUT (bootstrap $p = 0.007$), while the intra-democracy class shows a coefficient that is not distinguishable from zero (-0.58, bootstrap $p = 0.649$). OA Figure F1, in OA Appendix F, visualizes the same decomposition for both outcomes.

In this first-event decomposition the impact-year TFP contraction concentrates in the classes that change the regime type and is negligible where the regime type is preserved. The concentration is a property of the initial institutional shock, not of coups in general. The recurrent-event design of Section 6.1, which uses all clean episodes, finds the intra-democracy class contracts by about as much as the others and does not reject equality across classes, so the pattern here reflects the composition of first events, not an immunity of democracy-preserving coups. The democracy-to-autocracy contraction is -0.042 and tied for the largest of the four, while the within-autocracy and within-democracy classes show smaller and statistically weak effects. The autocracy-to-democracy cell returns a numerically identical -0.042, but it rests on two events and is not distinguishable from zero, so we do not read it as a second finding. The output-growth response concentrates less than productivity: the democracy-to-autocracy contraction is the largest of the four classes, but the within-autocracy class also contracts appreciably. The contrast between the two outcomes is part of the interpretation we develop in OA Appendix J.1.

6.4 Democracy-preserving coups as counterfactuals

The intra-democracy class (INTRA_DEM, $n = 18$) is the natural counterfactual to the focal democracy-destroying class. It contains 16 failed coup attempts in democracies and 2 successful coups that did not produce a regime change. In all 18 cases the formal institutional environment remains democratic at $t+1$, even when the coup attempt itself generates substantial short-run political uncertainty.

If coups affect productivity through generic political uncertainty alone, the INTRA_DEM class should show contraction comparable to the DEM → AUT class, since the political event is of the same class and the level of uncertainty generated is plausibly similar. The data do not support this prediction. The impact-year TFP coefficient in INTRA_DEM is -0.011 (bootstrap $p = 0.346$), economically and statistically negligible, compared with -0.042 in DEM → AUT. The output-growth coefficient in INTRA_DEM is -0.58 (bootstrap $p = 0.649$), far smaller than in the regime-changing classes. As noted in the decomposition, this counterfactual is specific to first events, and it identifies the composition of the initial institutional shock, not a general result that democracy-preserving coups leave productivity intact.

We report that contrast under the standard the paper adopts throughout, which is the 90% level, and under the inference method the paper treats as governing, which is the wild cluster bootstrap, not the asymptotic approximation. Only five countries contribute to the DEM → AUT coefficient at the impact year, well inside the regime in which cluster-robust asymptotics are unreliable (MacKinnon and Webb, 2018), so we do not lean on the asymptotic Wald test ($p = 0.039$). With Webb six-point weights, the null imposed and 2,000 replications, the bootstrapped contrast returns $p = 0.074$ (this and the other cross-class contrasts are in OA Table J1, Panel A. The class coefficients being contrasted are in Table 5). It rejects equality at the 90% level and not at the 95%, and we report it as exactly that.

The other comparisons establish less. The contrast between DEM → AUT and the intra-autocratic class does not reach conventional levels ($p = 0.113$), and the joint Wald test of equality across all four transition classes does not reject ($F = 1.71$, $p = 0.170$, OA Table J1, Panel A). The design is powered for the counterfactual the framework singles out, the comparison between coups that destroy democratic institutions and coups that leave them intact. The full four-way ordering of the classes lies beyond what it can adjudicate, and we make no claim on it.

The contrast between INTRA_DEM and DEM $\rightarrow$ AUT identifies the empirical pattern the option-value reading developed in Section 6.5 attributes to the dispersion of post-coup institutional outcomes. Section 7 places that attribution in a parameterisation and reports which of the channel loadings the data can and cannot separate. The contrast cannot rule out that generic political uncertainty plays some role, since the INTRA_DEM coefficient is negative and our analysis of the alternative $t+2$ regime-classification specification ('F2', OA Appendix J.1) returns the intra-democracy coefficient as negative but not statistically significant (-0.016, $p = 0.13$). What the contrast does show is that generic uncertainty about the immediate political environment alone is insufficient to generate the magnitudes observed in the focal cell. The two classes also differ in their calibrated institutional moments, which the structural layer does not separate.

6.5 The focal cell

Within the four transition categories of Table 5, the impact-year contraction in total factor productivity concentrates in coups that move a country from democracy to autocracy. The coefficient on the DEM $\rightarrow$ AUT indicator at $h = 0$ is -0.042 (standard error 0.011, clustered at the country level, $p < 0.001$ asymptotically and $p = 0.020$ under the governing wild cluster bootstrap), approximately twice the pooled first-coup average of Table 4. The corresponding effect in INTRA_AUT transitions is smaller (-0.016) and statistically indistinguishable from zero in the transition-decomposed specification, the effect in INTRA_DEM, discussed in Section 6.4, is both statistically and economically negligible (-0.011). At $h = +1$ the DEM $\rightarrow$ AUT coefficient is -0.030 ($p = 0.024$ under asymptotic inference) but does not survive the wild cluster bootstrap correction (boot. $p = 0.10$, OA Table I1). The effect is estimated precisely at impact and again at $h = +5$, but not at the intermediate horizons, so the point estimates trace a persistent level shift without a continuously significant path. The cumulative test of OA Appendix B.3 traces the persistence pattern in the point estimates but does not establish it, because the cumulative response is not distinguishable from zero under the governing bootstrap.

This configuration is most consistent with the institutional channel articulated by Bennett et al. (2025). In their account, the productivity-relevant consequences of coups operate through institutional restructuring: shifts in property-rights regimes, in judicial constraints, and in the formal post-coup environment alter conditions for entrepreneurs and firms in durable ways. Coups that displace a sitting democracy and install an autocratic regime are the class of events most plausibly associated with such restructuring. Failed coups and democracy-preserving successful coups (the INTRA_DEM cell) leave the formal institutional environment intact, even when they generate substantial short-run political uncertainty. The contrast between the two cells (-0.042 in DEM $\rightarrow$ AUT versus -0.011 in INTRA_DEM) is the central empirical pattern motivating the interpretation developed below. Under the wild cluster bootstrap that pairwise contrast returns $p = 0.074$ (Section 6.4), so the gap rejects equality at the 90% level though not at the 95%, and its economic magnitude is read alongside deliberately bounded statistical evidence.

The cell is small. The first-event-only sample contains $n = 14$ democracy-destroying coups, of which five have a complete impact-year local-projection cell and therefore contribute non-trivially to the point estimate: Brazil 1964, Chile 1973, Greece 1967, the Kyrgyz Republic 2010, and Uruguay 1973. The remaining nine events (Argentina 1961, Congo 1963, Grenada 1979, Honduras 1963, Myanmar 1962, Panama 1969, Seychelles 1977, Sierra Leone 1967, and Suriname 1980) remain in the panel, but their coup-year treated observations do not enter the $h = 0$ TFP coefficient because they lack the lagged-outcome cells the LP specification requires, generally because Penn World Table coverage is incomplete in the early period.

The five contributing events display substantial dispersion in the impact-year TFP response. Chile 1973 anchors the deepest contraction (the event-level coefficients are reported in OA Table H1), with the Kyrgyz Republic 2010 the next-deepest[8], Brazil 1964 and Greece 1967 at intermediate magnitudes, and Uruguay 1973 at the shallowest end. The cross-event variance in $\hat{\beta}_{\text{TFP}}$ is descriptive event-level heterogeneity that the option-value framework can accommodate but that the available within-cell continuous V-Dem variation does not identify. Within the focal cell the realized ΔV-Dem does not predict the event-specific impact-year TFP coefficients (OA Appendix B.1), so we do not read this dispersion as the empirical signature of variance in the post-coup institutional-outcome distribution across coup episodes. The same five events also exhibit the cross-block sign heterogeneity the two-block specification of OA Appendix B.2 identifies at the aggregate: at the event level, Greece 1967 pairs its mild TFP loss with an investment surge, while Chile 1973 pairs its deeper TFP loss with an investment contraction (both in OA Table H1). This cross-event dispersion shows that the focal-cell average is not generated by a single event and spans more than one historical era. Pre-coup income varies within the middle-income range represented by the five contributors. It is not used as a test of continuous institutional-dispersion scaling.

The interpretation of $\hat{\beta}_{\text{DEM}\to\text{AUT}}$ as a population-average effect must therefore be qualified by the limited number of events sustaining it, and Section 6.6 is designed to address the natural concerns: outlier country, regional outlier, alternative regime-classification timing, and joint pre-trend nullity.[9]

The pattern is consistent with a role for the dispersion of post-coup institutional outcomes, but the productivity-block estimates do not separately distinguish that channel from

[8]The Kyrgyz Republic 2010 event combines a DEM $\to$ AUT class assignment under the Bjørnskov–Rode coding with a mildly positive realized shift in the Varieties of Democracy (V-Dem) index from $t-1$ to $t+1$ (+0.18), reflecting the parliamentary phase of the Otunbayeva interim government before executive consolidation under Atambayev (2012 onward) and the subsequent reversion. The impact-year TFP contraction aligns with the medium-term institutional outcome captured by the BR coding not with the transitional V-Dem realization at $t+1$.

[9]Peru's 1992 *autogolpe* (Fujimori's executive self-coup of April 5) is the canonical Latin American reference for democracy-destroying self-coups but does not appear in the focal cell. Bjørnskov–Rode codes Peru as autocratic (regime = 3) from 1990 onward, so the 1992 event is classified as an intra-autocracy coup instead of a DEM $\to$ AUT transition. The same panel records a regime switch in 1990 without a coup event, which we discuss in OA Appendix K.3. Under V-Dem's Regimes of the World index Peru is classified as an electoral democracy in 1991 and as a closed autocracy in 1992, consistent with the historiographical treatment of the 1992 autogolpe as the regime-changing event (Levitsky and Way, 2002). We retain the BR coding for replication consistency and report the discrepancy here.

the class mean shift in the institutional environment. This reading is consistent with the option-value framework of Dixit and Pindyck (1994) formalized in Section 3 and given a structural interpretation in Section 7. The same empirical pattern is documented in Bennett et al. (2025) on the Groningen 10-sector labor-productivity panel. We document a consistent pattern in a broader country-coverage sample (110 first-coup events from 1960–2022) under the local-projection specification standard in modern macro-event analysis (Jordà, 2005; Plagborg-Møller and Wolf, 2021).

The magnitude is economically large. A coefficient of -0.042 on $\ln(\text{TFP})_t - \ln(\text{TFP})_{t-1}$ corresponds to approximately a 4.2 percent impact-year contraction in productivity. The point estimates do not recover materially over the horizons we examine: the $h = +5$ coefficient is -0.031 (Table 5). Under the governing wild cluster bootstrap the impact-year and the $h = +5$ coefficients are both distinguishable from zero at the level this paper adopts. The intermediate horizons are not, and neither is the cumulative five-year response. The level shift is therefore identified at the two ends of the horizon and estimated too imprecisely in between for the shape of the path to be claimed as an inferential result, not read from the point estimates alone. The output-growth response behaves differently: it contracts on impact but recovers as the post-coup environment stabilizes, a transitional flow disruption, not a durable level shift. That asymmetry is the stock-versus-flow shape the option-value mechanism implies (Section 3.3). It does not discriminate finely. A level that persists more than its own growth rate follows from any persistent productivity shock, so we read the shape as description, not as support for this framework over another. OA Appendix J.1 reports that both impact-year contractions survive the alternative F2 timing. It does not assess the full persistence paths under F2.

The 14 democracy-destroying focal events further allow three within-cell heterogeneity diagnostics that address the most natural critiques of the focal-cell composition. By coup type, 10 of 14 events are military-led under the Bjørnskov–Rode mode classification and 4 are civilian-led. By historical era, 13 of 14 occur before 1990 and only the Kyrgyz Republic 2010 falls in the post-Cold-War period. The Kyrgyz Republic event is therefore simultaneously the only civilian-led and the only post-Cold-War contributing observation, and it returns one of the deepest event-level estimates (-6.2 percent against a contributing-event mean of -4.2 percent). The same single observation is descriptively inconsistent with two of the most natural confounding interpretations: that the focal-cell estimate reflects a generic military-coup effect distinct from democracy-displacement, or that it reflects the geopolitical context of the Cold War instead of the institutional-replacement mechanism the option-value framework formalizes. Resting on one event, it cannot adjudicate either. By pre-coup income, the five TFP-contributing events span a middle-income range (reported by event in OA Table H1). The cross-event TFP coefficient does not display a monotone gradient in pre-coup income, with the richest contributing country (Chile) returning the most negative estimate and a similarly rich country (Uruguay) returning the smallest, which is descriptively inconsistent with pre-coup capital intensity being the operative source of cross-event heterogeneity.

These patterns are descriptive and are not used as evidence for the mechanism. Although the focal sample is concentrated before 1990, the point estimate is not entirely driven by the Cold-War contributors, and the one contributing observation that violates both of the natural confounds, civilian-led and post-Cold-War Kyrgyzstan, returns the second-deepest estimate in the cell and one well below the contributing-event mean. The external validity of the focal-cell magnitude is to democracy-destroying coups in middle-income countries roughly between 1964 and 2010, the temporal and income span of the five TFP-contributing events. Coverage of the early-decade dropped events (*e.g.*, Argentina 1961, Honduras 1963, Myanmar 1962) is constrained mainly by Penn World Table lagged-outcome availability. Coverage of later recurrent democracy-to-autocracy events is constrained by the first-event design.[10]

[10] The staggered comparison in OA Appendix K extends partial coverage of these events through subsequent-DA contributing switchers, and the pre-coup-regime decomposition in OA Appendix J.2 provides complementary evidence on focal-cell composition.

6.6 Robustness

We submit the DEM $\rightarrow$ AUT TFP coefficient to four robustness exercises. Each targets a specific concern that a reviewer would naturally raise given the limited cell size. Table 6 reports the results.

Table 6: Robustness architecture for the focal cell: TFP impact in democracy–to–autocracy coups

	$\hat{\beta}_{\text{DEM}\to\text{AUT}}$	SE	p (asym.)	p (boot.)
Baseline	-0.0423^{**}	(0.0110)	0.0001	0.0200
R1: Country jackknife (14 DEM–AUT countries dropped one at a time)				
coefficient range across 14 replications	[−0.0503, −0.0340]	—	—	—
N significant at $p < 0.10$ (boot.) / total	14/14	—	—	—
R2: Leave-one-region-out				
coefficient range across 7 region drops	[−0.0455, −0.0384]	—	—	—
N significant at $p < 0.10$ (boot.) / total	6/7	—	—	—
drop Latin America & Caribbean	−0.0455	(0.0123)	0.0002	$0.2525^{\dagger}$
R3: F2 regime classification (post-coup regime measured at t+2)				
$\hat{\beta}_{\text{DEM}\to\text{AUT}}$ under F2	-0.0369^{**}	(0.0126)	0.0033	0.0470
R4: Joint pre-trend test on T_{-3} and T_{-2} in DEM→AUT cell				
$\hat{\beta}_{-3}$	+0.0126	(0.0234)	0.591	—
$\hat{\beta}_{-2}$	+0.0250	(0.0147)	0.088	—
Joint Wald $F(2, 97)$	$F = 4.10$		0.019	—

Notes. Robustness on the DEM→AUT focal cell from Table 5. Baseline is the local-projection TFP estimate at $h = 0$ in the BBG-style decomposition, first-event-only sample truncated at the second coup year, with country and year fixed effects and country-clustered errors. The jackknife range $[-0.0503, -0.0340]$ holds across the 14 first-coup countries, of which five have a complete impact-year cell (TFP at t and $t-1$ plus controls) and identify the estimate. p (asym.) is the cluster-robust asymptotic approximation. p (boot.) is the wild cluster bootstrap (Webb six-point weights, null imposed, 2,000 replications), the adopted standard the stars follow: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. With five identifying countries the asymptotic p is unreliable, shown only for comparison. The pre-trend rows (R4) use a stacked specification with no bootstrap. † *Neither p-value here carries information, reported for completeness.* Dropping Latin America, which holds three of the five identifying countries, leaves two treated clusters and observations (Greece and the Kyrgyz Republic). The asymptotic 0.0002 reflects an error across 87 clusters with variation from two points, and the bootstrap 0.253 is not evidence against the effect either, since with two treated clusters the wild bootstrap has almost no power to reject (MacKinnon and Webb, 2018).

OA Figure F2, in OA Appendix F, displays the three coefficient re-estimation exercises as a coefficient plot, one row per displayed nontrivial modification of the baseline. The nine noncontributing-country drops, which reproduce the baseline, are omitted. The estimate stays negative in every replication and significant at the 90% level in every replication the design can identify, and the spread across them, from −0.034 to −0.050, brackets the baseline of −0.042.

The country jackknife is reported as 14 replications for completeness, but is effectively a five-country exercise: only 5 of the 14 nominal DEM → AUT countries (BRA, CHL, GRC, KGZ, URY, see OA Table H1) contribute non-trivially to the impact-year TFP point estimate. The other 9 events appear in the regression sample but lack the lagged-outcome cells the LP requires at $h = 0$, so dropping them produces a coefficient nearly identical to the baseline. The material sensitivity is therefore across the 5 contributing countries: the point estimate ranges from −0.034 to −0.050, and all five replications continue to reject at the 90% level this paper adopts. Chile (1973) is the most influential single country: dropping it attenuates the coefficient to −0.034 and moves the bootstrap p to 0.050, the only one of the five that does not clear the 5% level. The result is not driven by any single observation among the five TFP-contributing countries. The external check on identifying mass beyond the 5-country first-event set is the staggered comparison in OA Appendix K: the Callaway–Sant'Anna and de Chaisemartin–D'Haultfœuille estimators pull in subsequent-event DA switchers the first-event design omits (22 switchers satisfy the criteria and 20 enter the impact-year estimand, and 18 lie outside the baseline first-coup sample), and return post-treatment averages that bracket the baseline, while both impact-year estimates are smaller in magnitude. The leave-one-region-out exercise addresses the concentration of DEM–AUT events in Latin America: 8 of the 14 events occur in Latin America & the Caribbean and the remaining 6 are spread across East Asia & Pacific (1), Europe & Central Asia (2), Sub-Saharan Africa (3), with no events in the Middle East & North Africa, North America, or South Asia. By the same contributing-set logic, 3 of the 5 contributing countries (BRA, CHL, URY) are in LAC. Dropping LAC leaves the focal point estimate at −0.046, numerically larger than the baseline, but the replication is uninformative in either direction and we do not use it. Only Greece and the Kyrgyz Republic still contribute, which is two treated clusters and two treated observations.[11] The six nominal non-Latin-American focal events span three regions, but the two non-Latin-American countries contributing to the TFP coefficient are both in Europe and Central Asia. The F2 regime-classification check re-estimates the decomposition using the post-coup regime measured at $t+2$ instead of $t+1$, which provides the "settled" regime for cases in which the immediate post-coup year is itself transitional: under this alternative both focal impact-year contractions are attenuated only slightly from baseline and both remain significant at the 5% level under the wild cluster bootstrap (OA Table J2). Both focal contractions survive the alternative timing, as developed in OA Appendix J.1. The joint pre-event nullity test stacks $h = -3$ and $h = -2$ with horizon-interacted fixed effects and yields point estimates of +0.013 and +0.025 on the DEM $\rightarrow$ AUT indicator ($p = 0.591$ and $p = 0.088$, so $\hat{\beta}_{-2}$ is distinguishable from zero at the 90% standard this paper adopts), the joint Wald test returns $F(2, 97) = 4.10$ with $p = 0.019$, rejecting strict pre-event nullity at the 5% level but not at the 1%. The pre-event coefficients $\hat{\beta}_{-3} = +0.013$ and $\hat{\beta}_{-2} = +0.025$ are positive and small. Because the LP specification regresses $Y_{t+h} - Y_{t-1}$ on the shock indicator, a positive pre-event coefficient means $Y_{t+h} > Y_{t-1}$, so productivity was on net *declining* from $t-3$ toward the reference horizon $t-1$. We do not read that decline as favourable to the estimate. If the pre-coup slide would have continued absent the coup, part of the impact-year contraction is trend, not treatment, and the estimate overstates the effect. That reading is the one consistent with Bjørnskov (2022), whose finding that economic conditions predict coup occurrence makes the pre-coup decline a plausible confound. We cannot separate that confound from the anticipatory channel of Section 3.2, because the two imply a falling pre-event path with the same sign, which is why we do not read the pre-event coefficients as evidence for the model in either direction.

[11] The asymptotic p of 0.0002 that this replication returns is an artefact of a clustered standard error computed over 87 clusters when the identifying variation comes from two points, and the bootstrapped p of 0.253 is not evidence against the effect either, because with two treated clusters the wild bootstrap has almost no power to reject (MacKinnon and Webb, 2018). Across the six other region-out replications the coefficient ranges from -0.038 to -0.045 and the bootstrapped p from 0.013 to 0.096, all significant at the 90% level we adopt.

We therefore submit the coefficient to the honest sensitivity analysis of Rambachan and Roth (2023) under the relative-magnitudes restriction, which bounds the post-coup violation of parallel trends by a multiple M of the largest change between consecutive pre-coup coefficients ($\bar{\delta} = 0.025$ here, the step from $\hat{\beta}_{-2}$ to the normalised reference horizon). Both symbols are local to this sensitivity analysis: M here is the relative-magnitudes multiple, unrelated to the policy moment M_c of Section 3, and $\bar{\delta}$ is a pre-trend step, unrelated to the year fixed effect δ_t^h of Equation (3). The impact-year coefficient remains distinguishable from zero at the 5% level up to $M \approx 0.65$, and its confidence interval admits zero from $M \approx 0.70$ onward (Table 7). At the benchmark $M = 1$, where the post-coup violation is allowed to be as large as the largest pre-coup one, the interval admits zero. We report it as `honestdid` returns it and flag one irregularity in it: the sweep is not nested at $M = 1$. The relative-magnitudes set $\Delta^{RM}(M)$ grows with M, so the confidence intervals should widen weakly, and across the fifteen values of M we examine (fourteen adjacent transitions) they do so everywhere except at the two transitions around $M = 1$, where the intervals narrow from $M = 0.9$ to $M = 1.0$ and again from $M = 1.0$ to $M = 1.25$ (Table 7). The irregularity is not a property of our implementation: it survives pinning the test-inversion grid and is unchanged whether the sweep is run in one call or fifteen. We therefore do not lean on the $M = 1$ interval. The claim we make rests on the breakdown point, where the sweep is nested and behaves as the theory requires. The estimate is robust to trend violations up to roughly 65% of the largest observed pre-coup violation, and not beyond.

Table 7: Rambachan–Roth sensitivity of the democracy-to-autocracy impact-year TFP coefficient

M	$\Delta^{RM}(M)$ 95% robust CI	Excludes zero?
0	$[-0.067, -0.017]$	Yes
0.50	$[-0.080, -0.005]$	Yes
0.65	$[-0.084, -0.001]$	Yes
0.70	$[-0.086, +0.001]$	No
0.84	$[-0.111, +0.006]$	No
0.90	$[-0.115, +0.028]$	No
1.00	$[-0.097, +0.034]$	No
1.25	$[-0.107, +0.021]$	No

Notes. Robust confidence intervals for the DEM → AUT impact-year TFP coefficient under the Rambachan and Roth (2023) relative-magnitudes restriction $\Delta^{RM}(M)$, which allows a post-treatment violation of parallel trends up to M times the largest pre-treatment violation, solved by the conditional least-favourable method. The estimate excludes zero up to $M \approx 0.65$ and admits zero from $M \approx 0.70$. At the benchmark $M = 1$ the permitted post-coup violation equals the largest pre-coup violation. The sweep is nested in M so the intervals should weakly widen. Across the fifteen grid values (fourteen adjacent transitions), two transitions, both around $M = 1$, violate the expected nesting in the reported output (the $M = 1$ and $M = 1.25$ intervals shown above). The rows are selected values from that fifteen-point sweep. The full sweep is in the replication materials.

Among the four exercises specified in Section 5.5, the coefficient survives the country jackknife and the alternative regime-classification timing, under which both the TFP and growth contractions remain significant. The Latin-America exclusion leaves two treated clusters and is uninformative, not passed, and the joint pre-event nullity test rejects. The Rambachan–Roth sensitivity analysis is separate from the four and preserves significance up to the bound stated above. We carry the pre-event rejection instead of explaining it away.

Separately, we compare the first-event-only LP design with two staggered estimators that accommodate countries with multiple coups. The Callaway-Sant'Anna estimator under the absorbing-treatment assumption (cohort = first DEM → AUT year) returns an impact-year effect of −0.032 ($p = 0.008$) and a post-treatment average of −0.051. The de Chaisemartin-D'Haultfœuille (dCDH) non-absorbing estimator returns an impact-year effect of −0.015 ($p = 0.137$) and an average total effect of −0.039 across post-shock horizons. The two post-treatment averages bracket the baseline focal estimate of −0.042, though neither average is separately distinguishable from zero at the 5 percent level (both 95 percent intervals cross zero), and both impact-year estimates are smaller in magnitude than the baseline (OA Tables K2–K3). The dCDH joint test on the post-shock horizons does not reject the null ($p = 0.70$). The result is consistent with low power. Only 20 of 65 switch-on events enter the dCDH estimand for the impact-year effect: 22 satisfy the trajectory-stability and TFP-coverage requirements, and two are then lost to the estimator's internal control-availability check, with treatment reversals one source of the initial attrition. The non-rejection neither establishes an effect nor rules out the null. Countries that cycle repeatedly between democracy and autocracy within the panel window, Argentina, Nigeria and Honduras among them, lose events that way, though such a country can still contribute through a switch-on with a stable post-treatment trajectory. The dCDH average total effect of −0.039, rather than its impact-year estimate of −0.015, is aligned with the baseline. OA Appendix K documents how the staggered post-treatment averages bracket the baseline, the structural reason for the dCDH precision loss, and the recovery of five canonical events (Onganía 1966, Videla 1976, Velasco 1968, Buhari 1983, Prayut 2014, with Velasco and Buhari entering the estimand at $t+1$ relative to their canonical coup years) as contributing switchers under the non-absorbing design.[12]

The empirical pattern we document supports a layered set of claims of increasing specificity. At the aggregate level, the first coup observed in a country is associated with a

[12] Of the 65 switch-on events, 22 satisfy the formal trajectory-stability and TFP-coverage criteria, and 2 of those are lost to the estimator's internal control-availability check. Velasco and Buhari enter at $t + 1$ instead of the canonical coup year, because of the lag structure of the regime variable used to construct the switching indicator.

contemporaneous 2 percent contraction in total factor productivity and a 2.5-point contraction in output growth on the transformed scale, neither of which persists materially beyond one year in the pooled specification. At the decomposition level, the productivity contraction concentrates in coups that move a country from democracy to autocracy ($\hat{\beta}_0 = -0.042$, bootstrap $p = 0.020$), while the output-growth contraction appears in the DEM $\rightarrow$ AUT and INTRA_AUT classes and is negligible in INTRA_DEM, while the two-event AUT $\rightarrow$ DEM cell is uninformative. At the focal-cell level, the democracy-destroying TFP contraction survives leave-one-country drops and alternative regime-classification specifications. The joint pre-event nullity test, however, rejects at the 5% level, and we read the pre-coup decline as a potential confound, not as favourable to the estimate. The one region-drop that carries the identifying mass, Latin America, leaves two treated clusters and cannot be used in either direction. The TFP and growth responses diverge in persistence in a way that the stock–flow distinction of Section 3.3 naturally accommodates: TFP accumulates the disrupted-flow shortfall and does not recover, while growth is itself the flow and recovers as the post-coup environment stabilizes. The intra-democracy counterfactual (failed and non-regime-changing coups in democracies) produces no detectable productivity response at any reported horizon, consistent with the option-value reading in which the operative pathway is the dispersion of post-coup institutional outcomes, not the coup event itself or generic political uncertainty about the immediate political environment. Its investment response is not negligible. Impact-year investment contracts 2.9 points on the transformed gross-investment scale and rejects under the bootstrap, which is the sign reversal the structural layer of Section 7 takes up, not a null.

We do not claim that the data identify a structural causal mechanism. The analysis is associational, conditional on country and year fixed effects, lagged outcome, and lagged log GDP per capita. The robustness architecture disciplines the inference but does not provide quasi-experimental variation in coup occurrence. Within the limits of observational inference, the pattern is the most disciplined the data support, and it parallels the productivity finding of Bennett et al. (2025) in a broader country-coverage panel.

Section 7 places the option-value reading in a parameterisation, estimating the channel loadings under the headline calibration of post-coup institutional moments and checking them against alternative V-Dem calibrations, and extending the estimation to a two-block joint specification across TFP, growth, and gross investment that respects the theoretical distinction between the firm's direct decision (investment) and its downstream consequences (TFP, growth). The two-block estimation returns the uncertainty channel with the predicted negative sign in both blocks under the headline calibration, though that sign is not stable in the investment block across alternative institutional measures. It does not confirm the stock-versus-flow contrast, which we treat as illustration, not as a test, and no cumulative response is distinguishable from zero. What the structural layer adds is a parameterisation in which the positive investment response and the negative productivity response arise from the same two channels, carried by a separate investment-block coalition loading.

7 A structural interpretation of the option-value channel

This section does not add an independent test. It adds a parameterisation that disciplines the interpretation of the reduced-form patterns, and the reader should hold it to that standard, not to an evidentiary one. What it buys is a restriction the reduced form cannot impose: that all three outcomes load on the same two channels, with a sign structure that a single-loading account of "worse coups are worse" cannot reproduce. What it does not buy is identification. Under the preferred level specification all four channel loadings exclude zero at the 90% level. Only the investment-block coalition loading, however, remains sign-robust across weighting schemes and after concentrating out the common coup component. We state that here instead of at the end.

The reduced-form pattern concentrates the impact-year productivity contraction in coups that replace a democracy with an autocracy, and leaves democracy-preserving coups in democracies with a negligible effect. The structural layer asks what set of channel loadings would generate the impact-year productivity asymmetry and investment reversal from a single firm-level adoption decision. The persistence contrast is described separately, from the multi-horizon local-projection paths.

From this point onward all institutional moments and loadings are expressed in V-Dem units, and we suppress tildes for readability: $M_c \equiv \tilde{M}_c$, $\Sigma_c \equiv \tilde{\Sigma}_c$, $\alpha_0 \equiv \tilde{\alpha}_0$ and $\gamma_0 \equiv \tilde{\gamma}_0$. In this empirical convention institutional improvement raises M_c, the predicted productivity-block coalition loading is positive, and the predicted uncertainty loading remains negative. The investment block carries a separate coalition loading whose sign is not fixed by this mapping.

We estimate a two-block minimum-distance specification. The productivity block, total factor productivity and output growth, shares a coalition loading and an uncertainty loading, with a scale parameter absorbing the unit conversion between them. The investment block carries its own two loadings because the theoretical private-investment decision is direct, while productivity and growth are downstream consequences of the adoption flow that decision generates. The measured gross-investment aggregate is an imperfect proxy that also includes government and residential investment. The moments are the impact-year local-projection coefficients by transition class, and the institutional inputs are the class-average V-Dem shift and its dispersion.

Table 8: Joint minimum-distance estimates: two-block structural specification

Parameter		Estimate	90% bootstrap CI	95% bootstrap CI
Panel A: Two-block specification (5 parameters)				
Productivity block (TFP and growth share parameters)				
α_0	(coalition channel)	0.165	$[+0.012,\ +0.336]$	$[-0.021,\ +0.381]$
γ_0	(uncertainty channel)	-0.150	$[-0.281,\ -0.047]$	$[-0.311,\ -0.022]$
κ_{growth}	(growth scale vs. TFP)	100.43	$[+52.9,\ +186.7]$	$[+43.8,\ +223.7]$
Investment block (own parameters)				
$\boldsymbol{\alpha_0^{\text{inv}}}$	(coalition channel, investment)	$\mathbf{-33.97}$	$[\mathbf{-77.20},\ \mathbf{-0.73}]$	$[-89.37,\ +3.69]$
γ_0^{inv}	(uncertainty channel, investment)	-16.54	$[-33.13,\ -4.35]$	$[-36.66,\ -2.67]$
Panel B: Over-identification				
J_{tg} under optimal weighting ($\chi^2(3)$)			3.39	($p = 0.33$)
J_{inv} under optimal weighting ($\chi^2(1)$)			0.10	($p = 0.75$)

Notes. Generated-regressor two-step minimum distance on PWT 11.0 and the Powell–Thyne-reconciled coup panel. The country-cluster bootstrap (800 resamples) jointly resamples the impact-year local-projection coefficients and the calibrated V-Dem moments, so the intervals are bootstrap percentiles. Moments are calibrated on all eligible coups, first and subsequent. TFP and growth share (α_0, γ_0) with κ_{growth} absorbing units, and investment carries its own $(\alpha_0^{\text{inv}}, \gamma_0^{\text{inv}})$. Weighting is diagonal inverse-variance, preferred over full optimal because the productivity-block covariance is near-singular (condition number $\approx$ 22,000), and the two give the same points. Inference is at the 90% level. The investment coalition loading α_0^{inv} is the most robustly identified parameter, negative under identity, diagonal, and optimal weighting and after removing a common coup component (OA Appendix C.3). The uncertainty loadings are sign-identified in levels but not under that contrast. Panel B reports the optimal-weighted over-identification test, and neither block rejects. Calibration: V-Dem liberal-democracy index.

Table 8 reports the preferred specification: the class moments are calibrated on all eligible coups and the system is estimated under diagonal inverse-variance weighting. Under this scheme all four level loadings are sign-identified at the 90 percent level. The coalition loading on productivity is positive (+0.165) and the uncertainty loadings on both blocks are negative (−0.150 on productivity, −16.54 on investment), each with a 90 percent interval that excludes zero. The scale parameter κ_{growth} is a unit conversion, not a channel.

The one parameter we read as robustly identified is the investment-block coalition loading α_0^{inv}. It is negative (-33.97), its 90 percent interval excludes zero, and its sign survives every weighting scheme we try (identity, diagonal, and full optimal) and the removal of an outcome-specific common coup component through the contrast specification of OA Appendix C.3. The uncertainty loadings are more fragile: their negative sign holds in this level specification under scale-adjusted weighting but does not survive concentrating out the common component, which turns the productivity-block uncertainty loading positive. Cross-class institutional variation alone therefore does not separately identify the uncertainty channel, and we read its negative sign as suggestive, not established. The productivity-block coalition loading is likewise sensitive to the institutional measure: re-estimated on the first-event calibration across four V-Dem subindices it flips sign under rule of law (-0.015 against $+0.134$ under liberal democracy), while the investment-block uncertainty loading flips under private-property rights ($+4.75$ against -13.24). The details are in OA Appendix C.4. The over-identifying restriction is not rejected. We read the structural estimates as a sign-and-bound decomposition of the impact-year moments disciplined by the model, not as point-identified deep parameters, and the bootstrap propagates uncertainty in the reduced-form coefficients and in the calibrated institutional moments jointly, which is why the intervals are as wide as they are.

Investment carries the framework's most distinctive cross-class pattern, which concerns the ordering across classes, not the sign in one cell. Because the coalition channel can redirect capital toward favoured sectors, the separate investment loading permits investment to *rise* where a coup installs a new coalition and fall where the incumbent institutions survive. This is a directional point-estimate pattern the model allows, not a sign it derives. An account in which worse coups are worse predicts the opposite ordering, with investment more negative in the democracy-destroying class.

The point estimates show the reversal that the model permits. Impact-year investment is +3.26 in the democracy-destroying class and -2.85 in the intra-democracy class, and the intra-democracy coefficient is the one that rejects ($p = 0.049$ under the wild cluster bootstrap). The generic account does not produce that ordering even at the point estimate.

The inference does not carry it. The cross-class contrast returns $p = 0.124$ under the wild cluster bootstrap, with a 95% confidence interval of $[-1.20, +14.26]$ around a contrast point estimate of +6.12 (OA Table J1, Panel B), and only eight countries contribute to the democracy-destroying investment coefficient. That interval is wide enough that the estimate is close to uninformative about magnitude. What survives is the *sign* of the point estimates and the direction of the reversal, and even that rests on eight countries. We report it as directional evidence, not as a test.

The framework also implies a persistence contrast: productivity, a stock that integrates a disrupted adoption flow, should stay down, while growth, the flow itself, should recover. The point estimates trace that shape, with a productivity amplification ratio of 5.63 against a perfect-persistence value of six and a growth ratio of 0.66. We report it as illustration and not as a test. A level that persists more than its own growth rate is close to definitional, and no cumulative five-year response is distinguishable from zero under the bootstrap in any case. The cumulative estimates are in OA Appendix B.3.

The restriction that does carry content is the one the two-block model imposes across outcomes: that productivity and growth load on the same (α_0, γ_0) up to a scale factor. That is an over-identifying restriction a reduced-form description does not impose, and the data do not reject it ($\chi^2(1) = 0.002$, $p = 0.96$. OA Table J1, Panel C). We are careful about what that buys. Failing to reject an over-identifying restriction on a three-class moment system whose focal class contains thirteen calibration events is a statement about power as much as about fit, and it is not evidence *for* the restriction. It is the absence of evidence against it.

Because the annual growth rate is volatile, the same contrast is clearer in the log level of GDP per capita, a separately estimated level path shown in OA Figure F3 (OA Appendix F) as the level counterpart to the output-growth response. Productivity steps down at the impact year and remains depressed through the fifth year, the signature of a stock that integrates a disrupted flow. The output level dips at the impact year and returns toward its pre-event path within two years, the level counterpart of a growth flow that recovers once the post-coup environment stabilizes.

Table 8 is the paper's structural estimate. The appendix reports the derivation, the nested single-outcome step that motivates the two-block split, a more restrictive one-block alternative, and the V-Dem calibration checks that establish which of the loadings are stable (OA Appendix A for the derivation, OA Appendix B for the nested and cumulative steps, and OA Appendices C.1 and C.4 for the one-block and V-Dem specifications respectively). Those are nested alternatives and diagnostics, not competing estimates, and we do not ask the reader to choose among them.

8 Conclusion

Political instability carries a broad productivity cost. Following coups on a clean risk set reweighted to match coup and comparison countries on predetermined institutional, macroeconomic, and pre-event productivity characteristics, productivity contracts about three percent in the year of a coup and about four percent a year later. This immediate drop is the paper's most robust finding: it barely moves across control constructions, and it is much the same whether the coup ends a democracy, replaces one autocrat with another, or leaves a democracy standing. Balancing flattens most of the pre-event difference, though a longer pre-period leaves some residual sensitivity. The medium-run path is more design-sensitive, and longer horizons provide no robust evidence of permanent scarring.

The same design placed on the other macroeconomic outcomes locates the coup response in the productivity and real-activity block. Output growth and gross investment contract in the impact year alongside productivity, and government spending falls modestly, while trade, unemployment, net migration, foreign direct investment, and inflation show no robust response. Gross investment declines but is estimated imprecisely, so its point contraction is read as directional. The disturbance concentrates in real activity.

The regime a coup produces still shapes the initial institutional shock. Isolating first coups, the contraction looks larger where a coup destroys a democracy, so institutional destruction may deepen the initial blow. An option-value model of irreversible investment reads the shock as a coalition channel and an uncertainty channel. The clearest structural result lies in the investment block: the coalition loading is the only channel parameter whose sign survives the full set of weighting schemes and the removal of a common coup component.

These estimates are associational and, for the first-event and uncertainty channels, imprecise. A longer dynamic panel and a decomposition of investment into its private, public, and residential parts would show whether the medium-run gap is a lasting shortfall in technology adoption and where the coalition effect operates.

Declarations

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Ethics approval. Not applicable. The study uses secondary, publicly available, country-level aggregate data and does not involve human participants or animals.
Author contributions. C.C. conceived and designed the study, assembled the data, carried out the empirical and structural analysis, produced all tables and figures, and wrote the manuscript.
Data availability. The analysis uses publicly available data from the World Bank's World Development Indicators, the Penn World Table, the Bjørnskov–Rode Regime Dataset, the V-Dem dataset (liberal democracy index and the rule-of-law, judicial-constraints and private-property-rights subindices), the Powell–Thyne coup catalogue, the Geddes–Wright–Frantz regime dataset, and the Barro–Lee educational attainment data. The constructed panel dataset and the replication code are available from the author on reasonable request.

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Online Appendix

A Theoretical model: formal details

This appendix contains the formal derivations and specifications that support the theoretical framework of Section 3. The main-text discussion is restricted to the intuition and the predictions the empirical strategy adjudicates. The apparatus below is the underlying machinery.

A.1 The hazard process

Political condition is summarized by the instantaneous hazard of a coup d'état, λ_t, following a square-root (Cox–Ingersoll–Ross) mean-reverting diffusion,

$$d\lambda_t = \theta\,(\bar{\lambda}_r - \lambda_t)\,dt + \sigma\sqrt{\lambda_t}\,dW_t, \qquad 2\theta\bar{\lambda}_r \geq \sigma^2, \tag{5}$$

with mean-reversion speed $\theta > 0$, regime-specific long-run hazard $\bar{\lambda}_r > 0$, volatility $\sigma > 0$, and W_t a standard Brownian motion. The Feller condition keeps $\lambda_t > 0$: the state variable is a hazard rate. Coups arrive as a Cox process with intensity λ_t.

Each coup realises one of three outcomes $b \in \{F, SC, N\}$ (failed, self-coup, normal successful) with regime-conditional probabilities $(\pi^r_F, \pi^r_{SC}, \pi^r_N)$. Policy is a pair $s_t = (\tau_t, m_t)$: a capital tax τ_t and money growth m_t. A coup shifts the capital tax by a branch-specific draw,

$$\tau_{t^+} = \tau_t + \Delta_b + \varepsilon_b, \qquad \varepsilon_b \sim (0, V_b), \tag{6}$$

with $(\Delta_F, V_F) = (0, 0)$ for a failed coup, $(\Delta_{SC}, V_{SC}) = (0, V_{SC})$, $V_{SC} > 0$, for a self-coup (the same coalition, freed of constraints: dispersion but no mean shift), and $(\Delta_N, V_N) = (\Delta, V_N)$, $\Delta > 0$, $V_N \geq V_{SC}$, for a normal successful coup.

Mean and variance. Conditional on a coup, the policy change $\Delta\tau = \Delta_b + \varepsilon_b$ has, by the law of total variance,

$$M_r \equiv \mathbb{E}[\Delta\tau \mid \text{coup}] = \pi_N^r \, \Delta, \tag{7}$$

$$\Sigma_r = \underbrace{\pi_{SC}^r V_{SC} + \pi_N^r V_N}_{\text{within-branch dispersion}} + \underbrace{\pi_N^r (1 - \pi_N^r) \, \Delta^2}_{\text{coalition-turnover risk}} . \tag{8}$$

M_r is the coalition channel: the expected policy shift conditional on a coup. Σ_r is the uncertainty channel: the variance of policy outcomes conditional on a coup.

Regime calibration. Regime type r indexes the six Bjørnskov–Rode categories. The long-run hazard $\bar{\lambda}_r$ and the branch probabilities $(\pi_F^r, \pi_{SC}^r, \pi_N^r)$ are calibrated directly to BR v6.2:

OA Table A1: Calibration of the political environment to Bjørnskov–Rode v6.2

Regime r	Country-years	$\bar{\lambda}_r$	π_F^r	π_{SC}^r	π_N^r
Parliamentary democracy	2,873	0.015	0.61	0.00	0.39
Mixed democracy	1,322	0.023	0.42	0.06	0.52
Presidential democracy	1,690	0.047	0.57	0.03	0.40
Civilian dictatorship	3,255	0.049	0.52	0.03	0.45
Military dictatorship	1,985	0.111	0.56	0.04	0.40
Royal dictatorship	1,056	0.025	0.54	0.00	0.46

In parliamentary democracies and royal dictatorships $\pi_{SC} = 0$ in the data. The model collapses to a two-branch structure (F, N) in these regimes, with self-coups an off-equilibrium possibility.

A.2 Entrepreneur's irreversible-investment problem

A continuum of potential projects is indexed by a sunk investment cost f, distributed uniformly on $[0, \bar{f}]$. Investment is irreversible. An active project earns an after-tax flow $(1-\tau_t)\, a$. The entrepreneur chooses when to commit (Dixit and Pindyck, 1994). Let $P(\tau, \lambda)$ be the value of an active project and $\Omega(\tau, \lambda) \geq 0$ the option premium, the value of waiting in excess of investing now. A project of cost f is undertaken the first time $P(\tau, \lambda) - \Omega(\tau, \lambda) \geq f$. With f uniform and an interior threshold $0 < P(\tau, \lambda) - \Omega(\tau, \lambda) < \bar{f}$, the share of potential projects undertaken is linear in the threshold:

$$s_I(\tau, \lambda) = \frac{P(\tau, \lambda) - \Omega(\tau, \lambda)}{\bar{f}}. \tag{9}$$

Outside that range the share is clipped at 0 or 1 and the comparative statics below, which are the interior ones, do not apply. Aggregate investment is $I = i_F\, s_I$. The capital-channel expressions below use the per-project investment size relative to the capital stock, i_F/K.

The project value responds to the mean of coup risk, $\partial P/\partial\lambda < 0$: a higher hazard raises the chance of incurring the expected tax increase M_r. The option premium responds to the variance, $\partial\Omega/\partial\lambda > 0$: a higher hazard makes the policy jump Σ_r more imminent, and option value rises with the uncertainty of the underlying.

Proposition (Investment and the coup hazard, restating Proposition 1)**.** *For an interior investment threshold and an expected institutionally degrading policy jump $M_r > 0$, the investment share is strictly decreasing in the coup hazard,*

$$\frac{\partial s_I}{\partial \lambda} = \underbrace{\frac{1}{\bar{f}}\,\partial P/\partial\lambda}_{\textit{NPV / coalition channel}(\propto M_r)} - \underbrace{\frac{1}{\bar{f}}\,\partial\Omega/\partial\lambda}_{\textit{option-value / uncertainty channel}(\propto \Sigma_r)} < 0,$$

both terms negative. We impose, not derive, the ordering $|\partial FDI/\partial\lambda| > |\partial s_I/\partial\lambda|$: foreign capital is disproportionately exposed to expropriation, so we take it to load more heavily on both τ and Σ_r. The model carries no separate foreign-investment decision rule, so this is a *qualitative extension of the framework, not a result of it.*

The option premium Ω depends on both moments of the jump distribution, and its leading sensitivity is to the dispersion, scaling with $\sqrt{\Sigma_r}$: the value of waiting is the value of avoiding bad realisations, which scales with the dispersion of outcomes, not their mean. Because λ_t is observed and forward-looking, investment is a jump variable that responds to the hazard contemporaneously, with no stock-adjustment lag. The model therefore implies a declining pre-event path. We do not treat that implication as testable: it has the same sign as the reverse causality of Bjørnskov (2022), so the pre-event coefficients at $h = -3, -2$ cannot separate the two and are used only as a pre-trend diagnostic.

A.2.1 Closed form for the project value P

Treating λ as constant over the valuation interval, with $\bar{\lambda}_r$ used subsequently for the regime-specific calibration, and assuming the post-coup policy jump $\Delta\tau \sim (M_r, \Sigma_r)$ is drawn once at the next event,[13] the project value satisfies, with $\rho > 0$ the discount rate,

$$P(\tau, \lambda) = \frac{a(1-\tau)}{\rho} - \frac{a\,\lambda\,M_r}{\rho(\rho+\lambda)}. \tag{10}$$

The first term is the riskless-policy NPV $\int_0^\infty e^{-\rho t}(1-\tau)\,a\,dt$. The second is the expected discounted loss from the next coup: it arrives at time $T \sim \mathrm{Exp}(\lambda)$, so $\mathbb{E}[e^{-\rho T}] = \lambda/(\rho+\lambda)$, and conditional on arrival the tax rises by M_r in expectation, generating a flow loss $a\,M_r/\rho$. The project value P is linear in the policy jump and carries only the mean M_r. The variance channel enters solely through the option premium Ω of §A.2.2, computed separately.

The implied partial derivatives are

$$\frac{\partial P}{\partial \tau} = -\frac{a}{\rho} < 0, \qquad \frac{\partial P}{\partial \lambda} = -\frac{a\,M_r}{(\rho+\lambda)^2} < 0 \quad (\text{for } M_r > 0), \tag{11}$$

[13] This is a first-order approximation: the CIR process has $\mathbb{E}[\lambda_t] \to \bar{\lambda}_r$ on the mean-reversion time-scale $1/\theta$, which is short relative to the discount horizon $1/\rho$ in our calibration. The full state-dependent $P(\tau, \lambda)$ generalises the expression below by replacing $\bar{\lambda}_r$ with the conditional expectation of integrated λ. The implied moment structure is unchanged.

both negative for the case $M_r > 0$ of an institutionally degrading coup. The NPV channel runs through these derivatives.

A.2.2 Closed form for the option premium Ω

The option premium is the value of waiting for the next coup before sinking the investment cost. Standard real-options arguments (Dixit and Pindyck, 1994) give a closed form when the underlying state (τ_t) has a single jump-arrival rate and a Gaussian post-jump distribution. To leading order in ρ^{-1},

$$\Omega(\tau, \lambda) \approx \frac{\lambda}{\rho + \lambda} \cdot \frac{a}{\rho} \cdot \left[\sqrt{\frac{\Sigma_r}{2\pi}} \exp\left(-\frac{M_r^2}{2\Sigma_r} \right) - M_r \, \Phi\left(-\frac{M_r}{\sqrt{\Sigma_r}} \right) \right], \tag{12}$$

where Φ is the standard normal CDF. The bracketed expression is $\mathbb{E}[\max(-\Delta\tau, 0)]$ when $\Delta\tau \sim \mathcal{N}(M_r, \Sigma_r)$: the expected favourable component of the post-coup tax shift, capturing the upside the firm secures by waiting to invest only when the realised policy outcome is favourable ($\Delta\tau < 0$). The prefactor $\lambda/(\rho + \lambda)$ is the expected discount to the next jump, and a/ρ converts the policy-state shift into value units.

The leading-order scaling in the variance moment is

$$\frac{\partial \Omega}{\partial \sqrt{\Sigma_r}} > 0, \qquad \frac{\partial \Omega}{\partial \Sigma_r} \sim \frac{1}{\sqrt{\Sigma_r}} > 0 \quad \text{(at small } M_r\text{)}. \tag{13}$$

This is the central result of §A.2.2: the option premium scales as $\sqrt{\Sigma_r}$, not linearly in Σ_r, which is the standard option-pricing scaling, analogous to the Black–Scholes vega. The premium enters the coup only through the mean and the variance of the policy change it delivers, so the scaling is a property of Ω as a function of those two moments and does not depend on what the coup is conditioned on. Conditioning on the transition class c instead of the pre-coup regime r replaces (M_r, Σ_r) with the class-level moments (M_c, Σ_c) and leaves the functional form intact. The empirical structural specification of Section 7 is stated on that partition, and therefore writes outcome responses as linear in $(M_c, \sqrt{\Sigma_c})$ instead of in

(M_c, Σ_c).

A.2.3 TFP and growth from the adoption flow

This subsubsection derives the structural mapping between the option-value investment decision and the aggregate outcomes the empirical structural section adjudicates (TFP, output growth, gross investment). The derivation has five parts: the TFP–adoption stock relation, the growth–adoption flow relation, the cross-outcome proportionality restriction (with the within-block versus across-block refinement that motivates the two-block specification of Section 7), the cumulative implications used in the empirical test, and the sign-convention mapping between the model's policy state τ and the V-Dem empirical measure.

(a) TFP from the adoption flow. Productivity evolves through a continuum of potential technology upgrades indexed by adoption cost $x \in [0, \bar{x}]$, uniformly distributed. A firm with cost x adopts the upgrade the first time

$$P(\tau_t, \lambda_t) - x \geq \Omega(\tau_t, \lambda_t) \iff x \leq P(\tau_t, \lambda_t) - \Omega(\tau_t, \lambda_t) \equiv z_t. \tag{14}$$

The instantaneous adoption rate is $n_t \equiv F_X(z_t) = z_t/\bar{x}$, where F_X is the adoption-cost distribution with density $f_X = 1/\bar{x}$, distinct from the failed-coup branch F and the project cost f, with z_t the same threshold that drives aggregate physical investment in Equation (9). Each adopted upgrade raises log productivity by a small constant g, giving

$$d \ln A_t = n_t \, g \, dt + \mu_0 \, dt, \tag{15}$$

with μ_0 a non-coup-related baseline drift. Integrating from the year before a coup event at calendar time t^* to horizon h,

$$\ln A_{t^*+h} - \ln A_{t^*-1} = \int_{t^*-1}^{t^*+h} n_s \, g \, ds + (h+1) \, \mu_0. \tag{16}$$

Equation (16) is the stock relation of §3.3, Equation (2): total factor productivity is the integral of the adoption flow.

The local-projection coefficient at impact horizon $h = 0$ for transition class c, defined as

$$\beta^A_{c,0} \equiv \mathbb{E}[\ln A_{t^*} - \ln A_{t^*-1} \mid \text{trans} = c] - \mathbb{E}[\ln A_{t^*} - \ln A_{t^*-1} \mid \text{no coup}], \tag{17}$$

linearises around the pre-event policy state (τ_0, λ_0) as

$$\beta^A_{c,0} \approx g\, f_X(z_0)\, [\Delta P_c - \Delta\Omega_c], \tag{18}$$

with $\Delta P_c \equiv P(\tau_0 + \Delta\tau_c, \lambda_0) - P(\tau_0, \lambda_0)$ the project-value change the *realised* policy shift delivers, holding the hazard and the distribution of the next jump fixed, and $\Delta\Omega_c$ the option-premium change induced by the class's policy *dispersion* alone, evaluated at the same λ_0 and holding the mean jump fixed at its baseline. We define $\Delta\Omega_c$ as this dispersion-only partial change, $\Delta\Omega_c \equiv \Omega(\bar{M}, \Sigma_c) - \Omega(\bar{M}, \Sigma_0)$, so that the mean channel enters the response only through ΔP_c and the dispersion channel only through $\Delta\Omega_c$. The cross term $\partial\Omega/\partial M$ is not attributed to either coefficient. With $f_X(z_0) = 1/\bar{x}$ the adoption-cost density at the threshold, substituting (11)–(13) and using the calibrated class-specific moments $(M_c, \Sigma_c) \equiv (\mathbb{E}[\Delta\tau_c], \mathrm{Var}[\Delta\tau_c])$ yields the linear-in-M, linear-in-$\sqrt{\Sigma}$ decomposition. The conditioning changes here: the firm's problem above conditions on the pre-coup regime r, which is what the firm knows when it decides, while (M_c, Σ_c) condition on the transition class c, which is what the econometrician observes after the fact. The two are analogous conditional moments of $\Delta\tau$ under different conditioning sets. The empirical estimation uses the class-conditional pair.

$$\beta^A_{c,0} = \alpha_0\, M_c + \gamma_0 \sqrt{\Sigma_c}, \tag{19}$$

with

$$\alpha_0 = -\frac{g}{\bar{x}} \cdot \frac{a}{\rho} < 0, \tag{20}$$

$$\gamma_0 = -\frac{g}{\bar{x}} \cdot \frac{\lambda_0}{\rho + \lambda_0} \cdot \frac{a}{\rho} \cdot \frac{1}{\sqrt{2\pi}} < 0. \tag{21}$$

Both channel parameters are negative under the τ sign convention ($M_c > 0$ when the post-coup tax is higher): a higher mean tax jump lowers investment value (NPV channel), and a higher variance raises the option value of waiting (uncertainty channel). The two carry different discounting, and the difference is substantive, not an artefact. α_0 prices a tax shift the coup has already delivered, so the firm bears it immediately and no discount to a future event applies. γ_0 prices the option to wait for the *next* event, so it inherits the factor $\lambda_0/(\rho+\lambda_0)$ from the premium's own level at the prevailing hazard. Both are evaluated at λ_0. The sign mapping to the empirical V-Dem measure, which carries the opposite convention, is deferred to paragraph (e) below.

(b) Growth from the adoption flow. Output growth differs from TFP growth by the factor-accumulation channel. Under Cobb–Douglas production $Y_t = A_t\, K_t^{s_K}\, L_t^{1-s_K}$ with capital share s_K, log output growth decomposes as

$$\Delta \ln Y_t = \Delta \ln A_t + s_K\, \Delta \ln K_t + (1 - s_K)\, \Delta \ln L_t. \tag{22}$$

Labour supply is unmodelled and $\Delta \ln L_t \approx 0$ at the event-window horizons relevant for the test. Capital evolves through the same option-value investment threshold, so $\Delta \ln K_t \approx (i_F/K)\, \Delta n_t$, with i_F the average investment per project. The chain-rule expansion at impact horizon then gives

$$\beta^Y_{c,0} = \underbrace{g\, f_X(z_0)}_{\text{TFP-channel}}\, [\Delta P_c - \Delta\Omega_c] + \underbrace{s_K\, (i_F/K)\, f_X(z_0)}_{\text{capital-channel}} [\Delta P_c - \Delta\Omega_c]. \tag{23}$$

Both channels load on the same moment $\Delta P_c - \Delta\Omega_c$ because the option-value decision rule (14) drives both the adoption flow and the investment flow. Hence

$$\beta_{c,0}^{\text{growth}} = \kappa_{\text{amp}}\,(\alpha_0\, M_c + \gamma_0\,\sqrt{\Sigma_c}), \tag{24}$$

with

$$\kappa_{\text{amp}} = 1 + \frac{s_K\,(i_F/K)}{g}. \tag{25}$$

The scaling κ_{amp} is the structural amplification from the TFP channel to output growth. It is distinct from the estimated κ_{growth} of Section 7, which additionally absorbs the unit conversion from log TFP to the transformed growth scale. It is identified by the same data variation as (α_0, γ_0) and inherits the same sign as the TFP-channel coefficients.

(c) Cross-outcome proportionality. The two derivations above share a structural feature: the class-conditional response of any outcome y that depends on the option-value adoption rate is proportional to the same combination $\alpha_0\, M_c + \gamma_0\,\sqrt{\Sigma_c}$, with the proportionality constant κ_y determined by the mapping from adoption flow to that outcome's units. Formally, for any outcome whose change at the impact horizon takes the linearised form $\Delta y_{t^*} = \eta_y\,\Delta n_{t^*} + (\text{controls})$, the LP coefficient is

$$\beta_{c,0}^{y} = \eta_y\, f_X(z_0)\,[\Delta P_c - \Delta\Omega_c] = \kappa_y\,(\alpha_0\, M_c + \gamma_0\,\sqrt{\Sigma_c}), \qquad \kappa_y = \eta_y/g. \tag{26}$$

This is the cross-outcome restriction estimated in the joint minimum-distance step of Section 7.

When proportionality fails. Proportionality (26) holds for any outcome whose response is mediated by the same adoption-flow margin. It fails for outcomes that respond through additional channels not captured by Δn: a coalition-redistribution component (government redirects capital expenditure toward favoured sectors post-coup), a foreign-capital flight component (FDI responds to expropriation risk through a distinct channel), or an asset-price component (price effects on existing capital not captured by the flow). In each case the outcome's response decomposes as $\beta^y_{c,0} = \kappa_y(\alpha_0 M_c + \gamma_0\sqrt{\Sigma_c}) + \xi_{y,c}$, with $\xi_{y,c}$ the contribution from the non-adoption channel, the proportionality restriction is rejected when $\xi_{y,c}$ is statistically distinguishable from zero in at least one transition class.

(c.1) Within-block versus across-block proportionality. The proportionality restriction (26) applies most directly to outcomes that are downstream consequences of the aggregate adoption flow Δn. Investment, by contrast, is the firm's direct decision variable in the option-value framework, the same decision that drives Δn. The coalition channel operates through two distinct mechanisms: a direct effect of the expected policy shift M_c on the firm's project value P, which the investment-block loading α_0^{inv} captures, and an indirect effect on aggregate adoption that downstream outcomes inherit. These two mechanisms need not have the same magnitude or sign. In particular, a coup that redirects capital expenditure toward favoured sectors (state-owned enterprises, military procurement, public infrastructure) can produce a positive direct investment response even as it lowers expected returns to private adoption and hence reduces downstream productivity. We therefore distinguish two blocks: productivity outcomes (TFP, growth), which inherit the coalition channel only through downstream adoption, and direct investment, which loads on the coalition channel directly through firm-level policy considerations. Within the productivity block, proportionality (26) holds with shared (α_0, γ_0). Across blocks, separate $(\alpha_0^{\text{inv}}, \gamma_0^{\text{inv}})$ parameters are introduced to allow measured gross investment to contain non-adoption components not represented in the private-firm model. Section 7 reports this two-block specification as the primary structural model and reports the more restrictive one-block specification of (26) in OA Appendix C.

(d) Cumulative implications. The stock relation (16) implies a specific prediction for the cumulative LP coefficient $B^A_{c,H} \equiv \sum_{h=0}^{H} \beta^A_{c,h}$ as the horizon H grows. Suppose the adoption-flow shortfall Δn_c is incurred over the impact interval and the flow then reverts to its baseline with no subsequent catch-up, so that the coup leaves a persistent shortfall of $g\,\Delta n_c$ in the *level* of log productivity instead of a flow that stays depressed. This benchmark treats the level gap as durable. If adoption later runs above baseline, the cumulative coefficient below grows more slowly than the linear benchmark. Then

$$\beta^A_{c,h} = g\,\Delta n_c \quad \text{for all } h \in \{0, 1, \dots, H\}, \tag{27}$$

so the cumulative coefficient grows linearly in H,

$$B^A_{c,H} = (H+1) \cdot g\,\Delta n_c = (H+1) \cdot \beta^A_{c,0}. \tag{28}$$

The amplification ratio $B^A_{c,H}/\beta^A_{c,0} = H + 1$. With $H = 5$ (the empirical horizon), perfect persistence predicts amplification of 6.

For a flow outcome that recovers to baseline after the coup window (adoption rebounds and the level of y returns to its pre-event trend), the LP coefficient $\beta^y_{c,h}$ returns to zero for $h \geq h^*$, where h^* is the recovery horizon. The cumulative is

$$B^y_{c,H} = \beta^y_{c,0}\, r^y_{c,H}, \qquad r^y_{c,H} \equiv \frac{B^y_{c,H}}{\beta^y_{c,0}}, \tag{29}$$

where the recovery factor $r^y_{c,H}$ is the amplification ratio itself: it is close to 1 (only impact contributes) or smaller (rebound at intermediate horizons cancels part of the impact).

The empirical predictions are sharp: for TFP (stock) the amplification at $H = 5$ should approach 6 under perfect persistence, for growth (flow) it should be close to 1 under full recovery, and smaller when an intermediate rebound partly reverses the impact-year contraction. OA Appendix B.3 reports amplification ratios of 5.63 for TFP (DA), 0.66 for growth (DA), and 2.74 for investment (DA), as illustration, not as a test. Under the independence approximation the cumulative TFP loss is distinguishable from zero ($t = -4.17$,

$p < 0.001$) while the cumulative growth response is not ($t = -0.63$, $p = 0.53$), though under the governing wild cluster bootstrap neither cumulative effect is distinguishable from zero (TFP $p = 0.117$, OA Table B3 note). The contrast in point estimates is the predicted stock-vs-flow signature.

(e) Sign convention and the V-Dem mapping. The derivations above are stated with the policy state τ interpreted as a capital tax: τ rises when institutions degrade, so $M_c > 0$ signals an institutionally degrading coup and the channel parameters α_0, γ_0 are both negative. The empirical calibration uses the V-Dem liberal-democracy index as a proxy for institutional quality, with the opposite sign convention: V-Dem rises when institutions improve. Denote by $\tilde{M}_c$ the expected V-Dem jump in transition class c, then $\tilde{M}_c < 0$ in democracy-to-autocracy coups (institutional decline). To first order,

$$\tau_t = \tau_0 - \zeta \cdot \mathrm{VDEM}_t + \mathrm{const}, \qquad \zeta > 0, \tag{30}$$

so $M_c = \mathbb{E}[\Delta\tau_c] = -\zeta\,\tilde{M}_c$ and $\sqrt{\Sigma_c} = \zeta\,\sqrt{\tilde{\Sigma}_c}$. Substituting into the decomposition (19),

$$\beta^A_{c,0} = \alpha_0\,M_c + \gamma_0\,\sqrt{\Sigma_c} = (-\zeta\alpha_0)\,\tilde{M}_c + (\zeta\gamma_0)\,\sqrt{\tilde{\Sigma}_c} \equiv \tilde{\alpha}_0\,\tilde{M}_c + \tilde{\gamma}_0\,\sqrt{\tilde{\Sigma}_c}, \tag{31}$$

with $\tilde{\alpha}_0 = -\zeta\alpha_0 > 0$ and $\tilde{\gamma}_0 = \zeta\gamma_0 < 0$. The empirical estimates in V-Dem units therefore satisfy $\tilde{\alpha}_0 > 0$ and $\tilde{\gamma}_0 < 0$, the opposite signs of the τ-convention statement but the same economic mechanism. The minimum-distance point estimates of Section 7 carry both predicted signs: the primary two-block specification of OA Appendix B.2, the all-event calibration under diagonal inverse-variance weighting (Table 8), delivers $\hat{\tilde{\alpha}}_0 = +0.165$ in the productivity block (positive, the coalition channel, its 90% interval excluding zero but its sign not robust to removing a common coup component) and $\hat{\tilde{\gamma}}_0 = -0.150$ (uncertainty channel, negative and excluding zero at the 90% level in levels, though its sign likewise flips under that contrast). The load-bearing structural result is the investment-block coalition loading $\hat{\hat{\alpha}}_0^{\text{inv}}$, the one parameter whose negative sign survives every weighting scheme and the common-component contrast. The more restrictive one-block specification reported in OA Appendix C.1 pools across investment as well and yields $\hat{\hat{\alpha}}_0 \approx +0.14$ and $\hat{\hat{\gamma}}_0 \approx -0.16$. The two-block specification is adopted as primary on the within-block versus across-block distinction the option-value derivation supports, not because the data reject proportionality, which they do not on the three-class moment set. The opposite-sign reporting in the empirical results section does not contradict the theory. It is the same channel in the opposite-sign measurement of institutional quality.

A.3 The incumbent's seigniorage problem

The incumbent's horizon is not infinite: it holds office only until the next coup, a random time T exponentially distributed with parameter λ_t. The benefit of inflationary finance $B(m)$ is immediate, its cost $C(m)$ is borne each period the incumbent remains in office. The incumbent therefore solves

$$V(m, \lambda_t) = B(m) - \mathbb{E}\left[\int_0^T e^{-\rho s}\, C(m)\, ds\right], \qquad T \sim \text{Exp}(\lambda_t). \tag{32}$$

Taking the expectation over T, the survival probability $e^{-\lambda_t s}$ multiplies the discount factor $e^{-\rho s}$, and the cost term becomes $C(m)/(\rho + \lambda_t)$. The problem reduces to

$$\max_m \; B(m) - \frac{C(m)}{\rho + \lambda_t}, \qquad B' > 0,\ B'' < 0,\ C' > 0,\ C'' > 0, \tag{33}$$

with first-order condition $B'(m^*) = C'(m^*)/(\rho + \lambda_t)$.

Proposition 2 (Inflation and the coup hazard)**.** *Optimal money growth is increasing in the coup hazard, $\partial m^*/\partial \lambda > 0$: a higher hazard shortens the effective horizon, so the incumbent discounts the cost of inflation more heavily and chooses a higher rate.*

Level signature across branches. A failed coup and a self-coup both leave the *same* incumbent in office, with the same monetary preferences. The path of λ_t and hence of m^* is the same in both cases. A normal successful coup installs a new coalition whose monetary preferences need not match the incumbent's. The horizon channel therefore implies

$$\text{response}(F) = \text{response}(SC).$$

A normal successful coup may also shift inflation through the installed coalition's preferences. The model neither signs that shift nor requires it to be non-zero. The equality $F = SC$ is exact in the model. It holds because optimal inflation depends on the incumbent's horizon and not on Σ, the one margin that distinguishes a self-coup from a failed coup. The equality is diagnostic: if inflation responds alike to failed and self-coups but differently to normal coups, the operating channel is the incumbent's horizon. If F and SC diverge, the separation of the seigniorage and capital-tax decisions fails.

A.4 The crossing structure

For any outcome, write the three branch effects as $\beta_F, \beta_{SC}, \beta_N$. Two differences are estimable:

$$\text{gap}_1 \equiv \beta_{SC} - \beta_F, \qquad \text{gap}_2 \equiv \beta_N - \beta_{SC}.$$

A failed coup carries neither channel, and a self-coup adds policy uncertainty ($\Sigma > 0$) with the coalition held fixed ($\Delta_{SC} = 0$), so gap_1 is the effect of uncertainty alone. A normal successful coup adds the coalition change on top, but because the branch primitives impose only $V_N \geq V_{SC}$, it may also carry incremental uncertainty. So gap_2 combines the coalition shift with any $V_N - V_{SC}$, and it isolates the coalition channel only under the stronger restriction $V_N = V_{SC}$, which we do not impose. The full coup effect still decomposes exactly: $\beta_N - \beta_F = \text{gap}_1 + \text{gap}_2$.

Variance-driven vs. coalition-driven outcomes. Suppose the incremental uncertainty a normal coup adds over a self-coup, $V_N - V_{SC}$, is small relative to the uncertainty a self-coup adds over a failed one. Then the outcomes classified here as variance-dominated (foreign investment, total factor productivity, and output growth) have $|\text{gap}_1| > |\text{gap}_2|$: the self-coup behaves like a normal coup, and the decisive move is failed-to-self. Output growth is a positive scalar multiple of the same $\alpha_0 M_c + \gamma_0 \sqrt{\Sigma_c}$ combination as TFP, so it shares TFP's channel dominance up to scale. Outcomes that respond to the level of policy (inflation, government spending) have $|\text{gap}_2| > |\text{gap}_1|$: the self-coup behaves like a failed coup, and the decisive move is self-to-normal. The self-coup switches sides across the two kinds of outcome. That switch is the crossing. Absent a restriction on the relative size of the two uncertainty increments the model orders neither gap magnitude, since it constrains only $V_N \geq V_{SC}$ and not the increments themselves.

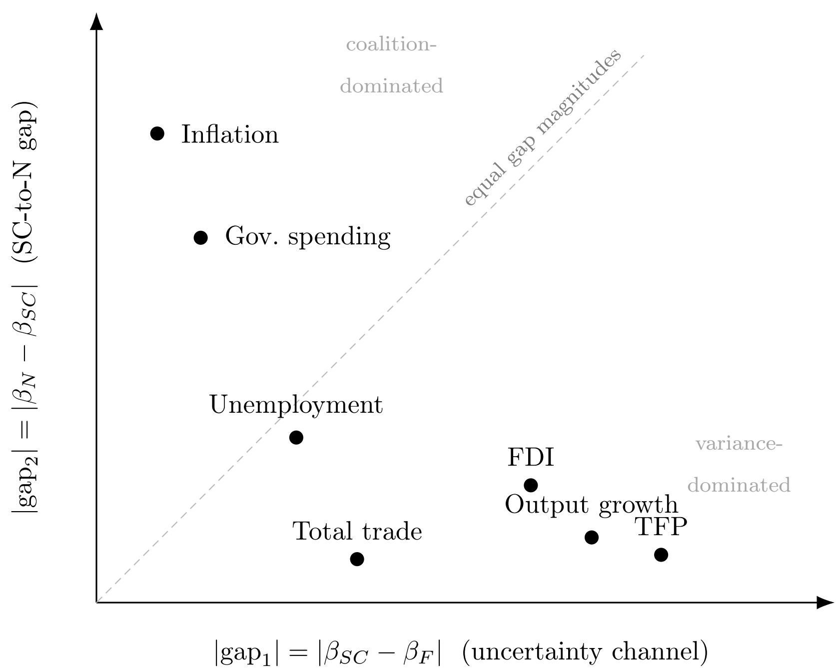
OA Table A2: Schematic channel classification by coup branch

Family	Outcome	β_F	β_{SC}	β_N	Dominant channel
Performance	Output growth	0	$--$	$---$	Uncertainty (gap_1)
	Total factor productivity	0	$--$	$---$	Uncertainty (gap_1)
	Foreign direct investment	0	$-$	$--$	Uncertainty (gap_1)
	Total trade	0	$-$	$-$	Uncertainty (gap_1)
Stability	Inflation	0	0	$\pm$	Coalition (gap_2)
	Government spending	0	0	$\pm$	Coalition (gap_2)
	Unemployment	0	$+$	$++$	Mixed

Qualitative implication of each coup branch: $-/+$ give the direction, 0 no effect, and $\pm$ a coalition-driven response whose sign the model does not pin down. Repeated symbols order magnitudes across branches, and that ordering is conditional on the restriction of §A.4: the model constrains only $V_N \geq V_{SC}$, not the relative size of the two uncertainty increments, so absent that restriction it ranks neither gap magnitude. For the level outcomes (inflation and government spending), all branches share a common response to the pre-coup hazard (§3.4) that is normalized out of the table. A 0 in these rows denotes no *additional* branch effect beyond that common response, and the equal failed and self-coup entries reflect $\beta_F = \beta_{SC}$, not the absence of a hazard response. The total-factor-productivity and output-growth implications follow from the formal option-value decomposition of §A.2.3. The foreign-direct-investment ordering is imposed, not derived, and the total-trade, unemployment, inflation, and government-spending rows are schematic qualitative extensions, in that the model delivers no separate behavioral equation for them and, for inflation and government spending, only the possibility of a coalition-driven move ($\pm$) that it does not sign. *The table is not the paper's claim set, and after the qualifications above it is not an unconditional implication set either.* The empirical design confronts three of these implications, all concerning the response to the *realised* regime change and set out in Section 3.6. It does not adjudicate the branch structure of this table. The failed / self-coup / normal ordering turns on the self-coup cell, which contains four events in the first-event-only sample, and the foreign direct investment row is a prediction the data do not support, which we report as such instead of reinterpreting. The rows are retained so that a reader can see the branch structure whole, with its conditions, before seeing which part we test.

Diagrammatic representation. The crossing can be read off a single diagram: plot each outcome in the plane of the two gap *magnitudes*, $|\text{gap}_1|$ horizontal and $|\text{gap}_2|$ vertical. The diagram ranks channel dominance, not the direction of either effect, whose signs are given in OA Table A2. Variance-dominated outcomes lie near the horizontal axis, coalition-dominated outcomes near the vertical, and mixed outcomes in the interior, on the diagonal when the two plotted gap magnitudes are equal. We classify each outcome into a qualitative dominance region before the data, and the plotted coordinates are schematic.

OA Figure A1: The crossing structure: schematic channel dominance. Axes are gap *magnitudes*. The signs of each effect are in OA Table A2.



In the first-event-only empirical specification we adopt, the self-coup category contains only four events, below the precision range required to test the crossing decisively. The crossing prediction is documented here for completeness. The durable institutional channel is separated from the transitional macroeconomic channel by the stock-versus-flow persistence contrast, and both focal contractions survive the F2-settling robustness of OA Appendix J.1, without requiring the self-coup cell.

B Structural estimation: full details

The main-text synthesis in Section 7 states the three sign-and-persistence predictions and the headline loadings. This appendix reports the full minimum-distance estimation behind them.

B.1 Single-outcome estimation

Throughout this appendix the institutional moments and loadings are in V-Dem units with tildes suppressed, following the convention stated in Section 7: institutional improvement raises M_c, so the predicted productivity-block coalition loading is positive and the predicted uncertainty loading is negative. The investment block's coalition loading may differ in sign.

The structural moment condition for impact-year TFP in each transition class $c \in \{\textsc{da}, \textsc{aa}, \textsc{dd}\}$ is, following the option-value derivation of OA Appendix A.2.3,

$$\beta^A_{c,0} = \alpha \cdot M_c + \gamma \cdot \sqrt{\Sigma_c} + u_c, \qquad \mathbb{E}[u_c] = 0, \tag{34}$$

where $\beta^A_{c,0}$ is the impact-year TFP coefficient on class c from the transition decomposition of Section 6.3, α is the loading on the coalition (NPV) channel, and γ is the loading on the uncertainty (option-value) channel. Three classes and two parameters yield one over-identifying restriction.

The single-outcome system is a nested, preliminary step. It is not a competing estimate, and we do not tabulate its full coefficient vector as one. OA Table B4 reports only calibration-specific diagnostics for $\hat{\gamma}$. The paper reports one set of structural estimates, the two-block system of Table 8. What the single-outcome step establishes is the identification asymmetry that motivates the two-block specification, and that is all we take from it. The option-pricing derivation in OA Appendix A.2.2 implies that the option premium scales as $\sqrt{\Sigma_r}$, not linearly in Σ_r, the standard Black–Scholes vega scaling for jump-diffusion. That derivation conditions on the pre-coup regime r, while the system estimated here conditions on the transition class

c. The premium depends on the coup only through the mean and the variance of the policy change conditional on it, so the same scaling holds on the class-level moments (M_c, Σ_c). We therefore adopt the $\sqrt{\Sigma_c}$ specification for the joint estimation reported in OA Appendix B.2 and for the cumulative test of OA Appendix B.3 on the theoretical motivation, not on an empirical functional-form test: with one over-identifying restriction and the small number of contributing events in the focal class, the $J = 0.30$ ($\sqrt{\Sigma}$) and $J = 0.83$ (linear Σ) statistics under Calibration B have limited power to discriminate between the two specifications, and we do not invoke that comparison as evidence in favor of either.

We calibrate (M_c, Σ_c) in two alternative ways measuring conceptually distinct objects. Calibration A uses the Bjørnskov–Rode regime transition matrix combined with the cross-panel V-Dem liberal democracy index averaged by regime category. It represents the *ex-ante* distribution of post-coup institutional outcomes that a forward-looking firm would have expected. Calibration B uses the within-event realized V-Dem change at $t+1$ relative to $t-1$ averaged within class. It represents the *ex-post* realized institutional shift. Calibration B is the one the estimation uses, and OA Table B1 reports the moments it supplies: a democracy-destroying mean shift of $M_{\text{DA}} = -0.178$ with a within-class dispersion of $\sqrt{\Sigma_{\text{DA}}} = 0.181$, against much smaller mean shifts in the two regime-preserving classes. The dispersion moment is a standard deviation and enters the moment conditions in that form, not as a variance. Computed on the single-outcome sample instead of the generated-regressor one, the same mean is -0.163, and we report both transparently following the parallel-calibration approach of Hansen and Heckman (1996). One tension remains. The BBG transition decomposition is categorical (BR regime label at $t-1$ vs. $t+1$) while Calibration B parameterizes Σ_c from continuous V-Dem magnitudes. Within the focal cell, Kyrgyz Republic 2010 carries a categorical DEM $\rightarrow$ AUT assignment under BR but a mildly positive realized ΔV-Dem (+0.18), and the within-cell regression of event-specific impact-year coefficients on the realized ΔV-Dem returns coefficients statistically indistinguishable from zero in each of the three outcomes the structural system uses: total factor productivity ($p = 0.98$), output growth ($p = 0.69$) and gross investment ($p = 0.83$). Six outcomes carry enough contributing events to run the regression. Of the three the system excludes, government spending is also indistinguishable ($p = 0.96$), while foreign direct investment ($p = 0.01$) and inflation ($p = 0.07$) reject at the level we adopt (OA Table J1, Panel D). We report those two instead of setting them aside. Neither enters the structural estimation, so the continuous-magnitude calibration is informative about the *average* institutional dispersion across DA events without predicting the within-cell heterogeneity that the categorical assignment captures in the estimated system.

OA Table B1: Institutional moments entering the structural estimation

	All-event calibration (headline)			First-event calibration		
Transition class	M_c	$\sqrt{\Sigma_c}$	n	M_c	$\sqrt{\Sigma_c}$	n
DEM → AUT	−0.143	0.130	53	−0.178	0.181	13
INTRA_AUT	−0.012	0.053	296	−0.019	0.063	63
INTRA_DEM	+0.031	0.118	64	+0.034	0.140	17

Notes. The two structural channels are calibrated from the realised change in `v2x_libdem` between $t-1$ and $t+1$ within each transition class. M_c is the class mean (coalition-channel moment). $\sqrt{\Sigma_c}$ is the class standard deviation (uncertainty-channel moment), entering in the square-root form the option-pricing derivation of OA Appendix A.2.2 implies. The headline estimation of Table 8 uses the all-event calibration, which pools first and subsequent coups within each class. The first-event calibration is shown for comparison and used in the robustness exercises. The data do not reject equality of the first- and subsequent-event moments within any class, though these tests are imprecise (OA Appendix C.3), so pooling is used to improve precision rather than treated as established equivalence. The two-step generated-regressor procedure recomputes both moments in every bootstrap replication, so the figures here are the point calibration with sampling error propagated. The reduced-form coefficients feeding the moment vector use the full local-projection cells. This table reports the *inputs*, with the estimates in Table 8. The AUT → DEM class (two events) is excluded from the primary specification and added as a robustness check in OA Appendix C.1.

Under either calibration $\hat{\gamma} < 0$ as predicted by the theory. The two calibration-specific estimates of $\hat{\gamma}$ are reported in the last two rows of OA Table B4: Calibration A (the BR transition matrix) is imprecise, reflecting the construction-imposed $(M, \Sigma) = (0, 0)$ for the intra-democracy class, which removes within-class identifying variation, while Calibration B (V-Dem realized) is the calibration the estimation adopts. The $\sqrt{\Sigma}$ specification we adopt for OA Appendix B.2 is motivated by the Black–Scholes vega scaling derivation in OA Appendix A.2.2, not by the empirical J-statistic comparison, which has limited discriminatory power at this sample size. Under the two-step estimation with the all-event calibration and diagonal weighting, $\hat{\gamma}_0 = -0.150$ (Table 8), with a 90% bootstrap interval of $[-0.28, -0.05]$ that excludes zero. We nonetheless read the uncertainty channel's sign as suggestive, not established, because it does not survive concentrating out an outcome-specific common coup component (OA Appendix C.3): cross-class institutional variation alone does not separately identify it. The coalition-channel parameter $\hat{\alpha}_0$ is positive and sign-identified at the 90% level in this specification, but with TFP moments alone (Calibration B, single outcome) it is not separately identified under either calibration.

The asymmetry (γ estimated with the predicted sign, α not, though neither is separately identified under the primary bootstrap) reflects the limited identifying variation in three transition-class moments for two structural parameters. The next subsection extends the moment set across three outcomes under the two-block specification motivated in OA Appendix A.2.3.

B.2 Two-block joint estimation

The option-value framework distinguishes the theory's private-investment decision from its downstream consequences (TFP, growth). Empirically, that block is estimated using the aggregate gross-investment series, an imperfect proxy that also contains government and residential investment. In the theory, investment is the contemporaneous decision variable: the firm chooses whether to sink the capital cost f in response to the option-value calculation $P(\tau, \lambda) - \Omega(\tau, \lambda)$, and the investment-block coalition loading α_0^{inv} enters investment through the direct effect of the expected policy shift M_c on the firm's project value P. TFP and growth are realised outcomes of the adoption flow Δn_t that prior firm decisions generate, the coalition channel reaches them only indirectly, through the aggregate adoption that downstream outcomes inherit. OA Appendix A.2.3 articulates the distinction in detail: a coup that redirects capital expenditure toward favoured sectors (state-owned enterprises, military procurement, public infrastructure) can produce a positive direct investment response even as it lowers expected returns to private adoption and hence reduces downstream productivity. The two-channel structure of the option-value derivation does not require the investment response and the productivity response to share the same α_0, the within-block proportionality between TFP and growth is the theoretically warranted restriction.

We therefore estimate a two-block specification. The productivity block, TFP and growth, shares the channel parameters (α_0, γ_0) with κ_{growth} jointly absorbing (i) the unit-conversion factor from raw lnTFP to the transformed growth scale ($\approx 100\times$) and (ii) the structural amplification $\kappa_{\text{amp}} = 1 + s_K\,(i_F/K)/g$ derived in OA Appendix A, Eq. (25), the two are not separately identified at the current scaling:

$$\beta_{c,0}^{\text{TFP}} = \alpha_0 M_c + \gamma_0 \sqrt{\Sigma_c} + u_c^{\text{TFP}}, \qquad \beta_{c,0}^{\text{growth}} = \kappa_{\text{growth}}(\alpha_0 M_c + \gamma_0 \sqrt{\Sigma_c}) + u_c^{\text{growth}}. \tag{35}$$

The investment block carries its own channel parameters:

$$\beta_{c,0}^{\text{invest}} = \alpha_0^{\text{inv}} M_c + \gamma_0^{\text{inv}} \sqrt{\Sigma_c} + u_c^{\text{invest}}. \tag{36}$$

The full parameter vector is $(\alpha_0, \gamma_0, \kappa_{\text{growth}}, \alpha_0^{\text{inv}}, \gamma_0^{\text{inv}})$, five parameters estimated from the 9×1 moment vector $\boldsymbol{\beta} = (\beta_c^{\text{TFP}}, \beta_c^{\text{growth}}, \beta_c^{\text{invest}})_{c \in \{\text{DA,AA,DD}\}}$. The system has $9 - 5 = 4$ over-identifying restrictions. The primary point estimates are computed by minimum distance under diagonal inverse-variance weighting on the all-event calibration (Table 8). The two inference layers are described in the next paragraph. The secondary fixed-moment layer reweights the moment conditions by $\hat{V}_{\boldsymbol{\beta}}^{-1}$, the inverse of the country-cluster bootstrap variance of the moment vector (500 resamples). For that efficiently-weighted estimator the correct asymptotic covariance is the standard expression $\widehat{\mathrm{Var}}(\hat{\theta}) = (\hat{G}'\hat{V}_{\boldsymbol{\beta}}^{-1}\hat{G})^{-1}$, where $\hat{G}$ is the 9×5 Jacobian of the moment vector evaluated at $\hat{\theta}$. This secondary layer is therefore distinct from the diagonal-weighted primary points and is not used for the channel-coefficient claims.

The outcome set (TFP, growth, gross investment) follows the theoretical derivation: each is mediated by the firm-level option-value adoption decision, with the units mapping handled by κ_{growth} in the productivity block and the investment-specific loading captured by the separate $(\alpha_0^{\mathrm{inv}}, \gamma_0^{\mathrm{inv}})$. Among the outcomes excluded from the structural system are FDI, inflation, government spending, unemployment, and total trade (OA Appendix G), together with net migration (Table 5), either because they are theoretically mediated by distinct channels (inflation through the incumbent's seigniorage problem of OA Appendix A.3, government spending through fiscal-channel responses outside the firm's investment decision) or because they lack a robust focal impact-year response suitable for the structural system. OA Appendix G reports the two exceptions to that reduced-form pattern: the intra-autocracy total-trade contraction and the single-country democracy-to-autocracy unemployment coefficient.

Table 8 reports the estimates. We report two inference layers throughout. The *primary* layer is a country-cluster bootstrap (800 resamples) that jointly resamples both the impact-year coefficients $\hat{\beta}_c^y$ and the calibrated moments $(M_c, \sqrt{\Sigma_c})$, propagating moment uncertainty into the structural standard errors. The *secondary* layer treats the calibrated moments as fixed and uses the asymptotic Jacobian-sandwich standard error. Because this layer ignores the moment uncertainty the joint two-step bootstrap is designed to incorporate, we do not report it in Table 8 and do not use it as the basis for any significance claim in the abstract, the introduction, or the conclusion.

Under the primary (bootstrap) layer, the investment-block coalition parameter $\hat{\alpha}_0^{\mathrm{inv}}$ is negative across its 90% bootstrap confidence interval ($[-77.20, -0.73]$), the inference standard we adopt throughout. At the 95% level the interval ($[-89.37, +3.69]$) includes zero (Appendix OA Table B2). The estimate is imprecise and only weakly identified, its bootstrap distribution heavy-tailed when the three-class moments are near-collinear, so the interval is sensitive to the resampling implementation. With that caveat, the negative sign at the 90% level is the structural finding the section leans on: it is what rationalises the positive cumulative investment response of OA Appendix B.3 without recourse to the uncertainty-channel parameter, and it is the directional pattern that distinguishes the option-value interpretation from a generic "democracy-destroying coups are worse" story, offered as a quantitative interpretation of the reduced-form pattern, not as a point-identified magnitude. The productivity-block coalition parameter $\hat{\alpha}_0$ is positive and its 90% bootstrap interval excludes zero under the preferred scale-adjusted weighting, but it crosses zero under identity weighting and under the common-component contrast, so we report its sign as directionally consistent rather than robustly identified. The uncertainty-channel coefficient $\hat{\gamma}_0$ is negative and its 90% interval, and its 95% interval, exclude zero in the level specification, but that sign does not survive concentrating out the common coup component (OA Appendix C.3), so cross-class institutional variation alone does not separately identify it. The investment-block uncertainty coefficient $\hat{\gamma}_0^{\text{inv}}$ behaves the same way: negative and level-identified, but not robust to the contrast or to the V-Dem subindex calibrations (OA Appendix C.4). We report these but lean only on the investment-block coalition loading, the one parameter whose sign is robust across weighting schemes and the common-component contrast. The over-identifying restrictions are not rejected: under optimal weighting the productivity block returns $\chi^2(3) = 3.39$ ($p = 0.33$) and the investment block $\chi^2(1) = 0.10$ ($p = 0.75$), though over-identification has limited power at this sample size.

OA Table B2: Inference robustness: 90% vs 95% intervals, and pairs vs wild cluster bootstrap

Panel A. Structural channel parameters: bootstrap confidence intervals

Parameter	point	90% CI	95% CI	identified?
α_0 (coalition, TFP)	+0.165	[+0.012, +0.336]	[−0.021, +0.381]	90% only
γ_0 (uncertainty, TFP)	−0.150	[−0.281, −0.047]	[−0.311, −0.022]	both
κ_{growth} (scale)	100.43	[+52.9, +186.7]	[+43.8, +223.7]	—
$\boldsymbol{\alpha_0^{\text{inv}}}$ (coalition, INV)	**−33.97**	**[−77.20, −0.73]**	[−89.37, +3.69]	**90% only**
γ_0^{inv} (uncertainty, INV)	−16.54	[−33.13, −4.35]	[−36.66, −2.67]	both

Panel B. Cumulative $B_{c,5}$ (TFP): pairs vs wild cluster bootstrap

Class	$B_{c,5}$	pairs 90% CI	pairs 95% CI	wild p	verdict
DA	−0.238	[−0.430, −0.008]	[−0.497, +0.038]	0.117	not significant
AGG	−0.101	[−0.233, +0.016]	[−0.258, +0.042]	0.214	not significant
DD (placebo)	+0.019	[−0.112, +0.154]	[−0.139, +0.198]	0.804	null (clean)

Notes. Master data: Penn World Table 11.0 TFP and the Powell–Thyne-reconciled coup panel (v2). Panel A reports the two-step generated-regressor minimum-distance estimates of Table 8 (all-event calibration, diagonal inverse-variance weighting, 800 country-cluster bootstrap resamples jointly resampling the impact-year LP coefficients and the calibrated V-Dem moments). At the 90% level the coalition and uncertainty loadings on both blocks exclude zero; at the 95% level only the uncertainty loadings do. The investment coalition loading α_0^{inv} is the one robust across weighting schemes and the removal of a common coup component (OA Appendix C.3). The paper adopts the 90% interval as its inference standard. Panel B compares the country pairs bootstrap with the wild cluster bootstrap for the cumulative TFP statistic. With only five contributing DEM $\to$ AUT episodes the pairs bootstrap over-rejects, while the wild bootstrap, which does not degenerate with few clusters (MacKinnon and Webb, 2018), gives $p = 0.117$ for the DA cumulative effect. The cumulative five-year response is not distinguishable from zero. The impact-year contrast returns $p = 0.074$ under the wild cluster bootstrap.

Notes on the cumulative panel. The interval and the p-value in each row come from *different bootstraps* and are not two readings of one object. The interval is an unrestricted, equal-tailed percentile interval from a pairs cluster bootstrap on the countries. The p-value is the wild cluster bootstrap with the null imposed (restricted, Webb six-point weights), which is the test this paper adopts. With five countries carrying the DA coefficient the two routinely diverge, and MacKinnon and Webb (2018) show that the restricted wild test is the reliable one in that regime. This is why the DA row can show an interval that excludes zero beside a p of 0.117 that does not reject: the p governs, and the verdict column follows it. The divergence is a property of few clusters, not an inconsistency.

The investment-block loading we report as suggestive rather than as identification of the channel in private firm investment: the Penn World Table gross-investment aggregate combines private business, government, and residential investment in a single series (and 6 of the 14 democracy-destroying countries have missing investment data at the event year or its lag, so the investment block is estimated on 8 of 14 events), the channel loading we recover therefore reflects the aggregate, not the firm-level decision variable the theory models. The coalition channel shows the predicted sign heterogeneity in the point estimates at the headline calibration (the productivity-block loading is not separately identified and reverses sign under the rule-of-law calibration of OA Appendix C.4): the productivity block loads positively (the firm's average expected policy shift enters the project value P positively in V-Dem units, where positive M_c corresponds to a shift toward greater liberal democracy), while the investment block loads negatively (consistent with the coalition-redistribution mechanism discussed in OA Appendix A.2.3 in which democracy-destroying coups raise investment in favoured sectors even as they lower private adoption returns), with the same aggregation caveat attached to the investment-block parameter as above.

We initially imposed full cross-outcome proportionality on all three outcomes, the more restrictive specification reported in OA Appendix C. The full-proportionality difference-in-J (ΔJ) test does not reject on the three-class moment set ($\chi^2(2) = 3.94$, $p = 0.139$), and rejects only marginally once the autocracy-to-democracy class is added ($\chi^2(2) = 5.53$, $p = 0.063$). The data therefore do not settle the specification choice. The theoretical reason to expect a departure from full proportionality is the within-block versus across-block distinction articulated in OA Appendix A.2.3: investment is the firm's direct decision and loads on the coalition channel directly, while TFP and growth inherit the channel only through the downstream adoption flow. The two-block specification adopted here relaxes that restriction and is motivated by the distinction the option-value derivation supports, not by a test that rejects the restriction. The estimates are reported in Table 8 of Section 7.

B.3 Implications and persistence

The stock-versus-flow prediction is illustrated by the cumulative response. Productivity is a stock that integrates a disrupted adoption flow, so its cumulative loss should approach the perfect-persistence value. Growth is the flow itself, so its cumulative response should be close to one impact-year effect or smaller once an intermediate rebound is allowed.

OA Table B3: Cumulative effect $B_{c,H} = \sum_{h=0}^{H} \beta_{c,h}$ and amplification ratio

Outcome	Class	$\beta_{c,0}$ impact	$B_{c,5}$ ($\mathrm{SE}_{\mathrm{ind}}$) cumulative	Wild p ($B_{c,5} = 0$)	Amplification $B_{c,5}/\beta_{c,0}$	Stock-flow direction
Panel A: Stock prediction (TFP) — amplification ratio → 6 under perfect persistence						
TFP	DA	-0.042	-0.238 (0.057)	0.117	5.63	✓
TFP	AA	-0.016	-0.086 (0.067)	0.505	5.25	
TFP	DD	-0.011	0.019 (0.040)	0.804	-1.74	
Panel B: Flow prediction (growth) — amplification ratio → 1 or smaller under recovery/rebound						
Growth	DA	-3.689	-2.428 (3.877)	0.296	0.66	✓
Growth	AA	-2.522	-3.656 (2.537)	0.229	1.45	
Growth	DD	-0.579	0.570 (4.069)	0.845	-0.98	
Panel C: Investment — directional pattern of the two-block specification						
Investment	DA	3.265	8.959 (4.873)	0.359	2.74	✓
Investment	AA	-0.510	2.102 (1.815)	0.554	-4.12	
Investment	DD	-2.855	-5.916 (3.169)	0.440	2.07	

Notes. $\beta_{c,h}$ are the impulse-response coefficients from the BBG transition decomposition (Section 6.3) at $h = 0, \ldots, 5$, and $B_{c,5} = \sum_{h=0}^{5} \beta_{c,h}$. The independence-approximation standard error $\mathrm{SE}_{\mathrm{ind}}(B_5)$ is reported for reference only and understates uncertainty with few clusters. The governing inference is the wild cluster bootstrap (Webb 6-point weights, restricted, 999 resamples, clustering on country, MacKinnon and Webb, 2018), reported in the "Wild p" column. Under it no cumulative five-year effect is distinguishable from zero in any cell (all $p > 0.10$). The amplification ratio $B_{c,5}/\beta_{c,0}$ is descriptive, approaching $H + 1 = 6$ under perfect TFP persistence and 1 or smaller under growth recovery with intermediate rebound. The impact-year contrast returns $p = 0.074$ under the wild cluster bootstrap.

The point estimates trace the predicted shape. The productivity amplification ratio is

5.63 against a perfect-persistence value of six, while growth amplifies at 0.66 and investment at 2.74. We report this as illustration, not as a test. A level that persists more than its own growth rate is close to definitional, and any model with a persistent productivity shock delivers it, so the amplification pattern discriminates weakly at best. Under the wild cluster bootstrap none of the cumulative five-year responses is separately distinguishable from zero in any case.

The two-block structural model produces one further empirical implication we can examine descriptively: the stock-versus-flow cumulative signature distinguishing TFP from growth. The productivity-block over-identification reported in OA Appendix B.2 is not rejected ($\chi^2(3) = 3.39$, $p = 0.33$), consistent with TFP and growth sharing a single productivity-block channel, the cumulative test reported here traces the channel's persistence signature at the horizon dimension.

Stock-versus-flow cumulative signature. OA Appendix A.2.3 derives the prediction that the cumulative LP coefficient $B^A_{c,H} = \sum_{h=0}^{H} \beta^A_{c,h}$ grows linearly in H for a stock outcome (TFP) under a persistent productivity-level shortfall generated by a temporary disruption to the adoption flow, and is close to one impact-year effect for a flow outcome (growth) that recovers quickly. It may be smaller when an intermediate rebound offsets the impact, or larger when recovery takes more than one period. The amplification ratio $B_{c,5}/\beta_{c,0}$ should approach $H + 1 = 6$ under perfect TFP persistence and approach 1 (or below, if rebound is present) under full growth recovery.

OA Table B3 reports the cumulative estimates and amplification ratios at the DA class. The point estimates are directionally consistent with the stock-versus-flow distinction, but no cumulative five-year effect is statistically distinguishable from zero under the table's governing wild cluster bootstrap (the inference layer appropriate to the five-episode setting), so we read the amplification ratios as directional, not significance, statements. The TFP cumulative loss is $\hat{B}^A_{\text{DA},5} = -0.238$, with amplification ratio 5.63 near the perfect-persistence value of 6 (wild $p = 0.117$, the independence approximation's $p < 0.001$ badly understates uncertainty with five contributing clusters). The growth cumulative response is $\hat{B}^{Y}_{\text{DA},5} = -2.43$ (amplification 0.66, wild $p = 0.30$), with a +2.1-point rebound at $h = 1$ partly reversing the −3.7-point impact on the transformed scale, the flow-recovery pattern the model predicts. The investment cumulative response is $\hat{B}^{I}_{\text{DA},5} = +8.96$ (amplification 2.74, wild $p = 0.359$, independence $p = 0.064$ reported for comparison). The point pattern thus matches the stock-versus-flow distinction (TFP stock-like, growth flow-like, investment positive in the DA cell) while the weight does not rest on the cumulative magnitudes. Under the two-block specification of OA Appendix B.2, the positive cumulative is rationalised by the investment-block coalition parameter $\hat{\alpha}_0^{\text{inv}}$ acting on the negative class average $M_{\text{DA}} = -0.178$: the negative coalition parameter, multiplied by the negative class-average institutional shift, delivers the positive predicted investment response in democracy-destroying coups. As reported in OA Appendix B.2, $\hat{\alpha}_0^{\text{inv}}$ is negative across its 90% bootstrap interval but not its 95%, imprecisely estimated and weakly identified under the headline calibration, with the negative *sign* stable across all four V-Dem subindex calibrations of OA Appendix C.4 (point estimates −20 to −51). The investment-block uncertainty-channel coefficient $\hat{\gamma}_0^{\text{inv}}$ carries the predicted negative sign, with 90% and 95% bootstrap intervals that exclude zero under the headline calibration (Table 8), but its sign is unstable across the V-Dem subindex calibrations of OA Appendix C.4, turning positive under private-property rights. We do not lean on it. The positive impact-year fitted response is driven by the coalition component and partly offset by the uncertainty component. The positive cumulative response is directionally rationalized by that impact-year decomposition but is not itself fitted by the structural system.

(a) TFP (stock) (b) Output growth (flow) (c) Gross investment

OA Figure B1: Impulse-response trajectory at the DEM $\rightarrow$ AUT cell

Local-projection coefficient $\beta_{\mathrm{DA},h}$ at horizons $h \in \{0, 1, 2, 3, 4, 5\}$ for TFP, output growth, and gross investment, with 95% country-clustered asymptotic confidence bands. The panel superscripts denote the outcome: A total factor productivity (as in Equation (1)), G output growth, and I gross investment. The focal cell contains only 14 democracy-to-autocracy events, the regime in which asymptotic cluster-robust inference has known small-sample bias (MacKinnon and Webb, 2018). Wild cluster bootstrap p-values with Webb weights are reported for the full grid of 54 outcome-by-class-by-horizon cells in OA Appendix I, OA Table I1. The headline descriptive summary of the stock-versus-flow pattern is the cumulative coefficient $B_{c,5} = \sum_{h=0}^{5} \beta_{c,h}$ reported in OA Table B3, which exploits all six horizons jointly and yields a TFP cumulative loss of -0.238 that is not distinguishable from zero under the governing wild cluster bootstrap ($p = 0.117$), the independence approximation's $p < 0.001$ understating uncertainty with few clusters, a growth cumulative not distinguishable from zero, and an investment cumulative of $+8.96$ (wild $p = 0.359$, independence $p = 0.064$). The qualitative pattern in the figure (TFP negative throughout, with a deeper estimate at $h = 2$ before partial recovery, growth $h = 1$ rebound that offsets over half of the impact, investment positive at five of the six horizons, with a small negative estimate at $h = +1$) matches the stock and flow predictions of the two-block specification of OA Appendix B.2 and the investment-block directional pattern the specification permits.

The cumulative investment estimate of $+8.96$ is directionally consistent with the separately estimated loading of the coalition parameter $\hat{\alpha}_0^{\mathrm{inv}}$ in the two-block specification, which is fit on the impact-year moments. The cumulative response is not itself fitted by the structural system. The magnitude carries three caveats. The first is measurement: the gross-investment aggregate combines private business, government, and residential investment in a single series, and the data do not permit disaggregation. The $+8.96$ magnitude plausibly reflects government investment surges (military procurement, state-led industrialization, public infrastructure) instead of private business investment, but the aggregate series cannot confirm this composition and we do not test it directly. The second concerns sample composition: of the 14 first-event democracy-destroying coup countries, 8 contribute to the investment cumulative regression (Brazil, Chile, Congo Republic, Greece, Honduras, Kyrgyz Republic, Myanmar, Uruguay), the remaining 6 (Argentina, Grenada, Panama, Seychelles, Sierra Leone, Suriname) have missing gross-investment coverage at the event year or its lag, so the +8.96 estimate is conditional on the contributing subsample. The third concerns pre-trends: the joint pre-event nullity test reported in Section 6.6 addresses pre-trends for TFP but has not been replicated for investment, if countries selected into democracy-destroying coups exhibit positive pre-event investment momentum, part of the cumulative could reflect that momentum rather than a coup-induced response. A direct pre-trend test for investment is left for future work.

The three caveats qualify the magnitude interpretation of +8.96 and of $\hat{\alpha}_0^{\text{inv}}$ specifically. They do not affect the structural decomposition itself. The two-block model includes and estimates a separate coalition parameter for investment. Its fitted sign and magnitude are properties of the current specification and do not establish structural identification. The mechanism the structural model encodes (OA Appendix A.2.3) is the direct contemporaneous loading of post-coup coalition redistribution on the investment flow. Refining the magnitude interpretation requires disaggregated private-versus-government investment data and an explicit pre-trend test, both beyond the present data.

B.4 Robustness

The inference reported in this section operates at three layers, each with its own appropriate standard-error treatment. We make the distinction explicit because treatment variation in the focal cell is concentrated in 14 treated countries, only five of which contribute to the impact-year TFP coefficient, making asymptotic cluster-robust inference unreliable (MacKinnon and Webb, 2018) despite the larger number of regression clusters, and we want to be transparent about which test depends on which inference layer.

Layer 1, individual-horizon LP coefficients $\hat{\beta}_{y,c,h}$. These are the building blocks of the decomposition. Asymptotic cluster-robust p-values for these coefficients are unreliable with few treated clusters, so we report wild cluster bootstrap p-values with Webb six-point weights (Webb, 2014) and 2,000 bootstrap replications alongside the asymptotic counterparts in OA Appendix I, OA Table I1. The headline impact-year coefficient on TFP in the DEM $\rightarrow$ AUT cell remains significant at the 5% level under bootstrap correction (bootstrap $p = 0.020$ vs. asymptotic $p < 0.001$), the investment coefficient at $h = 5$ remains significant at the 1% level (bootstrap $p = 0.004$).

Layer 2, cumulative coefficients $B_{c,H}$. These exploit the joint information across horizons by summing the $\hat{\beta}_{y,c,h}$. The analytical standard error of $B_{c,H}$ under independence is the square root of summed variances, a well-behaved object that is not subject to the small-cluster denominator-noise problem of ratio statistics. The cumulative test uses all six horizons jointly but is interpreted illustratively, consistent with the main text, and is reported in OA Appendix B.3, OA Table B3.

Layer 3, joint structural parameters $(\alpha_0, \gamma_0, \kappa_{growth}, \alpha_0^{inv}, \gamma_0^{inv})$. These are recovered from the two-block minimum-distance estimation of OA Appendix B.2. Inference is the country-cluster two-step bootstrap (800 resamples) that jointly resamples the impact-year coefficients $\hat{\beta}_c^y$ and the calibrated moments $(M_c, \sqrt{\Sigma_c})$, reported as percentile 90% and 95% confidence intervals in Table 8. This is the fully-propagated layer that does not treat the calibrated moments as fixed. The asymptotic Jacobian sandwich formula $(\hat{G}'\hat{V}_{\boldsymbol{\beta}}^{-1}\hat{G})^{-1}$, the covariance of the $\hat{V}_{\boldsymbol{\beta}}^{-1}$-weighted fixed-moment estimator that conditions on the calibrated moments, is the secondary layer, which we do not use for inference. This layer does not depend on the individual-horizon asymptotic SE problem of Layer 1 because the moment vector is the impact-year coefficient and the bootstrap is constructed from country-resampled regressions.

The layers agree. The cumulative test of Layer 2 traces the stock-versus-flow distinction, which we read as illustration, the joint estimation of Layer 3 recovers the point-estimate signs consistent with the directional interpretation, and the bootstrap of Layer 1 shows that the impact-year coefficient and the $h = +5$ coefficient, though not the intermediate horizons, survive small-cluster correction. OA Appendix B.2 is estimated from the nine impact-year coefficients. OA Appendix B.3 separately summarizes the six reported horizons through cumulative coefficients.

Three further robustness exercises bear on the estimates of this section. The first is a country leave-one-out over the 14 democracy-to-autocracy events in the first-event-only sample, of which 8 contribute non-trivially to the investment regression (the remaining 6 have missing investment data at the event year or its lag. The eight contributors are named in OA Appendix B.3). Dropping each contributing country one at a time and re-estimating $\hat{\alpha}_0^{\text{inv}}$ stays negative under every single-country drop, with Greece producing the smallest magnitude and no single country driving the result (the range is in OA Table B4). The second is the parallel calibration: the qualitative pattern (γ carrying the predicted negative point estimate in both blocks without being separately identified under fully propagated inference, α in the productivity block carrying a positive point estimate with an interval that crosses zero, cumulative TFP loss directionally consistent with the stock relation but not significant under the wild cluster bootstrap) is robust across both Calibration A (BR transition matrix, ex-ante) and Calibration B (V-Dem realized, ex-post), with magnitudes of $\hat{\gamma}$ differing by approximately a factor of four across the two as expected from the different scales of the calibrated Σ moments. The third is the persistence statistic itself: OA Appendix B.3 uses the cumulative $B_{c,5}$ statistic to illustrate the stock-versus-flow distinction, but an alternative ratio statistic $R_{y,c} = \beta_{y,c,h=5}/\beta_{y,c,h=0}$ has a noisy denominator and a bootstrap distribution with wide variance (standard deviations reported in OA Table B4, depending on how close $\beta_{y,c,0}$ is to zero). The cumulative statistic, summing variances instead of dividing by a noisy denominator, is the cleaner descriptive summary, the qualitative pattern in R is consistent with the cumulative summary, but the ratio diagnostic is too noisy to be informative.

Three further robustness exercises are reported in OA Appendix C: the more restrictive one-block specification (OA Appendix C.1), whose cross-outcome proportionality restriction the data do not reject on the three-class moment set and reject only at the 10% level when the AUT $\rightarrow$ DEM class is added, the per-outcome unrestricted specification (OA Appendix C.2), which is consistent with the within-block versus across-block heterogeneity by locating the coalition-channel sign-flip in the investment outcome, and the V-Dem subindex robustness exercise (OA Appendix C.4), which finds $\hat{\gamma}_0 < 0$ at the point estimate in all four V-Dem calibrations of the institutional dispersion (sign-stable, though the 95% bootstrap interval crosses zero in each) and finds $\hat{\gamma}_0^{\text{inv}}$ negative at the point in three of four calibrations, flipping positive under the property-rights subindex. The one-block specification does not reject the proportionality restriction on the three-class moment set ($\chi^2(2) = 3.94$, $p = 0.139$), and rejects it only at the 10% level once the autocracy-to-democracy class is added ($\chi^2(2) = 5.53$, $p = 0.063$), so the two-block specification is adopted as primary on theoretical, not test-based, grounds. The per-outcome unrestricted point estimates place the apparent departure from proportionality in the investment block: the uncertainty channel $\hat{\gamma}_y < 0$ is uniformly negative across the three outcomes, but the coalition channel $\hat{\alpha}_y$ is positive for TFP and growth and takes the opposite sign for investment, consistent with the within-block versus across-block distinction articulated in OA Appendix A.2.3.

A final robustness diagnostic concerns the choice of regime taxonomy. Alternative regime taxonomies based on the Geddes–Wright–Frantz dataset (Geddes et al., 2014) produce uncertainty-channel estimates with the same sign as the BBG-based specification reported above, but the overidentification tests are sensitive to GWF coverage gaps among small-state and civil-war-period autocracies (Yemen, Equatorial Guinea, Laos, Suriname, Comoros, Lebanon, Azerbaijan, Madagascar, Fiji during specific years are coded as "autocracy-unknown" in GWF v1.0 and remain unmapped in the extended GWF release). On the 13-class disaggregation the uncertainty channel carries the same negative sign as in the primary specification (OA Table B4, from a cross-class scaling regression without a constant), so the sign result is not an artefact of the four-class grouping. We treat the BBG-aligned transition decomposition as the primary structural specification and use GWF only as a diagnostic comparison.

OA Table B4: Structural-layer robustness diagnostics

Diagnostic	Value
$\hat{\alpha}_0^{\text{inv}}$, investment-block-only minimum distance (MD), full sample	-39.1
leave-one-country-out range (14 DA countries)	$[-46.8, -25.5]$
Persistence ratio $R_{y,c} = \beta_{c,5}/\beta_{c,0}$: bootstrap SD across 9 cells	0.3–48
GWF 13-class uncertainty loading $\hat{\gamma}$ (OLS, no constant)	-17.3 ($t = -2.22$)
Uncertainty loading $\hat{\gamma}$, Calibration A (BR matrix, linear Σ)	-0.51 ($p = 0.25$)
Uncertainty loading $\hat{\gamma}$, Calibration B (V-Dem realized, optimal weighting)	-0.13

Notes. Robustness checks on the structural layer. The leave-one-out check re-estimates the investment block alone by minimum distance (closed-form three-moment fit over the DA, AA, DD investment coefficients), whose full-sample loading -39.1 differs from the primary two-block estimate of Table 8 (-33.97) but bears on sign, not magnitude. The loading $\hat{\alpha}_0^{\text{inv}}$ stays negative when any single DEM $\rightarrow$ AUT country is dropped. The persistence ratio has wide bootstrap variance (standard deviations 0.3 to 48 across cells, depending on how close $\beta_{c,0}$ is to zero), which is why OA Appendix B.3 uses the cumulative statistic instead. The Geddes–Wright–Frantz 13-class taxonomy (Geddes et al., 2014) returns an uncertainty-channel loading of the same negative sign as the primary specification. The last two rows report $\hat{\gamma}$ under the calibrations of OA Appendix B.1: Calibration A (Bjørnskov–Rode transition matrix, linear-Σ) is imprecise because the intra-democracy class carries a construction-imposed $(M, \Sigma) = (0, 0)$, and Calibration B (V-Dem realized) is the adopted one. Both keep the negative uncertainty-channel sign.

C Restrictive specifications and robustness of the structural specification

The primary structural estimation in Section 7 uses the two-block specification: TFP and growth share a common (α_0, γ_0) with κ_{growth} absorbing the units mapping, and investment carries its own $(\alpha_0^{\text{inv}}, \gamma_0^{\text{inv}})$. The theoretical motivation is articulated in OA Appendix A.2.3: investment is the firm's direct decision variable and loads on the coalition channel through firm-level policy considerations, while TFP and growth inherit the channel only through the downstream adoption-flow. This appendix reports two alternative specifications (one strictly more restrictive than the two-block model, one strictly more flexible) and explains why neither is adopted as the primary specification.

C.1 One-block proportionality

The one-block specification imposes a single (α_0, γ_0) pair across all three outcomes, with outcome-specific scalings κ_y:

$$\beta_{c,0}^{y} = \kappa_y \cdot (\alpha_0 M_c + \gamma_0 \sqrt{\Sigma_c}), \qquad y \in \{\text{TFP}, \text{growth}, \text{investment}\}, \quad \kappa_{\text{TFP}} = 1.$$

The system has $9 - 4 = 5$ over-identifying restrictions. The joint MD estimates under the fixed-moment Jacobian-sandwich layer, with the country-cluster bootstrap entering only as the weighting matrix, are reported in OA Table C1: the uncertainty channel $\hat{\gamma}_0$ is negative and distinguishable from zero on this secondary layer, while the common coalition-channel loading $\hat{\alpha}_0$ is not. These fixed-moment p-values are not the primary inference layer for the channel coefficients.

OA Table C1: Fixed-moment inference layer, one-block proportionality specification

Parameter		Estimate	SE	z	p
α_0	(coalition channel)	0.136	0.087	1.57	0.115
γ_0	(uncertainty channel)	-0.159	0.060	-2.66	0.008
κ_{growth}	(growth scale)	96.13	25.8	3.72	< 0.001
κ_{invest}	(investment scale)	51.43	27.7	1.86	0.063

Notes. The fixed-moment inference layer for the *one-block* proportionality specification of OA Appendix C.1, whose four parameters are a single (α_0, γ_0) pair shared across all three outcomes and the two outcome scalings $\kappa_{growth}, \kappa_{invest}$. This is not the paper's primary specification. The primary estimation is the five-parameter two-block model of Table 8, which gives investment its own loadings instead of a shared (α_0, γ_0) with a scale. The layer treats the calibrated V-Dem moments $(M_c, \sqrt{\Sigma_c})$ as fixed and computes asymptotic Jacobian-sandwich standard errors. The country-cluster bootstrap enters only as the weighting matrix. On this fixed-moment layer the uncertainty channel $\hat{\gamma}_0$ is negative and distinguishable from zero, while the coalition channel $\hat{\alpha}_0$ is not.

The over-identification statistic is $J(5) = 7.67$ ($p = 0.175$), which does not reject. The cross-outcome proportionality difference-in-J (ΔJ) statistic against the unrestricted per-outcome model (OA Appendix C.2) is $\chi^2(2) = 3.94$ ($p = 0.139$), which also does not reject. On the three-class moment set, therefore, the data do not refute one-block proportionality.[14]

Adding the AUT $\to$ DEM class weakens the restriction but does not overturn it decisively. The four-class extension (two AD events in the LP regression sample, PRT 1974 and PRY 1989, with V-Dem realized $M_{AD} = +0.014$ and $\sqrt{\Sigma_{AD}} = 0.098$ computed over the four first-event AUT $\to$ DEM coups in the 1950–2025 V-Dem calibration sample (the Bjørnskov–Rode coup panel merged with V-Dem v14 realized LIBDEM, over which the class moments are calibrated), a wider universe than the 1960–2022 local-projection sample. Only PRT and PRY fall in that estimation window with Penn World Table coverage and enter the

[14]The restricted and unrestricted models are fitted under the same optimal weighting matrix, so the difference in their J-statistics is a valid nested test. Fitting the unrestricted model under per-outcome block-diagonal weights instead, as an earlier version of this exercise did, breaks the nesting and can return a negative ΔJ.

reduced-form decomposition) produces $J(8) = 15.27$ ($p = 0.054$), with the proportionality ΔJ rising to $\chi^2(2) = 5.53$ ($p = 0.063$), a rejection at the 10% level but not at the 5%. The bootstrap success rate is 438 of 500 reps (87.6%), and all 62 failed reps are cases in which the country-resampled panel contained no AD event, so $\hat{\beta}^{AD}$ could not be estimated. The parameter signs are preserved in the four-class extension ($\hat{\alpha}_0 = +0.033$, $\hat{\gamma}_0 = -0.254$).

The tests settle less than the specification choice requires. The proportionality restriction is not rejected on the three-class moment set and is rejected only marginally on the four-class set, so the data do not decide between one block and two. Section 7 adopts the two-block specification on the theoretical ground set out in OA Appendix A.2.3, that investment is the firm's direct decision and loads on the coalition channel through a different mechanism than the downstream productivity outcomes do, and the marginal four-class rejection is consistent with that reading without establishing it. What the restriction the two-block model imposes, that TFP and growth share a loading up to scale, is not rejected by the data ($\chi^2(1) = 0.002$, $p = 0.96$). The one-block estimates remain useful as a sensitivity check on the channel parameters: under the more restrictive specification the uncertainty-channel sign is preserved with fixed-moment significance ($\hat{\gamma}_0 < 0$, $p < 0.01$ under the Jacobian standard errors, not separately distinguishable from zero under the primary bootstrap), and the qualitative direction of the coalition channel is preserved ($\hat{\alpha}_0 > 0$ at the point estimate, though not distinguishable from zero).

OA Table C2 reports the three-class and four-class one-block estimates side-by-side.

OA Table C2: Joint minimum-distance estimates: AD-class inclusion sensitivity

	Baseline	**4-class with AD**
	3 classes: DA, AA, DD	4 classes: DA, AA, DD, AD
Panel A: Joint structural parameters		
$\hat{\alpha}$ (coalition channel)	0.136	0.033
(SE)	(0.087)	—
$\hat{\gamma}$ (uncertainty channel)	−0.159	−0.254
(SE)	(0.060)	—
$\hat{\kappa}_{\text{growth}}$	96.13	57.49
(SE)	(25.84)	—
$\hat{\kappa}_{\text{invest}}$	51.43	52.70
(SE)	(27.68)	—
Panel B: Specification tests		
Moments / parameters	9/4	12/4
Degrees of freedom	5	8
J-statistic	7.67 ($p = 0.175$)	15.27 ($p = 0.054$)
ΔJ (proportionality, χ^2_2)	3.94 ($p = 0.139$)	5.53 ($p = 0.063$)
Panel C: AD-class moments (4-class only)		
M_{AD} (V-Dem realized)	—	+0.014
$\sqrt{\Sigma_{AD}}$	—	0.098
$\hat{\beta}^{TFP}_{AD,0}$	—	−0.042
$\hat{\beta}^{growth}_{AD,0}$	—	−2.022
$\hat{\beta}^{invest}_{AD,0}$	—	−1.419

Notes. Joint minimum-distance estimation imposes the proportionality restriction $\beta^y_{c,0} = \kappa_y(\alpha M_c + \gamma\sqrt{\Sigma_c})$ with $\kappa_{TFP} = 1$. The 4-class column adds the AUT → DEM class ($M_{AD} = +0.014$, $\sqrt{\Sigma_{AD}} = 0.098$ from the V-Dem realized ΔLIBDEM over the four first-event AUT → DEM coups in the 1950–2025 calibration sample), with $\hat{\beta}^y_{AD,0}$ identified from the two, PRT 1974 and PRY 1989, in the 1960–2022 local-projection window with Penn World Table coverage. SEs for the 4-class column are not reported because the bootstrap V is dominated by the AD diagonal. The exercise documents only that the sign pattern is insensitive to the AD class, not that magnitudes are precise, and significance markers are omitted. No channel parameter in this one-block exercise (common α, γ with κ_{growth}, κ_{invest}) is distinguishable from zero under the primary country-cluster bootstrap. Under the secondary fixed-moment layer (OA Table C1) the uncertainty loading $\hat{\gamma}_0$ is distinguishable from zero, and in the primary two-block model of Table 8 the coalition and uncertainty loadings on both blocks exclude zero across their 90% bootstrap intervals, with the investment-block coalition loading the one robust across weighting schemes and the removal of a common coup component.

C.2 Unrestricted minimum distance

The unrestricted per-outcome specification estimates a separate (α_y, γ_y) pair for each outcome by minimum distance over the three class moments, with no cross-outcome restriction:

$$\beta^y_{c,0} = \alpha_y M_c + \gamma_y \sqrt{\Sigma_c} + u^y_c, \qquad y \in \{\text{TFP}, \text{growth}, \text{investment}\}.$$

The system has $9 - 6 = 3$ over-identifying restrictions. The six parameters are estimated jointly under the same optimal weighting matrix as the restricted model, so the over-identification statistic is a single $J(3)$, not one $J(1)$ per outcome. Under the V-Dem realized calibration the estimates are:

Outcome	$\hat{\alpha}_y$	$\hat{\gamma}_y$
TFP	0.139	-0.126
Growth	11.89	-13.23
Investment	-29.61	-19.34

The joint over-identification statistic is $J(3) = 3.73$, which does not reject. The unrestricted point estimates place the apparent departure from proportionality in the investment block. The uncertainty channel $\hat{\gamma}_y < 0$ is uniformly negative across the three outcomes, consistent with the option-value prediction. The coalition channel $\hat{\alpha}_y > 0$ is positive for TFP and growth (consistent with the shared productivity-block parameters) but takes the opposite sign for investment ($\hat{\alpha}_{\text{invest}} = -29.61$). The sign-flip on the coalition channel between investment and the productivity block is the empirical signature of the within-block versus across-block distinction articulated in OA Appendix A.2.3, and it is what makes the one-block specification's common $\hat{\alpha}_0$ a forced compromise between the productivity block's positive loading and the investment block's negative loading.

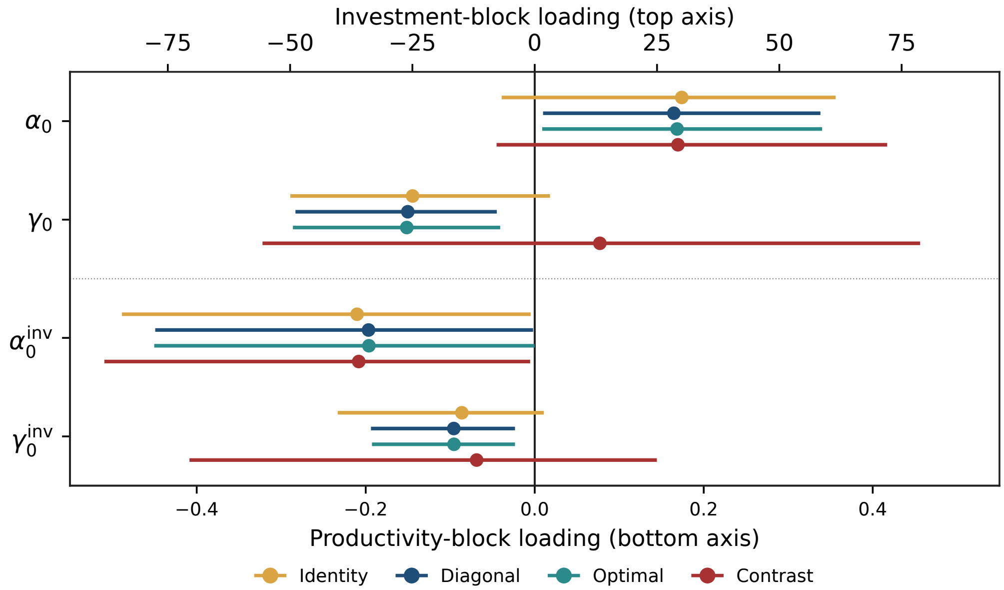
The unrestricted specification is not adopted as the primary model because it abandons the structural interpretation that gives the channels their economic content. With separate (α_y, γ_y) per outcome, α and γ are no longer coalition and uncertainty channels of a common firm-level adoption decision. They are outcome-specific reduced-form coefficients on M_c and $\sqrt{\Sigma_c}$. The two-block specification of Section 7 keeps the structural interpretation for the productivity block, restricting TFP and growth to share parameters (where the proportionality restriction is theoretically warranted and empirically not rejected, $\chi^2(1) = 0.002$, $p = 0.96$), and permits investment to load separately as a reduced-form allowance for its non-adoption components.

C.3 Weighting, the calibration sample, and a common-component contrast

Three choices in the primary specification of Table 8 deserve a diagnostic. The first is the calibration sample. The class moments there are computed from *all* eligible coups instead of first events only, which raises the event counts fourfold to fivefold and cuts the bootstrap standard deviation of the moments by roughly 57–60%. This is legitimate only if first and subsequent coups draw the same class-level institutional shift. We test it directly: within each transition class, neither the mean nor the dispersion of the realized institutional shift differs between first and subsequent coups at conventional levels (the smallest p-value, on the dispersion of the democracy-to-autocracy class, is 0.11). The all-event calibration is therefore a more precise estimate of the same class-level moment.

The second is the weighting matrix. Let Ω be the bootstrap covariance of the nine reduced-form betas. The productivity-block sub-block Ω_{tg} is near-singular (condition number $\approx$ 22,000), so inverting it for full optimal weighting is numerically fragile. Diagonal inverse-variance weighting, $W = \text{diag}(\Omega)^{-1}$, avoids that inversion and delivers nearly the same estimates and intervals as full optimal weighting ($\hat{\gamma}_0 = -0.150$ under diagonal against -0.151 under optimal, both excluding zero at 90%). Identity weighting couples the TFP and growth moments through their raw units, lets the growth moments dominate, and understates precision. Under either scale-adjusted scheme all four level loadings are sign-identified at 90%. We adopt the diagonal scheme as primary for this reason and report identity and optimal weighting as robustness.

The third is what identifies the sign of each channel. To separate the part of each loading that comes from cross-class institutional *heterogeneity* from the part that comes from a common coup component shared by all classes, we re-estimate the system on class contrasts (each class relative to intra-democratic coups): $\Delta\beta^y = \alpha_y \, \Delta M + \gamma_y \, \Delta\sqrt{\Sigma} + \Delta u^y$. This is algebraically the same model with an outcome-specific common component concentrated out. It introduces no new mechanism. Under the contrast system the investment-block coalition loading α_0^{inv} stays negative and excludes zero, but the uncertainty loadings lose their negative sign, the productivity-block γ_0 moves to $+0.08$ to $+0.14$ and its interval straddles zero. The negative sign of γ_0 in the level specification therefore comes largely from the common $\approx 2\%$ coup component instead of from cross-class institutional variation, which is why we describe the uncertainty channel as sign-identified in levels but not separately identified by cross-class variation, and describe α_0^{inv} as the one structural parameter robust across weighting schemes and across the levels-versus-contrasts distinction. OA Figure C1 collects the four loadings under all four schemes.



OA Figure C1: Structural loadings across weighting schemes, all-event calibration. Productivity-block loadings (α_0, γ_0) read against the bottom axis, and investment-block loadings $(\alpha_0^{\text{inv}}, \gamma_0^{\text{inv}})$ read against the top axis. The two axes share zero. Bars are 90% bootstrap percentile intervals. The investment-block coalition loading α_0^{inv} is negative and excludes zero under every scheme, including the common-component contrast. The productivity-block uncertainty loading γ_0 is negative and excludes zero in levels but flips positive and crosses zero under the contrast diagnostic. α_0 is sign-identified only under scale-adjusted level weighting.

C.4 V-Dem subindex robustness

The headline two-block estimates calibrate the class-level moments $(M_c, \sqrt{\Sigma_c})$ from the V-Dem liberal-democracy aggregate index (`v2x_libdem`). This is the calibration we adopt on theoretical grounds: the dispersion object Σ_c in the option-value model is the uncertainty a firm faces over the *broad* post-coup institutional environment, and `v2x_libdem` is the most aggregate V-Dem index, spanning the electoral, liberal, and rule-of-law components jointly, the natural empirical counterpart to the model's Σ. The sub-indices examined below each isolate a single institutional dimension, a narrower slice of that environment. We report them to test whether the structural channel parameters are properties of the firm's option-value problem instead of artifacts of the particular measure, recognising that no single sub-index is as close to the model's broad-dispersion object as the aggregate. If the parameters are structural, the estimates should be stable across alternative V-Dem sub-indices. This appendix re-estimates the two-block specification under three alternatives: `v2x_rule` (rule of law), `v2x_jucon` (judicial constraints on the executive), and `v2clprptym` (private property rights for men). For each sub-index we recompute class-level M_c and $\sqrt{\Sigma_c}$ from the within-event $\Delta\text{sub-index}_{t+1,t-1}$ on the same first-event-only sample and re-run the generated-regressor two-step bootstrap (500 replications, joint resample of the LP coefficients and the calibrated moments) of OA Appendix B.2. The reduced-form LP coefficient vector is identical across sub-indices, so only the calibration changes.

OA Table C3 reports the comparison. The `v2x_libdem` row is the first-event, identity-weighted calibration of these four channel-loading estimates, reported as a robustness benchmark; it is not the all-event diagonal specification of Table 8. The scale κ_{growth} is not displayed. The productivity-block uncertainty channel $\hat{\gamma}_0$ carries the predicted negative sign at the point estimate under all four calibrations (range -0.084 to -0.189), but its 95% bootstrap interval crosses zero in every calibration. The robust finding is the *sign-stability* of the point estimate across institutional dimensions, not significance, consistent with the sign-and-bound reading of the primary specification. Second, the investment-block coalition channel $\hat{\alpha}_0^{\text{inv}}$ is negative at the point estimate under all four calibrations (range -19.5 to -51.3). Its 95% bootstrap interval includes zero in every calibration (OA Table C3), so this first-event exercise speaks to the *sign-stability* of the coalition loading across institutional dimensions, not to interval significance. The 90% interval significance of $\hat{\alpha}_0^{\text{inv}}$ is a property of the headline all-event diagonal specification (Table 8), not of this first-event robustness calibration. The productivity-block coalition loading $\hat{\alpha}_0$ is positive at the point estimate under three of the four calibrations and negative under rule of law (-0.015), with a 95% interval that crosses zero throughout. The cross-block sign contrast therefore rests on the investment-block loading $\hat{\alpha}_0^{\text{inv}}$, negative at the point in all four, not on a separately-identified productivity-block sign. Third, the investment-block uncertainty channel $\hat{\gamma}_0^{\text{inv}}$ carries the predicted negative sign under the `v2x_libdem`, `v2x_rule`, and `v2x_jucon` calibrations (-13.2, -24.9, -10.0) but flips to positive under the property-rights calibration ($+4.8$). Its sign is therefore not stable across sub-indices and its bootstrap interval crosses zero throughout, so we do not lean on $\hat{\gamma}_0^{\text{inv}}$ in the structural argument.

Two explanations for the property-rights outlier operate in parallel. Conceptually, the property-rights subindex measures a substantive equilibrium outcome (the security of private property *after* the coup) instead of the institutional environment within which firms make their pre-coup investment commitments. The option-value mechanism this paper models is defined over uncertainty about the post-coup institutional environment, which the first three subindices (liberal democracy, rule of law, judicial constraints) capture more directly. Statistically, the property-rights subindex has much less informative cross-class variation than the other three: the intra-autocracy class has $\sqrt{\Sigma} = 0.612$ for `v2clprptym` versus 0.063 for `v2x_libdem`, and the intra-autocracy class mean is positive ($M_{AA} = +0.183$) rather than near zero, so the calibration matrix ($\hat{M}_c$, $\sqrt{\hat{\Sigma}_c}$) for `v2clprptym` carries less identifying power. The non-confirmation reflects both effects. We cannot cleanly separate them.

OA Table C3: V-Dem subindex robustness under the first-event, identity-weighted calibration

	Productivity block		Investment block		
Calibration (V-Dem subindex)	$\hat{\alpha}_0$	$\hat{\gamma}_0$	$\hat{\alpha}_0^{\text{inv}}$	$\hat{\gamma}_0^{\text{inv}}$	$J(4)$
v2x_libdem (liberal democracy, *first-event calibration*)	+0.134	−0.111	−31.64	−13.24	3.54
	[−0.38, +0.39]	[−0.54, +0.06]	[−106.7, +3.1]	[−76.0, +4.5]	[$p = 0.47$]
v2x_rule (rule of law)	−0.015	−0.189	−51.30	−24.92	2.01
	[−0.60, +0.40]	[−0.58, +0.02]	[−181.3, +10.8]	[−82.4, +2.3]	
v2x_jucon (judicial constraints)	+0.119	−0.102	−37.76	−10.01	1.98
	[−0.60, +0.56]	[−0.54, +0.01]	[−167.5, +14.2]	[−65.0, +6.9]	
v2clprptym (private property rights)	+0.142	−0.084	−19.51	+4.75	5.17
	[−0.89, +1.55]	[−0.42, +0.23]	[−147.3, +213.5]	[−60.1, +36.6]	

Notes. Each row re-estimates the generated-regressor two-block joint MD of OA Appendix B.2 on PWT 11.0 and the Powell–Thyne-reconciled coup panel under the *first-event calibration with identity weighting*, using a different V-Dem subindex to calibrate $(M_c, \sqrt{\Sigma_c})$. This is a robustness exercise on the institutional measure, not the headline specification of Table 8, which uses the all-event calibration with diagonal weighting and 800 resamples. Brackets give each calibration's 95% country-cluster two-step bootstrap interval (500 resamples). Three of the five parameters hold their sign across all four calibrations: $\hat{\alpha}_0^{\text{inv}}$ negative, $\hat{\gamma}_0$ negative, and $\hat{\kappa}_{\text{growth}}$ positive throughout. Two do not: $\hat{\alpha}_0$ turns negative under rule of law, and $\hat{\gamma}_0^{\text{inv}}$ turns positive under private property rights. Read the table as establishing which signs are stable across institutional measures under the first-event calibration, not as pinning down magnitudes. The over-identification statistic $J(4)$ ($\chi^2(4)$) uses efficient (cluster-bootstrap) weighting, valid as a χ^2 test but not reproducible by hand, and is not rejected in any calibration.

D Pre-trend identification ladder for the recurrent-event design

The recurrent-event impact in Section 6.1 is estimated on an entropy-balanced clean risk set. This appendix reports the full ladder of control constructions behind that choice. The designs were fixed in advance and are not selected by their pre-trend p-value: we report every rung, and the pre-trend of the preferred design whatever it is. All estimates use the stacked design with episode-by-unit and episode-by-year fixed effects, clustered by country, with restricted Webb wild cluster bootstrap inference.

Two facts organise the table. First, the impact-year and one-year coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$ are extraordinarily stable: they sit near -0.03 and -0.04 to -0.05 under every control construction, including hard stratification and a deeper eight-year holdout. Second, the pre-event leads are not flattened by changing the control pool. Moving from never-treated controls to the clean risk set barely moves the joint pre-trend ($p = 0.089 \rightarrow 0.074$), and nearest-neighbour matching does not help. Only comparability restrictions flatten the leads: hard stratification (R3) does so but destroys power, turning the medium-run response insignificant and the long-run positive. Pooled entropy balancing with corrective weights (R8-Q) flattens the leads while preserving the impact and a detectable medium run.

OA Table D1: Pre-trend identification ladder, recurrent-event stacked design

Design	Control / balancing	Treated (ep/ctry)	ESS	Pre-trend wild p	$\hat{\beta}_0$	$\hat{\beta}_1$	$p_{2:5}$	$\hat{\beta}_5$	$\hat{\beta}_8$
R0 $[-3,+8]$	never-treated	58 / 45	—	0.089	-0.030	-0.049	0.004	-0.024	-0.022
R1 $[-5,+8]$	never-treated	45 / 35	—	0.076	-0.032	-0.049	0.047	-0.027	-0.024
R2	risk set	45 / 35	—	0.074	-0.033	-0.048	0.057	-0.020	-0.010
R3	risk set + hard strata	45 / 35	—	0.829	-0.029	-0.033	0.423	$+0.015$	$+0.022$
R4	risk set + NN match (20)	45 / 35	—	0.039	-0.033	-0.045	0.032	-0.012	$+0.006$
R5	risk set + region×year FE	45 / 35	—	0.060	-0.029	-0.038	0.148	$+0.001$	$+0.016$
R8	risk set + pooled EB	44 / 35	1491	0.131	-0.031	-0.044	0.007	-0.009	$+0.006$
R8-Q	**risk set + pooled EB + Wing**	**44 / 35**	**1491**	**0.114**	$\mathbf{-0.030}$	$\mathbf{-0.044}$	**0.011**	-0.006	$+0.009$
R9	risk set + IPW	45 / 35	235	0.066	-0.038	-0.058	0.029	-0.058	-0.069
Holdout $[-8,+8]$, entropy bal.		(deeper)	943	extrap. $\{-6,-7,-8\}$: $p = 0.35$		-0.036	0.253	$+0.010$	$+0.030$

Notes. R8-Q is the preferred design used in Section 6.1. The pre-trend column is the joint wild-bootstrap test on the pre-event leads. For R8 the weights are built to balance pre-event TFP at $h = -1, -3, -5$, so the flatness at those horizons is partly mechanical. The independent checks are the holdout leads not used in the weights: within the primary window $\{-4, -2\}$ do not jointly reject ($p = 0.25$), and in a deeper eight-year design the extrapolation leads $\{-6, -7, -8\}$ do not jointly reject ($p = 0.35$), though the deeper window shows some residual sensitivity and its medium-run response is no longer detectable ($p_{2:5} = 0.25$). Entropy balancing is global over country-episode observations, not within each stack. The effective sample size (ESS) of the control weights is 1491 (about 45% of the control country-episode observations), the maximum control weight is small, and no single control dominates. The IPW weights (R9) do not achieve the same balance (ESS 235) and leave the leads inclined. $p_{2:5}$ is the joint wild test that the year-two-through-five coefficients are zero. Designs were fixed in advance and are not chosen by their pre-trend p-value.

D.1 Aggregation, local balancing feasibility, and sample composition

As a further diagnostic we implemented, for TFP, the two stages that covariate-balanced weighted stacked designs separate, in the spirit of Wing, Freedman, and Hollingsworth (Wing et al., 2024): stack-specific covariate balancing and corrective across-stack aggregation. Three findings follow, reported for TFP on the frozen $[t-5, t+8]$ window and covariates.

First, the corrective aggregation weights are empirically immaterial here. Adding them to the pooled design, moving from R8 to R8-Q, leaves the impact and the leads essentially unchanged ($\hat{\beta}_0 = -0.031 \rightarrow -0.030$, $\hat{\beta}_1 = -0.044$, pre-trend $p = 0.131 \rightarrow 0.114$). The corrective aggregation is empirically immaterial for these estimates, and we include the weights in the preferred specification.

Second, stack-specific balancing is support-limited in this application. Under the broad donor pool, entropy balancing succeeds in only 32 of 44 event stacks, the remaining 12 falling back to unweighted controls. Under region-by-regime exact stratification it succeeds in none of the 39 stacks that retain enough controls, and five episodes have too few same-region, same-pre-regime controls to enter at all. Where balancing does succeed the balanced moments match by construction (maximum standardized imbalance below 0.002). These specifications are feasibility diagnostics rather than successful locally balanced designs, because local overlap is insufficient for entropy balancing in many stacks: the larger pre-event coefficients reflect that infeasibility, not a verdict on the design. This is why we use pooled entropy balancing with corrective stack weights rather than a within-stack implementation.

Third, restricting the pooled design to the 39 stacks that survive the strata keeps the balancing moment at $h = -5$ near zero but raises the joint pre-trend over all leads to $p = 0.0085$, so part of the pre-event attenuation in the full sample reflects episode composition, not weighting alone. In the full prespecified risk-set sample, pooled entropy balancing substantially attenuates the pre-event differences, but this does not by itself validate parallel trends.

E The recurrent-event design across all outcomes

We apply the preferred recurrent-event estimator (R8-Q: pooled entropy balancing with corrective stack weights, window $[t-5, t+8]$, same predetermined covariates) to every outcome, and report the impact under R8-Q alongside the range of the impact-year coefficient across the full control-construction ladder R0–R9 of OA Table D1. The pattern is concentrated. The coup response lives in the productivity and real-activity block: total factor productivity and output growth contract sharply on impact, gross investment contracts and recovers, and government spending falls modestly. The remaining outcomes, trade, unemployment, net migration, foreign direct investment, and inflation, show null or imprecise responses. Within each outcome the impact-year coefficient is stable across control constructions, as the range column shows.

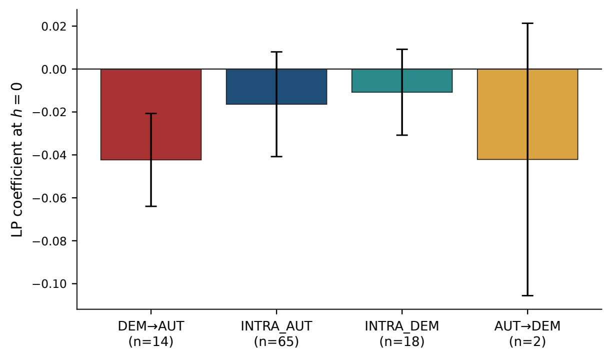
OA Table E1: Recurrent-event responses across outcomes (R8-Q), and impact-year stability across the R0–R9 ladder

Outcome	Treated episodes	$\hat{\beta}_0$ (R8-Q)	$\hat{\beta}_1$ (R8-Q)	Pre-trend wild p	$p_{2:5}$	$\hat{\beta}_0$ range (R0–R9)
TFP (log)	44	−0.0303	−0.0436	0.114	0.011	[−0.039, −0.029]
Output growth	40	−2.84	−0.284	0.125	0.032	[−2.93, −2.30]
Gross investment	27	−0.994	−1.84	0.717	0.362	[−1.03, −0.82]
Gov. spending	31	−0.346	−0.528	0.941	0.049	[−0.47, −0.32]
Trade	35	+4.12	+2.91	0.315	0.698	[+4.05, +4.80]
Unemployment	15	+0.138	−0.086	0.028	0.624	[+0.08, +0.21]
Net migration	44	−0.0094	+0.0006	0.926	0.234	[−0.010, −0.009]
FDI	32	+0.0005	−0.000	0.038	0.197	[+0.000, +0.001]
Inflation	33	+1.34	+1.43	0.378	0.825	[+1.34, +1.82]

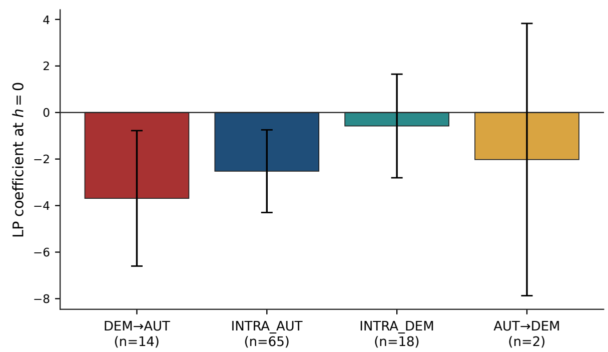
Notes. All columns use the R8-Q design (pooled entropy balancing on region, decade, pre-coup regime, lagged GDP per capita, and pre-event TFP at $h = -5, -3, -1$, plus corrective stack weights), with weights re-estimated within each outcome's estimation sample, which is why the treated-episode counts differ. Inference is the restricted Webb wild cluster bootstrap. The pre-trend column is the joint test of the four pre-event leads ($h = -5$ to -2); the holdout test cited for output growth in Section 6.1 uses only the leads excluded from the balancing, which reject for growth. Units follow the paper's conventions: TFP is in log points, output growth, investment, government spending, trade, unemployment, and inflation are scaled by 100, and net migration and FDI enter as shifted logs, so magnitudes are not comparable across rows. The $\hat{\beta}_0$ range column reports the minimum and maximum impact-year coefficient across the eight control constructions R0, R2–R5, R8, R8-Q, and R9. Within each outcome the impact-year coefficient is stable across designs. The coup response is concentrated in the productivity and real-activity block; trade, unemployment, migration, FDI, and inflation are null or imprecise, and some carry pre-event trends of their own.

F First-event decomposition and focal-cell figures

This appendix collects the figures for the first-institutional-transition (Design B) results. They are referenced from Section 6 and Section 7 but placed here so that the main text leads with the higher-support recurrent-event evidence of Figures 2 and 3. OA Figure F1 gives the impact-year transition decomposition, OA Figure F2 the focal-cell robustness architecture, and OA Figure F3 the productivity-versus-GDP persistence contrast.

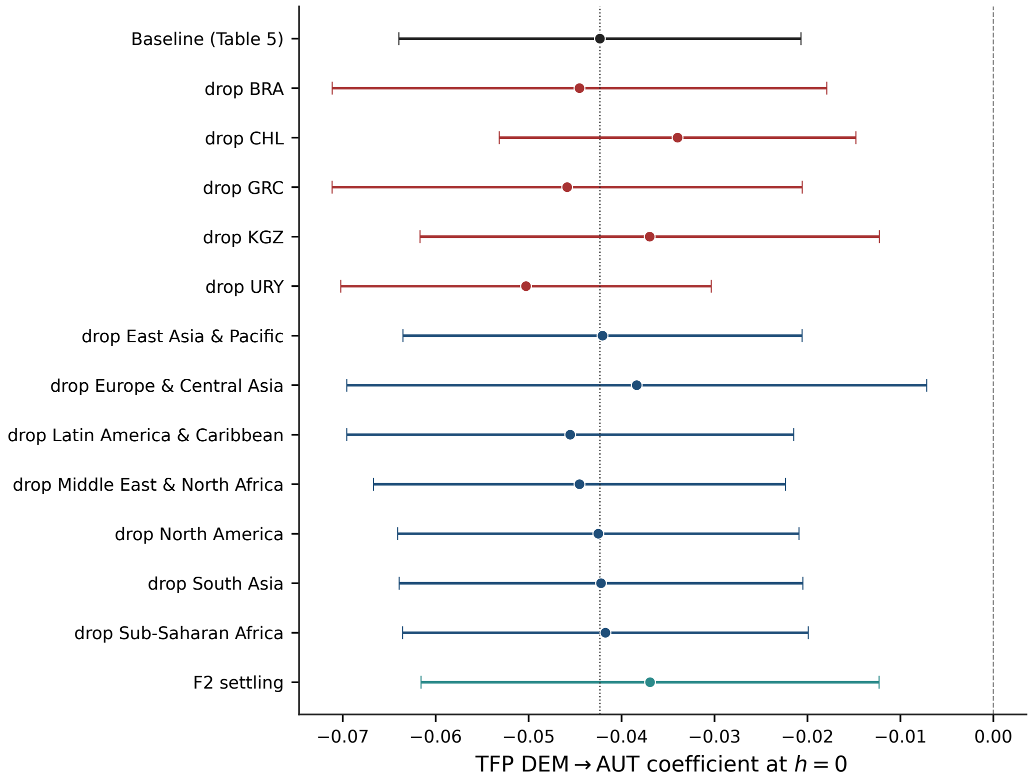


(a) Total factor productivity

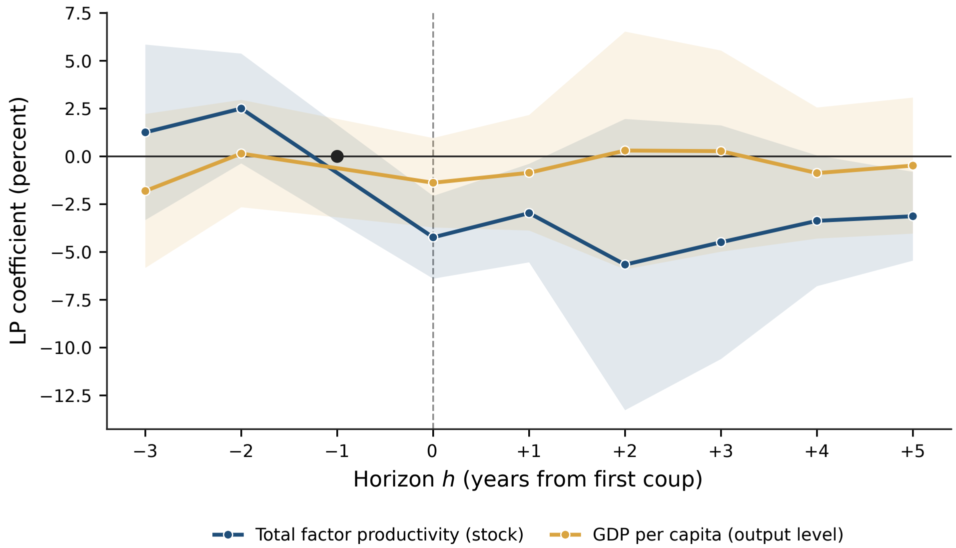


(b) Output growth

OA Figure F1: Impact-year decomposition by post-coup regime trajectory. Bars are LP coefficients at $h = 0$ with 95% confidence intervals from country-clustered standard errors. The intervals are asymptotic and descriptive. Governing wild-bootstrap significance for the covered cells is reported in OA Table I1.



OA Figure F2: Robustness architecture for the focal cell. Each row reports the DEM $\rightarrow$ AUT TFP coefficient at $h = 0$ under a single modification of the baseline. The nine noncontributing-country drops, which reproduce the baseline, are omitted. Horizontal bars are 95% confidence intervals from clustered standard errors.



OA Figure F3: Productivity persistence and GDP-per-capita level recovery in the democracy-to-autocracy class. Local-projection paths for total factor productivity (a stock) and the log level of GDP per capita, a separately estimated level path shown as the level counterpart to the output-growth response. Coefficients in percent, with 95% confidence bands from country-clustered standard errors. The reference horizon $h = -1$ is normalized to zero.

G Non-focal outcomes: FDI, total trade, inflation, government spending, unemployment

The local-projection responses for the five non-focal outcomes are reported in OA Table G1. Four of the five, foreign direct investment, total trade, inflation and unemployment, show no statistically detectable impact-year response in the first-event-only sample, and for those four the aggregate specifications confirm the absence of a detectable response at $h = 0$ through $h = +2$. Government spending is the exception: it contracts at the impact year, and only at the 10% level. The BBG-style transition decomposition is reported at $h = 0$ for each of the five outcomes to verify that a weak or absent aggregate response is not concealing a transition-specific effect. In the decomposition, and with the exceptions noted below, the coefficients are also statistically zero.

There are two exceptions in the decomposition. For total trade the INTRA_AUT coefficient at $h = 0$ is statistically significant ($\hat{\beta}_0 = -0.079$, $p < 0.05$), indicating that autocracy-to-autocracy coups specifically are associated with a trade contraction. This is a secondary, exploratory finding, reported here as suggestive of a sub-population pattern, not as a focal claim. Second, for unemployment the DEM $\rightarrow$ AUT coefficient at $h = 0$ is -0.335 with an asymptotic p of 0.005, and exactly one country contributes to it. A wild cluster bootstrap cannot be computed for a single-cluster cell. That coefficient is the clearest demonstration in the paper that asymptotic significance in the small transition cells is not self-certifying, since it can be produced by a single country, and it is why we rest the focal productivity claim on the bootstrap, not on the asymptotic p-value. We do not interpret the unemployment estimate as evidence of anything.

The foreign direct investment null is the one that bears on the framework. Proposition 1 imposes that foreign capital, being disproportionately exposed to expropriation, loads more heavily than domestic investment on both channels. A realised regime change moves those same channels, so the ordering motivates a qualitative indirect comparison, but it does not strictly order the realised FDI coefficient against the aggregate gross-investment coefficient, which can carry different loadings and offsetting channels. FDI does not respond at any reported horizon. We report this as an indirect reduced-form check that does not support the FDI ordering, and we do not reinterpret the null as consistent with the framework. It is not a direct test of Proposition 1, whose content is the response to the coup *hazard*, an object this design cannot reach for the reason given in Section 3.6. A second qualification is about the series, not the check: the World Bank FDI measure is nominal, volatile, and dominated by a handful of large flows, so the exercise is inconclusive, not adverse.

OA Table G1: Non-focal outcomes: aggregate and BBG-transition local projections

	Aggregate LP			*BBG-transition LP at $h = 0$*			
Outcome	$h = 0$	$h = +1$	$h = +2$	DEM→AUT	AUT→DEM	INTRA_AUT	INTRA_DEM
FDI	−0.020	+0.028	+0.053	+0.094	−0.034	−0.028	−0.045
	(0.041)	(0.056)	(0.064)	(0.111)	(0.128)	(0.053)	(0.114)
Total trade	−0.041	+0.045	+0.098	−0.015	+0.006	−0.079**	+0.032
	(0.027)	(0.059)	(0.067)	(0.095)	(0.011)	(0.038)	(0.028)
Inflation	+3.866	−0.662	−1.164	+7.278	+1.213	+4.490	−0.462
	(2.636)	(2.328)	(1.808)	(5.149)	(3.060)	(4.253)	(0.838)
Gov. spending	−0.454*	−0.192	+0.132	−0.324	−0.360	−0.535	−0.023
	(0.253)	(0.457)	(0.405)	(0.400)	(0.412)	(0.333)	(0.131)
Unemployment	−0.124	−0.264	+0.000	−0.335***	+0.000	−0.110	−0.062
	(0.243)	(0.369)	(0.381)	(0.118)	(0.000)	(0.354)	(0.143)

Notes. Coefficients on the first-coup indicator (aggregate) and on the four transition indicators (decomposition), for the five outcomes that carry no focal claim in this paper. The table is not null everywhere: government spending contracts at the impact year in the aggregate, total trade contracts in the intra-autocracy class, and unemployment falls in the democracy-to-autocracy class. Those three cells are the exceptions and are discussed in the text. Specifications match those in Tables 4 and 5: first-event-only sample truncated at the second coup, country and year fixed effects, lagged outcome and lagged log GDP per capita, standard errors clustered by country. AUT→DEM has $n = 2$ events and is reported for completeness only. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

H Democracy-to-autocracy events

OA Table H1 lists the 14 first-coup events classified as democracy-to-autocracy transitions in the first-event-only sample. The five countries marked with a dagger are those with a complete impact-year local-projection cell, which requires TFP at t and $t - 1$ together with the specification's controls, and therefore contribute non-trivially to the focal estimate. The remaining nine events remain in the estimation panel, but their coup-year observations do not enter the $h = 0$ coefficient because they lack the lagged-outcome cells the LP specification requires, generally because Penn World Table coverage is incomplete for these countries in the relevant period.

OA Table H1: Democracy-to-autocracy first-coup events ($n = 14$)

Country	ISO	Coup year	TFP $\hat{\beta}_0$	Invest. $\hat{\beta}_0$	Pre-coup GDP p.c.
Argentina	ARG	1961	—	—	—
Brazil†	BRA	1964	−0.033	−0.11	3,623
Chile†	CHL	1973	−0.076	−4.86	8,520
Congo, Rep.	COG	1963	—	—	—
Greece†	GRC	1967	−0.027	+35.71	7,570
Grenada	GRD	1979	—	—	—
Honduras	HND	1963	—	—	—
Kyrgyz Republic†	KGZ	2010	−0.062	+1.37	4,076
Myanmar	MMR	1962	—	—	—
Panama	PAN	1969	—	—	—
Seychelles	SYC	1977	—	—	—
Sierra Leone	SLE	1967	—	—	—
Suriname	SUR	1980	—	—	—
Uruguay†	URY	1973	−0.012	+13.55	7,643

Notes. The 14 first-coup events classified as democracy-to-autocracy (DEM → AUT) in the first-event-only sample of the BBG-style decomposition (Section 6.3). Pre-coup regime is measured at $t-1$ using the Bjørnskov–Rode classification, post-coup regime at $t+1$ with a $t+2$ fallback. † marks the five countries with non-missing TFP at the coup year, which identify the focal estimate. For those five, the last three columns give the event-level local-projection coefficient at $h = 0$ for TFP (log points) and gross investment (transformation of Section 4), and pre-coup ($t-1$) GDP per capita in constant USD. The five TFP coefficients average -0.042, the focal estimate $\hat{\beta}_0$ (Section 6.5). An em dash in the investment column for a non-daggered event means its coefficient is not reported here, not that it fails to enter the investment regression, whose eight contributors appear in OA Appendix B.3. The remaining nine countries stay in the panel but their coup-year observations do not enter the $h = 0$ coefficient, lacking the lagged-outcome cells the LP TFP specification requires, generally from incomplete early Penn World Table coverage.

I Wild cluster bootstrap inference for individual LP coefficients

The asymptotic cluster-robust p-values reported alongside the LP coefficients in Sections 6 and 7 are computed under the standard sandwich formula clustered at the country level. MacKinnon and Webb (2018) document that asymptotic cluster-robust inference is unreliable when the number of treated clusters is small. The focal DEM $\rightarrow$ AUT cell has 14 treated countries, only five contributing at the impact year, so asymptotic validity fails despite the larger number of regression clusters.

OA Table I1 reports wild cluster bootstrap p-values with Webb (2014) six-point weights and 2,000 bootstrap replications, alongside the asymptotic counterparts, for the LP coefficient $\hat{\beta}_{y,c,h}$ at each combination of outcome $y \in \{\text{TFP}, \text{growth}, \text{investment}\}$, transition class $c \in \{\text{DA}, \text{AA}, \text{DD}\}$, and horizon $h \in \{0, 1, 2, 3, 4, 5\}$. Several individual coefficients remain significant after the bootstrap correction: TFP DA at $h = 0$ remains significant at the 5% level (bootstrap $p = 0.020$ vs. asymptotic $p < 0.001$), investment DA at $h = 5$ remains significant at the 1% level (bootstrap $p = 0.004$), growth AA at $h = 0$ remains significant at the 1% level (bootstrap $p = 0.007$). Bootstrap p-values are larger than the asymptotic counterparts in almost every cell, as expected under MacKinnon–Webb, and the qualitative pattern is preserved.

OA Table I1: Wild cluster bootstrap p-values for LP coefficients, by outcome, class, and horizon

		Horizon h					
Outcome	Class	0	1	2	3	4	5
TFP							
	DA	-0.042**	-0.030	-0.057	-0.045	-0.034	-0.031*
	p	0.020 (0.000)	0.101 (0.024)	0.276 (0.145)	0.214 (0.150)	0.151 (0.053)	0.061 (0.008)
	AA	-0.016	-0.037*	-0.003	-0.014	-0.009	-0.007
	p	0.207 (0.187)	0.091 (0.071)	0.898 (0.894)	0.671 (0.632)	0.812 (0.796)	0.853 (0.848)
	DD	-0.011	0.010	0.014	0.014	0.008	-0.016
	p	0.346 (0.290)	0.676 (0.514)	0.715 (0.463)	0.533 (0.466)	0.635 (0.603)	0.344 (0.335)
Growth							
	DA	-3.689**	2.086	0.161	0.331	-0.643	-0.674
	p	0.017 (0.013)	0.319 (0.169)	0.934 (0.938)	0.870 (0.828)	0.735 (0.667)	0.650 (0.609)
	AA	-2.522***	-0.998	0.045	-0.950	0.400	0.369
	p	0.007 (0.005)	0.408 (0.391)	0.965 (0.962)	0.484 (0.454)	0.680 (0.682)	0.691 (0.682)
	DD	-0.579	1.758	2.988*	0.367	-0.809	-3.156
	p	0.649 (0.610)	0.340 (0.280)	0.092 (0.051)	0.845 (0.829)	0.332 (0.299)	0.272 (0.226)
Investment							
	DA	3.265	-0.556	1.167	2.707	0.350	2.026***
	p	0.403 (0.321)	0.706 (0.666)	0.848 (0.573)	0.317 (0.212)	0.812 (0.795)	0.004 (0.002)
	AA	-0.510	-0.385	0.409	0.687	0.718	1.183
	p	0.231 (0.212)	0.673 (0.654)	0.650 (0.619)	0.414 (0.408)	0.347 (0.327)	0.103 (0.090)
	DD	-2.855**	-1.769	-1.259	0.162	0.128	-0.321
	p	0.049 (0.040)	0.357 (0.257)	0.536 (0.420)	0.888 (0.874)	0.890 (0.907)	0.867 (0.748)

Notes. Top number in each cell is the LP coefficient $\hat{\beta}_{y,c,h}$ from the BBG-style transition decomposition of Section 6.3. Beneath it is the wild cluster bootstrap p-value, with the conventional asymptotic cluster-robust p-value in parentheses. Stars on coefficients refer to the bootstrap p-value, which is the governing inference throughout. The bootstrap uses the Webb (2014) six-point distribution with 2,000 replications, addressing the MacKinnon–Webb (2018) finding that asymptotic cluster-robust inference is unreliable with few treated clusters (the focal DA cell has 14 treated countries, five contributing at the impact year). Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

J Additional focal-cell robustness

This appendix collects two robustness exercises referenced from the focal-cell results of Section 6.5: the robustness of both focal contractions to the alternative (F2) regime-classification timing, and a secondary heterogeneity decomposition that classifies coups by pre-coup regime alone. OA Table J1 also collects, for reference, the cross-class contrast, proportionality, and within-cell test statistics reported in the text.

OA Table J1: Cross-class contrast, proportionality, and within-cell test statistics

Test	Statistic	Asym. p	Boot. p
Panel A. Impact-year TFP contrasts ($h = 0$)			
DA = DD	$F = 4.39$	0.039	0.074
DA = AA	$F = 2.55$	0.113	—
Four-class omnibus (DA=AD=AA=DD)	$F = 1.71$	0.170	—
Panel B. Impact-year investment contrast ($h = 0$)			
DA − DD point estimate = +6.12	—	—	0.124
wild bootstrap 95% CI [−1.20, +14.26]			
Panel C. Structural proportionality restriction			
TFP and growth share loadings up to scale (ΔJ)	$\chi^2(1) = 0.002$	0.96	—
Panel D. Within-cell regression on realized Δ V-Dem ($h = 0$)			
Total factor productivity	—	0.98	—
Output growth	—	0.69	—
Gross investment	—	0.83	—
Government spending	—	0.96	—
Foreign direct investment	—	0.01	—
Inflation	—	0.07	—

Notes. Statistics reported in the text and collected here for reference. Panel A gives the impact-year TFP pairwise and omnibus Wald tests of equality across BBG transition classes. The asymptotic p is the cluster-robust Wald value and the bootstrap p is the Webb six-point wild cluster bootstrap that governs the focal inference. Panel B is the democracy-destroying minus intra-democracy investment contrast at $h = 0$ ($3.265-(-2.855) = 6.12$ on the transformed gross-investment scale). It is reported as directional evidence, not as a test, and its interval is wide. Panel C is the cross-outcome nested difference-in-J for the single restriction that TFP and growth share channel loadings up to scale. Panel D regresses the event-specific impact-year coefficient on the realized ΔV-Dem within the focal cell, outcome by outcome. Six outcomes carry enough contributing events to run. All values are from the replication materials.

J.1 TFP–growth asymmetry

A robustness exercise on the alternative regime-classification timing shows that both focal contractions survive measuring the post-coup regime at $t + 2$ instead of $t + 1$: each F2 impact-year coefficient stays close to its baseline and rejects zero at the 5% level under the wild cluster bootstrap (OA Table J2). The qualitative conclusions for both responses are unchanged by the timing of the regime measurement.

OA Table J2: Alternative regime-classification (F2, t+2) timing: focal impact-year contractions

Outcome, $\hat{\beta}_{\mathrm{DA}}(h{=}0)$	Baseline (t+1)	F2 (t+2)	SE (F2)	Wild p (F2)
TFP (log)	−0.042	−0.037	(0.013)	0.047
Output growth (transformed scale)	−3.69	−3.45	(1.64)	0.045

Notes. The F2 classification measures the post-coup regime at t+2 rather than t+1, so a coup followed by a two-year settling period is coded by its eventual regime. Both focal impact-year contractions survive: each F2 coefficient is close to its baseline and rejects zero at the 5% level under the wild cluster bootstrap (Webb six-point weights, clustering on country). The exercise covers the two focal contractions and does not re-estimate the later horizons under F2.

The asymmetry between TFP and growth is one of persistence, and it is that persistence contrast, not any difference in timing sensitivity, that the stock–flow distinction of Section 3.3 (Equation 2) predicts. TFP accumulates the cumulative shortfall in technology adoption over the disrupted-flow window and does not recover when the flow recovers, while growth is itself the flow and recovers as the post-coup environment stabilizes.

Under the option-value reading of the focal cell developed in Section 6.5 and parameterized in Section 7, the same asymmetry follows naturally: option-value-driven shortfalls in irreversible technology adoption accumulate into the productivity stock, while flow dynamics like output growth respond to the realized post-coup environment and recover as it stabilizes. We report the asymmetry as an empirical observation that the option-value reading accommodates, not as a separate conceptual contribution.

A related observation concerns the intra-democracy cell. Under the F2 classification the INTRA_DEM TFP coefficient is -0.016 ($p = 0.13$), where under the baseline it was -0.011 and statistically zero. Both are small and statistically insignificant. It is consistent with the interpretation that democracy-preserving coups in democracies can themselves generate some institutional outcome dispersion (expectations effects, elite polarization, partial repression) even when formal regime classification remains democratic. The pattern is consistent with the option-value reading: the magnitude of the TFP response scales with the variance of post-coup institutional outcomes, from near zero in stable democracies through small negative in intra-democracy coup attempts to large negative in democracy-destroying events.

J.2 Secondary heterogeneity by pre-coup regime

For completeness we report a parallel decomposition that classifies coups by the pre-coup regime alone, ignoring the post-coup trajectory. OA Table J3 shows the impact-year coefficients on TFP, growth, and migration for three regime categories: democracies ($n = 32$), civilian dictatorships ($n = 48$), and military dictatorships ($n = 9$). The TFP impact is concentrated in democracies ($\hat{\beta}_0 = -0.022$, $p = 0.011$), the growth impact in civilian dictatorships ($\hat{\beta}_0 = -2.71$, $p = 0.013$), and military dictatorships have insufficient first-event sample to support precise inference. Joint Wald tests of equality across the three categories do not reject at the 10% level for any focal outcome.

OA Table J3: Pre-coup regime heterogeneity (secondary / exploratory)

	$h=-3$	$h=-2$	$h=+0$	$h=+1$	$h=+5$
TFP					
Democracies (n=32)	$+0.008$	$+0.012$	-0.022^{**}	-0.004	-0.021^{*}
	(0.012)	(0.011)	(0.008)	(0.012)	(0.012)
Civilian dict. (n=48)	$+0.056^{**}$	$+0.047^{**}$	-0.010	-0.037	$+0.003$
	(0.027)	(0.019)	(0.015)	(0.026)	(0.042)
Military dict. (n=9)	-0.078^{*}	-0.084	-0.015	-0.027	-0.036
	(0.046)	(0.058)	(0.013)	(0.022)	(0.026)
Growth					
Democracies (n=32)	-0.549	-0.488	-2.374^{**}	$+1.905^{*}$	-1.816
	(1.248)	(1.603)	(1.041)	(1.108)	(1.339)
Civilian dict. (n=48)	-2.421	-0.293	-2.712^{**}	-1.211	-0.479
	(1.582)	(1.212)	(1.091)	(1.404)	(0.995)
Military dict. (n=9)	-1.078	-2.108^{*}	-1.734	-3.507^{***}	$+0.193$
	(0.786)	(1.171)	(1.414)	(1.354)	(0.683)
Migration					
Democracies (n=32)	$+0.010$	-0.004	-0.022	-0.023	$+0.069$
	(0.031)	(0.028)	(0.031)	(0.030)	(0.074)
Civilian dict. (n=48)	-0.057	-0.065	-0.274	-0.176	$+0.312$
	(0.073)	(0.059)	(0.316)	(0.208)	(0.235)
Military dict. (n=9)	$+0.001$	$+0.095$	$+0.001$	-0.000	$+0.103$
	(0.027)	(0.071)	(0.105)	(0.068)	(0.119)

Notes. Local projections with three interactions of the first-coup indicator with pre-coup regime category from Bjørnskov–Rode (measured at $t-1$). Democracies = parliamentary, mixed, presidential. Royal dictatorships (n=25 overall, n=10 as first-event) and transitions/occupation enter through a separately estimated fourth term whose coefficient is not reported. Joint Wald tests of regime equality fail to reject at 10% in all focal outcomes (reported in text). Estimates therefore presented as *suggestive heterogeneity*. Significance stars in this table follow the *asymptotic* cluster-robust p-value: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

K Multi-period staggered estimators

The primary specification in Section 6.5 restricts attention to the first coup in each country and treats the country panel as truncated at the year of the second coup. This first-event design rests on an econometric foundation, and on a looser correspondence with the option-value framework of Section 3. The correspondence is that a single shock indicator per country isolates a firm facing a hazard that has not yet realised, which is the cleanest counterpart to the option-value experiment. It is not that the framework excludes later shocks: coups arrive there as a Cox process, so the theory speaks to recurrence and does not by itself select the first event. The econometric foundation does the work: with the Bjørnskov and Rode (2020) panel admitting up to 8 coup events per country over the 1960–2022 window, post-event horizons of the first coup overlap with pre-event horizons of subsequent coups in any country with two or more events, contaminating the estimated β_h in the absence of a staggered-event correction.

This appendix complements the primary first-event specification with two multi-period staggered estimators that admit later democracy-to-autocracy treatment starts as additional treatments. Under the non-absorbing construction, some re-entries occur without a contemporaneous coup. The Callaway and Sant'Anna (2021) estimator assumes treatment is absorbing (each unit can be treated once). The de Chaisemartin and D'Haultfœuille (2024) estimator allows treatment to switch on and off, the empirically appropriate specification for democracy–autocracy cycling. Both are reported as robustness for the primary first-event design, not as replacements, on theoretical grounds (option-value first-shock correspondence) and on empirical grounds documented in Subsections K.1–K.2 below.

K.1 Cycling diagnostics

We define the multi-period treatment indicator D_{it} as the cumulative-DA-event flag: $D_{it} = 1$ if country i has experienced at least one DEM $\rightarrow$ AUT coup on or before year t *and* is currently in an autocracy ($\text{regime}_{it} \in \{3, 4, 5\}$ in the Bjørnskov–Rode coding), and zero otherwise. The indicator switches off when a country returns to democracy after a previous DA event and switches back on at a subsequent autocratic re-coding, a construction that can switch on without a contemporaneous coup, as with Peru 1990. OA Table K1 summarizes the resulting switch counts across the panel.

OA Table K1: Cycling diagnostics for the multi-period DA treatment indicator

Quantity	Count
Total switch-on events ($0 \rightarrow 1$)	65
Total switch-off events ($1 \rightarrow 0$)	47
Countries that switch on at least once	40
Countries that never switch	164
Total country-years in treated state ($D_{it} = 1$)	758
Total country-years in control state ($D_{it} = 0$)	12,094

Notes. 40 of the 204 countries in the merged panel of Section 4 experience at least one DEM $\rightarrow$ AUT event over 1960–2022, and the remaining 164 never switch. Across those 40 countries, the panel records 65 switch-on events. The difference of $65 - 40 = 25$ reflects countries that cycle (return to democracy and re-enter autocracy through a subsequent DA treatment switch-on). 47 switch-off events occur where a previously-treated country returns to democracy. The 204 countries are the merged panel, not the estimation sample. A productivity regression needs productivity data on the controls as well as on the treated, and only 119 of the 204 carry Penn World Table TFP coverage: 24 of them experience a democracy-to-autocracy transition and 95 never do. The staggered estimators are therefore identified off at most those 119 countries, which is why their samples are smaller than the merged panel and why they differ from one another.

The cycling rate is substantial. Argentina, Nigeria, and Honduras each cycle three times within the 1960–2022 window. Bolivia, Ecuador, Ghana, Guatemala, Niger, and Pakistan each cycle twice. This pattern violates the absorbing-treatment assumption underlying the Callaway–Sant'Anna estimator and motivates the de Chaisemartin–D'Haultfœuille non-absorbing alternative as the correctly-specified comparison.

OA Table K2 compares the three estimators on the impact-year TFP effect for DEM $\rightarrow$

AUT events.

OA Table K2: Three-estimator comparison: impact-year TFP effect

Estimator	Treatment process	$\hat{\beta}$	SE	p-value
First-event LP (baseline, §6.5)	Single-event per country	−0.042	0.011	0.020 [< 0.001]
Callaway–Sant'Anna (`csdid`)	Absorbing staggered	−0.032	0.012	0.008
de Chaisemartin–D'Haultfœuille (`did_multiplegt_dyn`)	Non-absorbing switching	−0.015	0.010	0.137
		Avg. effect	SE	95% CI
Callaway–Sant'Anna post-treatment average	Absorbing	−0.051	0.028	[−0.107, +0.004]
dCDH avg. total effect across post-shock horizons	Non-absorbing	−0.039	0.025	[−0.088, +0.010]

Notes. The first-event LP coefficient is the DEM → AUT BBG-decomposition estimate at $h = 0$ reported in Section 6.5. Callaway–Sant'Anna uses the first DA year per country as the cohort variable and includes the lagged log of GDP per capita as the time-varying covariate. Controls are the never-switched countries that carry productivity data: of the 164 never-switched countries in the merged panel, 95 have TFP coverage and can serve as counterfactuals in a TFP regression. dCDH uses the non-absorbing switching indicator D_{it} defined in Subsection K.1 with five post-treatment effects, three placebo periods, and country-cluster standard errors. The p column is not one object across rows. For the first-event LP baseline it gives the wild cluster bootstrap value that governs this paper, with the asymptotic value in brackets. For Callaway–Sant'Anna it is that estimator's own z-test. The dCDH p-value is the two-sided normal approximation computed from the reported estimate and standard error, because `did_multiplegt_dyn` reports an interval rather than a p-value. Standard errors are each estimator's own default: the influence-function multiplier bootstrap of Callaway and Sant'Anna (2021) for `csdid`, and the analytic cluster-robust errors of de Chaisemartin and D'Haultfœuille (2024) for `did_multiplegt_dyn`, both clustered on country.

OA Table K3: Event-study paths under the staggered estimators (democracy-to-autocracy TFP)

	Coefficient	SE	Switcher obs.
First-event focal $\hat{\beta}_{\text{DA}}(h{=}0)$ (baseline)	−0.042	(0.011)	—
Panel A: Callaway–Sant'Anna event study (absorbing treatment)			
Pre-treatment average	−0.007	(0.008)	—
Impact year (h=0)	−0.032	(0.012)	—
h=1	−0.044	(0.030)	—
h=2	−0.068	(0.040)	—
h=3	−0.063	(0.035)	—
h=4	−0.053	(0.032)	—
h=5	−0.047	(0.032)	—
Post-treatment average	−0.051	(0.028)	—
Panel B: de Chaisemartin–D'Haultfœuille (non-absorbing)			
Impact year (h=0)	−0.015	(0.010)	20
h=1	−0.034	(0.021)	19
h=2	−0.042	(0.028)	19
h=3	−0.037	(0.026)	19
h=4	−0.035	(0.023)	19
Average total effect	−0.039	(0.025)	96
Placebo ($h = -1$)	+0.023	(0.011)	20
Joint nullity of the five post-treatment effects: $p = 0.70$			

Notes. Outcome is log TFP, lagged log GDP per capita as control, 1960–2022 panel. Panel A is the Callaway and Sant'Anna (2021) estimator (first-cohort grouping, absorbing treatment). Panel B is the de Chaisemartin and D'Haultfœuille (2024) estimator, which admits treatment reversal and is the correctly-specified comparison for the recurrent setting of OA Appendix K.1. The switcher-observations column counts switch-on events entering each dCDH effect: of the 65 switch-ons, 22 meet the trajectory-stability and TFP-coverage criteria and two are lost to the internal control-availability check, leaving 20, while the average-effect row's 96 sums switcher observations across horizons. The two post-treatment averages (−0.051, −0.039) bracket the baseline −0.042, both impact-year estimates are smaller, and the low-power joint test (p = 0.70) neither establishes an effect nor rules it out. The dCDH placebo at h=−1 (+0.023, standard error 0.011) is itself nominally significant at the 5% level and points in the same direction as the baseline pre-event coefficients. We carry it, a further reason to read the staggered comparison as inconclusive, not confirmatory.

The two staggered post-treatment averages bracket the baseline impact estimate, while both staggered impact-year estimates are smaller in magnitude. The first-event LP returns -0.042, csdid returns -0.032 at the impact year (average post-treatment effect -0.051), and dCDH returns -0.015 at the impact year (average total effect -0.039). The two post-treatment averages, -0.051 and -0.039, bracket the baseline, and all five share its sign. The full event-study paths under both estimators, the switch-on counts entering each dCDH horizon, the placebo, and the joint nullity test are in OA Table K3.

K.2 Identification–power trade-off

The Callaway–Sant'Anna estimator gains identification power by exploiting the 95 never-switched countries that carry TFP coverage, drawn from the 164 never-switched countries in the merged panel against each first-DA cohort, but at the cost of treating DA as absorbing in a setting where Argentina, Nigeria, and Honduras each cycle three times. The de Chaisemartin–D'Haultfœuille estimator specifies the treatment process correctly, allowing D_{it} to switch on and off, but pays for that correctness in identification power. Of the 65 switch-on events in the panel, only 20 enter the dCDH estimand for the impact-year effect. The estimator requires that the post-shock treatment trajectory remain stable for the duration of the dynamic horizon (here, five years), so switch-on events whose treatment status flips back to zero within the window are dropped. The attrition falls on events, not on countries: a high-cycling country can lose one switch-on event to the stability requirement and still contribute another, which is why countries such as Argentina, Nigeria and Honduras appear among the contributing switchers in OA Table K4 despite cycling repeatedly between regime types. The control pool is unaffected (95 never-switched countries with TFP coverage provide the counterfactual), but of the switch-ons twenty-two satisfy the trajectory-stability and TFP-coverage criteria, and twenty enter after the estimator's internal control-availability check.

This is an identification–power trade-off, not evidence that the underlying effect is null.

The dCDH joint nullity test on the post-shock horizons returns $p = 0.70$, but the point-estimate magnitude (-0.039 averaged across horizons, with individual horizon estimates ranging from -0.015 to -0.042) is aligned with the first-event LP baseline of -0.042 and with the csdid post-treatment average of -0.051. This magnitude alignment, alongside the dispersion of precision, is the pattern expected under the trade-off: the correctly-specified estimator preserves the signed-magnitude signal but loses statistical power because the non-absorbing structure of the empirical treatment process reduces the effective treated sample.

K.3 dCDH sample composition

OA Table K4 lists the 22 switchers that satisfy both the post-shock trajectory stability requirement (treatment remains on through $t+4$) and the TFP coverage requirement (non-missing $\ln \mathrm{TFP}$ at all of $t-1$ through $t+4$). The dCDH estimator uses 20 of these in the impact-year identification, the remaining two being lost on a control-availability check that the estimator applies internally.

OA Table K4: Composition of the dCDH contributing sample

Country	Switch year	DA event index	First DA event?	Canonical recovered
ARG	1966	2	No (subsequent)	**Onganía**
ARG	1976	3	No (subsequent)	**Videla (*Proceso*)**
BRA	1964	1	Yes	Castelo Branco
BDI	1997	1	No	—
BDI	2015	2	No (subsequent)	—
CAF	2003	1	Yes	—
CHL	1973	1	Yes	Pinochet
ECU	1964	1	No	—
FJI	2000	1	Yes	Speight
GRC	1967	1	Yes	Colonels
HND	1973	2	No	—
MRT	2009	1	No	—
NGA	1966	1	Yes	Aguiyi-Ironsi
NGA	1984	2	No	**Buhari (lag-1)**
NGA	2003	2	No	—
PER	1969	1	No	**Velasco (lag-1)**
PER	1990	1	No	Artificial switch (see notes)
SDN	1989	2	No	Bashir
THA	2014	3	No	**Prayut**
TUR	2016	2	No (subsequent)	—
URY	1973	1	Yes	Bordaberry
ZMB	1997	1	Yes	—

Notes. "DA event index" counts the country's democracy-to-autocracy coup events up to the switch year. A treatment re-entry without a contemporaneous coup retains the index of the most recent DA coup. "First DA event?" records whether the switch year is the country's first democracy-to-autocracy event under the multi-period coding, which keys on *any* coup. This is not the baseline criterion: the baseline first-event design keys on the country's *first coup of any type*. Four of the eight "Yes" rows (Brazil 1964, Chile 1973, Greece 1967, Uruguay 1973) are also that country's first coup and therefore overlap the baseline DEM → AUT cell. The other four (Central African Republic 2003, Fiji 2000, Nigeria 1966, Zambia 1997) are first DA events at *later* coups, because those countries' first coups were not democracy-destroying: the first coups of the Central African Republic, Fiji and Zambia began from autocracy, and Nigeria's first coup in the reconciled panel is a failed attempt dated October 1962 by Powell–Thyne (1963 in Bjørnskov–Rode, re-timed as described in Section 4.2), after which the democracy remained in place. A switch can also post-date its DA event by one year, because the treatment state requires the country to be currently autocratic and the regime code sometimes registers that only at $t + 1$. Such a row's switch year is not itself a DA-event year and it is marked "No". Where the lagged event is the country's first DA coup the row carries index 1 (Peru 1969), and where it is a later one the index is higher (Nigeria 1984). These are the *lag-1* cases flagged in the last column. The 22 country-year switchers satisfy the dCDH inclusion rule over five post-event horizons: $D_{i,t-1}$=0, D_{it}=1, $D_{it'}$=1 for $t' \in (t, t$+4], plus non-missing ln TFP at t−1 through t+4. Two are lost in the internal control-availability check (dCDH reports 20). Four overlap with the baseline first-event sample, the remaining eighteen are switch-on observations outside the baseline first-coup sample. Five canonical post-1960 democracy-destroying coups appear as subsequent-event switchers: Onganía 1966 (Argentina), Videla 1976 (Argentina), Velasco 1969 (Peru, lag-1 relative to the 1968 event), Buhari 1984 (Nigeria, lag-1 relative to 1983), and Prayut 2014 (Thailand). Other canonical events (García Meza 1980, Argentina 1961, Honduras 1963, Guatemala 1963/1982, Ghana 1972/1982, Pakistan 1977/1999, Mali 2012/2021) are excluded by the cycling diagnostic of §K.1: each is followed by a return to democracy within five years. PER 1990 enters via a BR regime-code transition unaccompanied by a coup event. The Fujimori 1992 *autogolpe* is classified as intra-autocracy under the same coding (footnote 9 in §6.5).

The dCDH joint nullity test on the five post-shock event-study coefficients returns $p = 0.70$, which under a default reading would be interpreted as failure to reject the absence of a treatment effect. With the available precision, the test cannot distinguish the null from the reported negative alternative. The joint test allocates power across five horizons identified from 20 effective switchers, with non-zero point estimates at every horizon (ranging from -0.015 to -0.042) and a clustered standard error of roughly 0.01–0.03 per horizon, the test cannot distinguish a joint -0.030 alternative from zero with the available identification mass. The average-total-effect p-value on the -0.039 average over $[-0.088, +0.010]$ is the same observation in a single-parameter form.

The structural reason for the precision loss is documented in Subsections K.1 and K.2. The baseline impact-year estimate (-0.042) lies between the two staggered estimators' post-treatment averages (csdid -0.051, dCDH -0.039). This is a descriptive cross-estimand magnitude comparison, not three estimates of the same parameter. The dispersion of p-values is consistent with differences in treatment-process assumptions and effective treated-sample size. It does not by itself distinguish the null from a negative effect.

The baseline focal estimate (-0.042) and the two post-treatment averages (-0.051 for Callaway–Sant'Anna, -0.039 for dCDH) all lie within $[-0.051, -0.039]$, though neither average is separately distinguishable from zero. The impact-year estimates from the two staggered estimators are smaller in magnitude (-0.032 and -0.015). No estimator provides evidence that the underlying effect is null. The first-event LP retained as the primary specification is justified on the avoidance of overlapping event windows (Section 5.2) and on the observation that the non-absorbing DA treatment process leaves the correctly-specified multi-period estimator under-powered relative to it. Callaway–Sant'Anna provides the more powerful but mis-specified absorbing benchmark, returning a magnitude consistent with the LP; de Chaisemartin–D'Haultfœuille provides the correctly-specified but less powerful non-absorbing benchmark, returning the same signed magnitude with a wider band.